BEST 'O' LEVEL BIOLOGY REVISION NOTES ®

'O' LEVEL BIOLOGY TOPICAL NOTES

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BEST 'O' LEVEL BIOLOGY NOTES

7.0. SCOPE AND SEQUENCE

TOPIC	FORM 3	FORM 4
7.1 Safety, Careers and Branches in Biology	• Safety in the laboratory • Branches of Biology • Careers	• Safety labels and symbols
7.2 Chemicals of life	• Constituents and identification of : - Water - Carbohydrates - Proteins - Lipids - Nucleic acids	Classification ,chemical structure and uses: - Carbohydrates - Proteins - Lipids - Nucleic acids
7.3 Cells and cellular activities	• Plant and animal cell structure • Cell specialization • Cellular transport	
7.4 Enzymes	• Nature and properties of enzymes • Mode of action	• Industrial application of enzymes
7.5 Plant Science	• Nutrition • Productivity • Transport	• Reproduction • Coordination and response
7.6 Animal Science	• Nutrition • Gaseous exchange • Respiration • Transport • Immunity • Sexual reproduction in humans	• Productivity • Homeostasis • Coordination and response • Endocrine system • Skeletal system
7.7 Microbiology and Biotechnology	• Characteristics Types and economic importance of microorganisms	• Recombinant Gene Technology
7.8 Genetics	• Chromosomes and Genes • Monohybrid inheritance • Mutations	• Variation • Selection
7.9 Biodiversity	• Classification	• Threats and conservation measures
7.10 Ecosystems	• Ecosystems • Natural systems • Artificial systems	• Management of ecosystems
7.11 Health and Diseases	• Health • Diseases (Infectious and non-infectious)	• Drug use and abuse

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TOPIC 1: SAFETY, CAREERS AND BRANCHES IN BIOLOGY

FORM 3: SAFETY IN THE LABORATORY

KEY CONCEPT	LEARNING OBJECTIVES Learner should be asble to	CONTENT (Attitudes, Knowldege and Skills)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.1.1 Safety in the laboratory	• identify causes of accidents in the laboratory • outline laboratory safety rules • perform fire drills • memorise the local emergency numbers • make a fire guard around the laboratory • demonstrate use of the First Aid Kit	• Causes of accidents in the laboratory - Fire - Fumes - Acids and strong bases - Handling of micro organisms - Improper handling of apparatus - Electricity • Laboratory safety rules - Fire drills - Local emergency call numbers	• Identifying causes of accidents in the laboratory. • Listing accidents that may occur during experiments. • Demonstrating proper handling of apparatus. • Discussing electrical hazards. • Discussing laboratory safety rules. • Performing mock fire drills. • Demonstrating emergency call. • Preparing a fire guard around the laboratory. • Demonstrating use of the contents of the First Aid Kit.	• Charts showing electrical hazards • ICT tools • Brail software/Jaws • Fire extinguishers • Fire blankets • Sand buckets • Fire alert bells • Fume cupboards • Goggles and masks • Gloves • First Aid Kit

SAFETY IN THE LABORATORY

A **laboratory** is a special room or building designed for carrying out experiments. A laboratory has:

1. Large windows to allow enough air and light for better ventilation.
2. Spacious to allow enough visibility.
3. Shelves – for keeping chemicals, specimens, apparatus and models.
4. Supply of gas, electricity and water.
5. Working benches.
6. An emergence door in case if danger occurs.
7. Preparation room.
8. First Aid Kit.
9. Fire extinguishers.

THE BIOLOGY LABORATORY SAFETY RULES

- Biology laboratory has sophisticated instruments which need to be handled with special care. Most laboratory chemicals are harmful and need to be handled with care.
- The following laboratory rules should be adhered to:

1. Do not enter the laboratory without permission from the teacher.
2. ALWAYS observe normal laboratory safety practices.
3. Long hair MUST always be tied securely at the back.
4. Do not play, or run unnecessarily in the laboratory i.e. no **horseplay** in the lab.
5. NOTHING must be tasted, eaten, smoked or drunk in the laboratory. (Do not eat, drink or smoke in the laboratory. Never test laboratory substances using the sense of taste).
6. Never smell substances, specimens, chemicals or gases directly.
7. Do not store food in the laboratory.
8. Do not preserve food and drink in laboratory refrigerators.
9. Do not use laboratory glassware for eating or drinking.
10. Do not steal from the laboratory.

11. DO NOT use or handle any equipment, chemicals or specimens unless permitted to do so by the teacher.
12. Clean the apparatus and benches after the experiment.
13. Do not dispose of wastes in the sink.
14. Never use mouth suction for pipetting or starting a siphon.
15. Never use human tissue, blood or fluids in experiments.
16. Return apparatus and chemicals to their normal positions after use.
17. Avoid unnecessary noise in the laboratory.
18. Use equipment only for its designed purpose.
19. Any accident or damage of apparatus must be reported immediately.
20. Label chemicals and specimens to avoid confusion.
21. Always keep flammable substances away from flames.
22. Turn off water and gas taps after use.
23. Never point the open end of the test tube to your fellow or yourself when heating.
24. Do not play with electrical power sockets.
25. Monitor instruments while they are operating.
26. Follow the safety precautions provided by the manufacturer when operating instruments.
27. Wash hands immediately with soap after an experiment with toxic chemicals.
28. Clean up any spillage immediately. Use appropriate materials for each spillage.
29. Unauthorized persons should be kept out of a laboratory. Visitors should always be accompanied by authorized personnel.
30. Avoid working alone. If you must work alone, have someone contact you periodically.
31. Use forceps, tongs, or heat-resistant gloves to handle hot objects.
32. Learn emergency first aid.
33. Seek medical attention immediately if affected by chemicals and use first aid until medical aid is available.
34. Learn what to do in case of emergencies e.g. fire, chemical spill etc.
35. Use a fume cupboard when handling concentrated acids, bases, and other hazardous chemicals.
36. All electrical, plumbing, and instrument maintenance work should be done by qualified personnel.
37. Access to emergency exits, eye-wash fountains and safety showers must not be blocked.
38. Do not dispose of any hazardous chemicals through the sewer system. These substances might interfere with the biological activity of waste water treatment plants, create fire or explosion hazards, cause structural damage or obstruct flow.
39. Use personal safety equipment as described below.
    - (a). Body protection: laboratory coat and chemical-resistant apron.
    - (b). Hand protection: gloves, particularly when handling concentrated acids, bases, and other hazardous chemicals.
    - (c). Dust mask: when crushing or milling/grinding samples, etc.
    - (d). Eye protection: safety glasses with side shields. Persons wearing contact lenses should always wear safety glasses in experiments involving corrosive chemicals.
    - (e). Full-face shields: wear face shields over safety glasses in experiments involving corrosive chemicals.
    - (f). Foot protection: proper footwear should be used. Do not wear sandals in the laboratory.

HAZARDS AND ACCIDENTS IN THE LABORATORY

What is a hazard?

– A hazard is a **source of danger** which may **cause accidents** in the laboratory.

What is an accident?

– An accident is something unexpected that may cause injury and sometimes lead to illness or death. Some accidents may be difficult to predict and prevent.

Below is a list of some common accidents in the laboratory:

- Burns caused by flames and hot liquids.
- Burns caused by corrosive chemicals such as concentrated acids and strong alkalis.
- Slipping and falling due slippery floors.
- Cuts and scratches caused by sharp objects such as scalpel blades and broken glass.
- Electric shock due to faulty electric sockets.
- Poisoning caused by eating, drinking or smelling toxic substances.
- Outbreak of fire due to leaking gas tapes and flammable liquids and gases.
- Explosions
- Loose hair and clothing catching fire.

SOME COMMON HAZARDS OR CAUSES OF ACCIDENTS IN THE LABORATORY

1. Toxic chemicals.
2. Improper handling of apparatus.
3. Not observing lab rules.
4. Fire, flammable liquids and gases.
5. Hot liquids.
6. Fumes.
7. Acids, bases and oxidants.
8. Improper handling of microorganisms.
9. Horseplay.
10. Not following instructions.
11. Sharp instruments.
12. Electricity etc.

SAFETY BEGINS AT HOME

– Pupils should be aware that **SAFETY BEGINS AT HOME**.
– The safety basics learnt at home should also transcend to the school and laboratory environment.

WAYS OF MAINTAINING SAFETY IN THE LABORATORY

Ways of preventing accidents in the laboratory include the following:

- Wear protective clothing where necessary. Gloves, goggles and lab coat are examples of protective clothing.
- Poisonous chemicals should be kept under heavy lock and key/ Poisonous chemicals should be kept out of reach of children.
- Pupils should be monitored closely when doing experiments.
- Sharp objects like broken bottles, scalpel blades should be well disposed of.

- Laboratory chemicals should be labelled appropriately.
- Dangerous chemicals and equipment should be labelled with hazard/warning symbols.
- Students should observe and adhere to laboratory rules.
- Flammable substances should be kept away from flames.
- Laboratory doors should open outwards for easy exit in case of fire. An emergence door is also necessary for this case.
- Turn off all the gas taps and water taps after an experiment.

TRY THESE QUESTIONS

1) What do you understand by the term accident?
2) State some common accidents that can happen in the laboratory.
3) What do you understand by the term hazard?
4) State some common hazards in the laboratory
5) Outline ways of preventing accidents in the laboratory.

FIRE

– Fire can start in the laboratory due to:

1. Electrical faults
2. Flammable liquids and gases near naked flames or sparks.
3. Leaking gas cylinders, gas taps and gas pipes.
4. Oxidants – substances which liberate oxygen readily.

- The no naked / open flame sign is often put where flammable substances are stored such as flammable liquids and gases. Examples of flammable substances include methylated spirits, ethanol (ethyl alcohol) and Bunsen gas.

DIAGRAM: No naked flames symbol

FIRE FIGHTING – Putting out a fire in the laboratory

Necessary items and equipment:

1. Fire blanket

- Fire blankets are blankets which are manufactured using fire resistant materials.
- Fire blankets are used **to extinguish fires in the laboratory, mainly fats, oils, waste bin and clothing fires**.
- Covering with a fire blanket cuts the supply of oxygen to the burning contents and works by smothering the fire.

DIAGRAM: Fire blanket symbols.

- **Ways To Use A Fire Blanket:**

a) Fire blankets are used to extinguish fires due to fats and oils. The first thing you must do (if possible) is to turn off the heat source, then stretch the blanket out so that it is large enough to cover the whole flame. Place the fire blanket over the burning object to smother the flames, leaving it in place for at least 30 minutes to make sure that the fire is out. If you were unable to switch off the heat source before you started then switch it off as soon as you can safely do so. Then leave the room, close the door and make a telephone call to the fire brigade.

b) Fire blankets are also useful if a person’s clothes catch fire. Surround them in the blanket (making sure to keep your hands safely wrapped in the blanket) and encourage the person to roll on the floor until the flames have been extinguished.

c) Another great use for a fire blanket is to act as protection if you need to walk through a burning room. Wrap yourself, baby, child or someone else safely in the fire blanket as you pass through the fire affected area.

2. Sand bucket

- A fire sand bucket is used **to extinguish fires by filling it with sand, and throwing the sand over the fire**.
- Typically, a fire sand bucket is painted bright red, with the word “Fire” printed in white.

PICTURE: Fire sand bucket.

3. Fire hose and Fire hydrant

- A **fire hose** is a high-pressure hose that **carries water or other fire retardant (such as foam) to a fire to extinguish it**. It is attached either to a fire engine, fire hydrant, portable fire pump or to the plumbing system of a building.

DIAGRAMS: Fire hose and fire hydrant symbols.

- **Fire hydrants** are devices for **extracting water from pipelines and water distribution systems for the purposes of fighting fire**.
- In the event of a fire outbreak, a fire hydrant can assure fast water supply by connecting a fire hose or a fire truck to it.

4. Fire extinguisher

- Types of fire extinguishers:

(a) CO_2 fire extinguishers.
(b) Halon fire extinguisher, aka B.C.F., Bromochlorodifluoromethane, (haloalkenes / halogenated hydrocarbons).
  - CO_2 and halon extinguishers can be used without causing damage to electrical equipment. The extinguishing power of halon is about 6 times that of CO_2!

(c) Water fire extinguisher: water has the disadvantage that it conducts electricity.
(d) Powder extinguishers (containing salts) cause damage to instruments.

Fire extinguisher symbol.

How to use a fire extinguisher.

Response Actions to fire

- When fire is detected stay calm, try to oversee the situation and watch out for danger.
- Take immediate response actions in the following order (use the acronym **REACTS** to help you remember what to do in case of an emergency)**:**

  R – **Rescue** and **relocate** anyone in immediate danger.

  E – **Emergency call**; call for help, call 911 from a SAFE location

  A – **Alert** others by shouting and activating the building fire alarm. **Ask** them to evacuate immediately.

  C – **Confine** the emergency by closing the doors and windows of the room where the fire is.

  T – **Take action**; Switch off electricity and/or gas supply. Use an extinguisher if you can do so safely.

  S – **Show/tell** the Fire Department the exact location of the fire.

- Persons with burning clothing should be wrapped in a fire blanket on the floor, sprayed with water or be pulled under a safety shower. A CO_2 fire extinguisher can also be used, but do not spray in the face.
- When using fire extinguishers it is important that the fire is fought at its base and not in the middle of the flames.
- If gas cylinders are present there is the danger of explosion by overheating. If they cannot be removed, take cover and try to cool them with a fire-hose. When the situation looks hopeless, evacuate the building. Let everybody assemble outside and check if no one is missing. To practice this, a regular fire drill (once a year), should be held.

FIRST AID

- First aid is emergency care given to a patient before regular medical aid can be obtained.
- First aid is an immediate help which is given to a sick or injured person before sending him/her to hospital for further treatment.

DIAGRAM: First Aid Symbols.

Importance of First Aid

1) Saves life.
2) Reduces fear of death.
3) Brings hope and encouragement to the patient.
4) Relieves the victim‘s pain.
5) Prevents the illness or injury from becoming worse.
6) Helps a person to recover from shock.
7) It shows spirit of helping each other.

FIRST AID KIT

DIAGRAMS: First Aid Kit Symbols

- First aid kit is a small container consisting of a set of components for giving first aid.
- First Aid kit is a small box which is used to keep instruments and chemicals for First Aid.
- The first Aid kit should be placed in a safe and accessible place.

Components of the First Aid Kit and their Uses

- *A pair of scissors:* is used for cutting dressing materials such as gauze, bandages·
- *Gauze:* is used to cover the wound to prevent dirt and micro-organisms from entering.
- *Assorted bandages*: are used for securing an injured part in order to protect and support it.
- *Adhesive plaster:* are used for covering minor wounds/cuts and grazes·
- *Cotton wool:* is used for cleaning and drying wounds and applying medicine·
- *New razor blade:* used for cutting any flaps of skin when cleaning the wound.
- *Gentian Violet (GV):* is used as an antiseptic to clean wounds.
- *Petroleum jelly or Vaseline:* used for treatment of burns
- *Safety pins:* used for holding/securing bandages.
- *Iodine tincture or spirit:* used for cleaning wounds to reduce bleeding
- *Soap:* is used for washing wounds, hands and medical facilities.
- *Analgesics (pain killers)* e.g. *Panadol and paracetamol:* used to reduce pain.
- *Sterilized forceps and pins:* are used for removing splinters and grit from wounds.
- *Liniment*: used to reduce muscle pains.
- *A pair of tongs:* used for holding pieces of bandages when cleaning the wounds.
- *Antibiotic solution:* is used for applied to wounds to kill micro-organisms.

FIRST AIDER

- A **First Aider** is a specialist who gives first aid / emergency care.

Qualities of a first Aider

1. S/he should have the ability to quickly assess the problem and give appropriate help immediately.
2. S/he must be vigilant and able to act quickly, quietly, calmly.
3. S/he should be sympathetic to the victim.
4. S/he should be able to recognize dangerous signs of injury and give immediate help, for example detecting immediately if
   - breathing has stopped or is failing
   - there is severe bleeding
   - poisoning
   - fractures, etc.
5. S/he should be able to help the injured person without unnecessary movements.

Precautions to be observed by the First Aider

- The First Aider should keep himself/herself safe to avoid dangers from the patient. Some of the dangers that s/he may face include infection by pathogens such as viruses and bacteria. So they should:
  - Wear protective gloves to avoid contact with blood.
  - Wear eye protection.
  - Wear masks and gowns.

FORM 3: BRANCHES OF BIOLOGY

KEY CONCEPT	LEARNING OBJECTIVES Learner should be able to	CONTENT (Attitudes, Knowldege and Skills)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.1.2 Branches of Biology	• identify various branches of Biology	• Branches - Cytology - Anatomy - Physiology - Ecology - Genetics - Biotechnology - Microbiology - Zoology - Botany	• Discussing the various branches of Biology	• Relevant reference materials • ICT tools

WHAT IS BIOLOGY AND WHY IS IT IMPORTANT?

- Biology is the science that studies living organisms and their environments.

THE IMPORTANCE OF STUDYING BIOLOGY

1. Increases our understanding of the environment and ways of conserving it.
2. Increases our understanding of living organisms and ways to conserve them.
3. Helps us to understand ourselves better since we are living things.
4. Enables us to apply biological skills and knowledge to other scientific fields such as medicine, agriculture, nutrition etc.
5. Enables us to acquire food, shelter, clothing and other useful products from living organisms.
6. Helps us to avoid taboos, superstitions, magical beliefs and other traditional beliefs.
7. Helps us to understand and treat diseases.
8. Increases our understanding of genetics and inherited characteristics e.g. albinism, haemophilia, sickle cell anaemia, etc.

BRANCHES OF BIOLOGY

- Biology has many branches. The two major branches are **botany** and **zoology.**
  - **Botany** is a branch of biology which deals with the study of plants. A person who studies botany is called a *botanist.*
  - **Zoology** is a branch of biology which deals with the study of animals. A person who studies zoology is called a *Zoologist.*
- **Some of the other branches of biology are given below.**
  - **Anatomy** is the study of the structure of living organisms. A person who studies anatomy is called an *anatomist*.
  - **Biotechnology** is the study of the use of microorganisms to perform specific industrial

processes. A person who studies biotechnology is called a *biotechnologist*.

- **Cytology** is the study of microscopic structures of plant and animal cells. A person who studies cytology is called a *cytologist*.
- **Ecology** is the study of relationships among living things and between organisms and their surroundings. A person who studies ecology is called an *ecologist.*
- **Genetics** is the study of genes, heredity and variation in living organisms. A person who studies genetics is called a *geneticist*.
- **Evolution** is the study of the origin of life and organism and changes of species over time. A person who studies evolution is called an *evolutionist*.
- **Microbiology** is the study of microorganisms (organisms that can only be seen with a microscope) e.g. bacteria, viruses, some fungi and some protists. A person who studies microbiology is called a *microbiologist*.
- **Mycology** is the study of fungi. A person who studies mycology is called a *mycologist.*
- **Morphology** is the study of the form and structure of organisms. A person who studies morphology is called a *morphologist*.
- **Physiology** is the study of functions and activities of cells and parts of body. A person who studies physiology is called a *physiologist*.
- **Bacteriology** is the study of bacteria. A person who studies bacteriology is called a *bacteriologist.*
- **Virology** is the study of viruses. A person who studies virology is called a *virologist.*
- **Immunology** is the study of the body‘s defense against diseases and foreign substances. A person who studies immunology is called an *immunologist.*
- **Entomology** is the study of insects. A person who studies entomology is called an *entomologist*.
- **Parasitology** is the study of parasites and their effects on living organisms. A person who studies parasitology is called a *parasitology.*
- **Dermatology** is the medical study of the skin and its diseases. A person who studies dermatology is called a *dermatologist.*
- **Taxonomy** is the study of the classification of organisms. A person who studies taxonomy is called a *taxonomist*.
- **Agriculture** is concerned with production of crop plants and livestock through farming systems. Genetically modified organisms (GMOs) in agriculture ensure better quality, early maturity and high yield products.
- **Forestry** is the science of managing forest resources for human benefit. The practice of forestry helps maintain an adequate supply of timber and management of such valuable forest resources such as water, wildlife, grazing areas and recreational areas.

FORM 3: CAREERS IN BIOLOGY

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (ATTITUDES, SKILLS AND KNOWLEDGE)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.1.3 Careers in Biology	• list Biology related careers	• Careers such as: - Medicine - Research scientists - Ecologists - Conservationists - Dieticians	• Researching on careers related to various branches of Biology.	• ICT tools • Braille software/Jaws • Career guidance booklet • Resource persons

CAREERS IN BIOLOGY

RESEARCH AND TEACHING CAREERS

- Researchers work for laboratories, clinics, hospitals and government agencies.
- To be a professional researcher in biology and/or to teach at a college or university, you will need to earn a master's and possibly a doctoral degree.
- Research and teaching careers include:
  - **forensic biologists** -- work with law enforcement agencies to help solve crimes by applying scientific methods to uncover evidence;
  - **cytotechnologists** -- use microscopic interpretation of cells to detect diseases;
  - **medical technologists** -- perform tests on body fluids using state-of-the-art medical equipment;
  - **biotechnologists** -- help develop and improve medical treatments and environmental products;
  - **biochemists** -- conduct cell research in laboratories, clinics, hospitals and government agencies, or create new and beneficial products for humanity and the environment such as medicine, agricultural chemicals, cosmetics, food and beverages, environmentally friendly cleaners and biodegradable plastics;
  - **business biologists** -- work with chemical or pharmaceutical companies and other organizations to research and test new products.

HEALTHCARE CAREERS

- A major in biology can also prepare you to enter dental, medical or veterinary school or a training program to become a healthcare professional such as an optometrist or pharmacist.
- Healthcare careers include:

- **Medicine –** the science and art of preserving health and treating diseases and injuries.
  - A person who practices is called a ***medical doctor***.
  - Medicine is a science because it is based on knowledge gained through careful study and experimentation. It is an art because its success depends on how skilfully medical practitioners apply their knowledge in dealing with patients. The goal of medicine include saving lives, relieving suffering and maintaining the dignity of sick people. Biological knowledge helps the doctors, surgeons and nurses to diagnose, treat and prescribe the right medicine to cure a disease.
  - Biological knowledge will also help them to offer education to the patients on how to prevent themselves from the diseases e.g. purifying drinking water, vaccination against polio, measles and other diseases.

- **Veterinary science (Veterinary medicine)** – the branch of **medicine** that deals with the diagnosis and treatment of diseases and injuries of animals (especially domestic animals).
  - Doctors that treat animals are called *veterinary doctors* or *Veterinarians*.
  - Veterinarians are trained to prevent, diagnose and treat diseases in animals. Their work is valuable because many animal diseases can be transmitted to human beings e.g. rabies, tuberculosis, tularemia (rabbit fever) anthrax etc.
  - Basic knowledge of biology is required for successful study of veterinary science.

- **Pharmacy** is the profession concerned with the preparation, distribution and use of drugs.
  - Members of this profession are called *pharmacists* or *druggists (A pharmacist is* a health professional trained in the art of preparing and dispensing drugs).
  - Pharmacy also refers to a place where drugs are prepared or sold.
  - The drugs are made depending on the chemical composition of the body of an organism and how they can react with such medicines.
  - **Pharmacology** is another medical science related to pharmacy.
  - **Pharmacology** is the science or study of drugs, their preparation and properties and uses and effects and possible remedies to be taken.
- **Nutrition** is the science which deals with food and how the body uses it.
  - A specialist in the study of nutrition is called a ***nutritionist*** or ***dietician***.
  - People, like all living things need food to live. Food provides substances that the body needs to build and repair its tissues and to regulate its organs and systems. Food also supplies energy for every action we perform. Knowledge of biology helps to identify the type of food required by an individual based on its quality and quantity.

OTHER CARREERS IN BIOLOGY

- An **ecologist** is a person who studies ecology.
- A **conservationists** someone who works to protect the environment from destruction or pollution.
- An **agronomist** is an expert in soil management and field-crop production.

FORM 4: SAFETY LABELS AND SYMBOLS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (ATTITUDES, SKILLS AND KNOWLEDGE)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.12.1 Safety labels and symbols	• identify various safety labels and symbols • explain the meaning of various safety labels and symbols	• Labels and symbols	• Identifying various safety labels and symbols • Discussing the meaning of the various labels and symbols	• Containers with labels and symbols • Print media

SAFETY / HAZARD SYMBOLS

- Laboratory safety symbols are placed around the lab. They are usually located on chemical containers, cabinets, walls, and equipment.
- Laboratory safety symbols are visual reminders of potential hazards.
- Knowing the meaning of each symbol enables you to recognize potential dangers and hazards.
- Understanding the hazards will allow you to take proper safety precautions when working

with a particular chemicals or equipment.

Flammable Substance


- Flammable substance – easily catches fire.
- Flammable substances must not be handled near oxidizing agents, naked flames or sparks.
- Small containers of flammable substances should be stored in separate storage facilities inside the laboratory, but large containers should be stored in special security storage cabinets outside the laboratory.
- Some examples of flammable chemicals are ethanol, isopropanol, methylated spirits, hydrogen gas and petrol.

Oxidizing Substance


- Oxidizing substances are oxidizing agents that rapidly provide oxygen to another substance and can cause fire to burn fiercely.
- Oxidizing agents usually transfer oxygen to flammable substances and cause them to burn. Therefore, oxidizing substances should always be stored separately from flammable substances!
- Oxidizing agents may also set you and your clothing on fire if you‘re not careful—so don‘t forget your gloves, eye protection, and lab coat as precautions.
- Examples of oxidizing substances are hydrogen peroxide, potassium permanganate, chlorine and fluorine.

Corrosive Substance


- Corrosive substances are strong chemicals that can corrode your skin or other substances.
- Examples of corrosive substances are strong acid solutions (e.g. sulphuric acid) and strong alkali solutions (e.g. sodium hydroxide).
- One drop of these corrosive substances can cause you serious eye damage! When working with corrosive substances, non-corrosive gloves, eye protection, and lab coats are all essential.
- You should also make sure you know where the eye-wash station is, in the event of any accidents.

Toxic Substance

- Toxic substances are poisonous substances that can cause severe damage or death if swallowed, inhaled, or absorbed through the skin.
- Examples of toxic chemicals are potassium cyanide, mercury and sodium azide.
- You should always use eye protection, gloves, and a face mask to prevent inhalation when working with toxic substances. And don't forget: handle the chemical *inside a fume cupboard*.

Harmful

- A substance that may cause harm in some way.
- A substance which if it is inhaled, ingested or penetrates the skin may cause limited health risks.
- Examples of harmful substances are iodine and copper sulphate.

Irritant

- Irritants are substances can irritate your eyes and skin, causing itchiness, soreness, redness, and blistering. They can also cause toxicity if inhaled or swallowed.
- Examples of irritants are ordinary smoke, iodine, calcium chloride, sulphur dioxide and ammonia.

- Always wear protective clothing when working with irritants.

Health Hazard

- These chemicals can cause serious health damage such as respiratory problems, reproductive toxicity, mutation of gametes, and cause cancer.
- For example formaldehyde, ethidium bromide, phenol, chloroform and acrylamide belong to this category.
- Phenol is a reproductive toxin and its vapour is also corrosive to the eyes, skin, and respiratory tract.
- Acrylamide is carcinogenic and neurotoxic too.
- When working with these chemicals always wear appropriate PPE (personal protective equipment) such as eye protection, nitrile gloves, a lab coat, and a face mask.

Environmental Hazard

- These chemicals damage or pollute the environment.
- They can contaminate soil and water and kill aquatic animals and trees if not properly disposed of.
- You should be very careful while disposing of these substances!
- For example, bromoform and phenol are environmental hazards.

Explosive

- An explosive is a substance that may explode if it comes in contact with a flame or heat. It may also explode due to friction or shock.
- Examples of explosives substances are ammonium nitrate in contact with heat, sodium and potassium elements in contact with water, mixture of hydrogen and oxygen in contact with a fire/spark.
- Sodium is usually stored in liquid paraffin to exclude water and air that may cause it to explode.

Compressed Gas

- Gas cylinders and aerosol cans are compressed gases that should be treated with caution.
- When accidentally heated, gas cylinders and aerosol cans can explode.
- Compressed gas cylinders are normally stored separately from the main lab in special safety cabinets.

Glassware Hazards

- Glassware has the tendency to become a health hazard, especially if broken and contaminated with toxic and infectious substances.
- Serious cuts can occur due to broken glassware.

Biohazard

- Biohazards are living microorganism that may cause infection or disease.
- This safety sign warns you that you are entering an area in which human samples (e.g. cells, tissues and fluids), bacterial samples, used syringes etc, are present.
- This sign can be found on the doors or waste bins of your lab to indicate the presence of biohazardous materials which are highly contagious. The sign also indicates where to discard waste associated with living organisms.
- The biohazardous materials should not escape into the environment as they could contaminate it and also cause diseases.
- When working with these chemicals always wear appropriate PPE (personal protective equipment) such as a lab coat, gloves and a face mask.
- Also make sure you know what bins you should use for disposal of biohazardous wastes.
- In clinics, hospitals and medical laboratories, biohazardous wastes are usually disposed of by burning them in incinerators.

Carcinogen Hazard

Carcinogen hazard – It pertains to human carcinogens (cancer-causing chemicals) such as methylene chloride, formaldehyde, and benzene. This sign suggests that you have to wear proper personal protective equipment.

Laser hazard

- Lasers can harm your eyes, causing severe injury; you should be well protected before entering labs that use lasers.

Ionizing Radiation

- This safety sign is associated with x-ray rooms and labs that use radioactive isotopes.
- Radioactive substances produce ionizing radiation which can damage cells, cause cancer and mutations.
- Examples of radioactive substances include x-rays, radioactive isotopes such as cobalt-60 and uranium-235. These produce radiation in the form of alpha, beta and gamma rays.
- Always wear protective clothing such as lead coated lab coats and eye protection when working with radioactive substances.
- Radioactive isotopes should be stored in lead boxes, since the radiation cannot penetrate lead.

Electrical Hazard

- Electrical hazard – Electrical hazards in the lab that can cause mild tingling and death. Devices labelled as electrical hazards should always be turned off when not in use.

High Voltage Hazard

- This sign is associated with laboratory appliances that use high voltage.
- Always switch off the electrical appliances after use.
- Never operate electrical appliances in damp conditions because:
  (a). water is conductive and can cause electrocution of the user.
  (b). water reduces the resistance of the human body leading to electric shock.

Hot Surface Hazard (Hot Temperatures hazard)

- The **hot surface hazard** sign warns of the possibility of burns from hot surfaces.
- This symbol is found in lab equipment that produces extreme heat such as electric hot plates, autoclaves and lab ovens for heating and sterilizing purposes.
- Always handle hot objects with tongs and heat resistant gloves to avoid nasty burns.

Low Temperature Hazard (Cold Temperatures hazard)

- The **cold temperature hazard** symbol pertains to a very low temperature hazard (cryogenic hazard) inside the lab such as:
  (a). cold storage areas where chemicals like liquid nitrogen are stored.
  (b). –80°C freezers and liquid nitrogen refrigerators used to freeze and preserve samples.
- Special cold-resistant protective gloves are for handling very cold objects and samples from these freezers to avoid frostbite.

Form 3: CELL SPECIALISATION

SUB TOPIC	LEARNING OBJECTIVES Learner should be able to:	CONTENT (Attitudes, Knowledge and Skills)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.3.2 Cell Specialisation	• identify specialized cells • relate cell structure to function	• Palisade – photosynthesis • Root hair – absorption • Neurone – transmission of impulse • Red blood cell – transport of oxygen • White blood cell – immunity • Muscle cell – contraction • Sperm –fertilization and motility • Ovum – fertilization and food reserve	• Relating cell structure to function • Examining specialised cells using bioviewers, microscopes and photomicrographs • Drawing labeled specialised cells	• Biosets • Bioviewers • Photomicrographs • Print media • Microscopes • ICT tools • Braille software/Jaws

CELL SPECIALISATION

- As an organism develops differentiation occurs.
- Differentiation is when a cell changes to become specialised for its job.
- Cells change as they develop in order to carry out specific functions.
- The cells change and become adapted to their functions. Such cells are called specialised cells.

Specialised cells

- A specialised cell is a modified cell which is designed to do a particular job
- Examples of specialised cells and their specialisation:

  - **Red blood cells** are specialised for carrying oxygen.
  - **Muscle cells** are specialised for contraction.
  - **Palisade cells** are specialised for photosynthesis and gaseous exchange.
  - **Root hair cells** are specialised for absorption of water and mineral salts.
  - **White blood cells** are specialised for immunity.
  - **Sperm cells** are specialised for motility and fertilisation.
  - The **ovum** is specialised for fertilisation and food reserve.
  - The **neurone** is specialised for transmission of impulses.

How different types of cells are adapted to carry out their functions

1) Red blood cells (a.k.a. erythrocytes)

Red blood cells are found in blood.

- **Function:** carry/transport oxygen from lungs to respiring tissues.

Structure of red blood cells

– Flexible circular, disc shape (doughnut shape/biconcave shape).

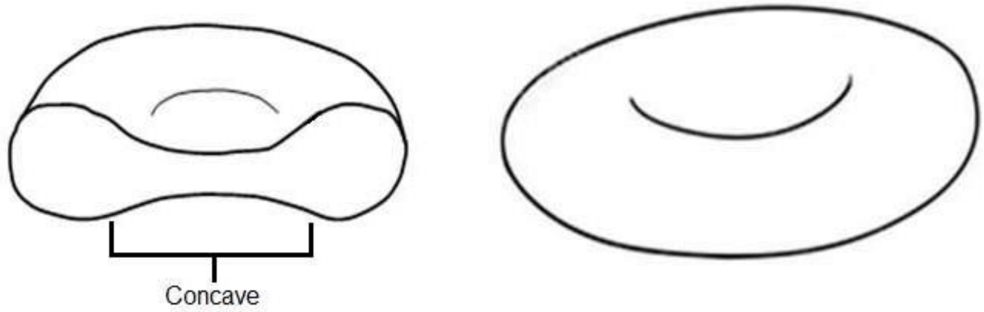
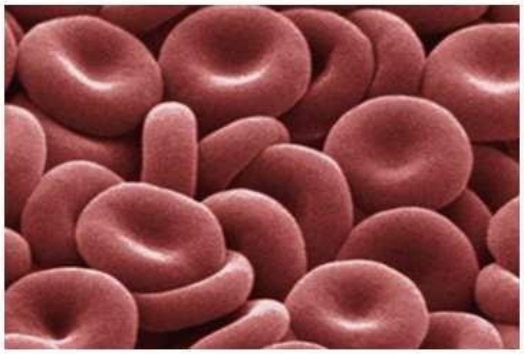
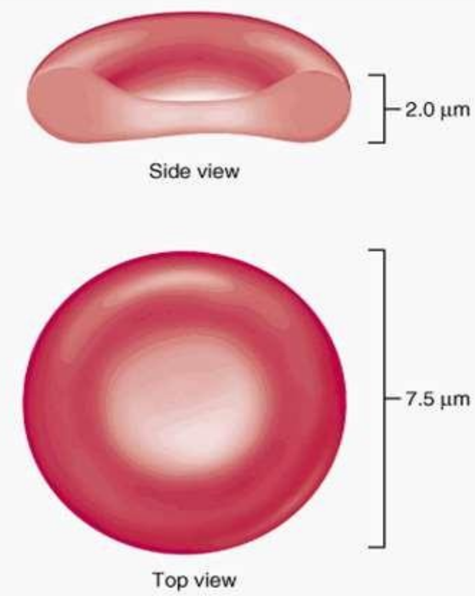


DIAGRAM: Red blood cells are biconcave shaped.



Red blood cells are disc shaped (doughnut shaped or biconcave shaped). The biconcave shape increases surface area for gaseous exchange.



Adaptations of red blood cells

1. Contain **haemoglobin** (red pigment) which **combines with oxygen**.
2. **No nucleus** and **organelles** to **make space** for **more haemoglobin**.
3. **Biconcave shape** to **increase surface area** to **volume ratio** for faster diffusion of oxygen.
4. **Flexible** so can **squeeze through narrow capillaries**.

2) White blood cells (a.k.a. leucocytes)

- White blood cells are found in blood.
- Function: White blood cells fight infection.
- White blood cells are part of the immune system.
  - White blood cells are produced in the bone marrow.
  - White blood cells can weaken and kill pathogens including viruses, fungi, bacteria and protists.
- Two main types of white blood cells are:
  - **Phagocytes** - which engulf and then break down pathogens. They have a multi-lobed nucleus.
  - **Lymphocytes** - which produce antibodies which attach themselves to pathogens to make it easier for the phagocytes to engulf them and produce antitoxins to cancel out the effects of toxic chemicals made by the pathogens. They have a spherical

nucleus.

Adaptations of white blood cells

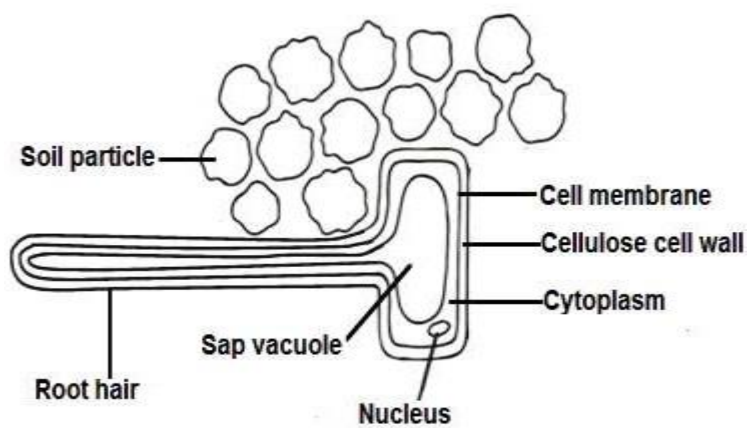
(a) Irregular shape, they can change shape to squeeze out of blood vessels and get to the site of infection.
(b) Others have cytoplasm which can flow making it possible for the cell to change shape, surround and engulf bacteria.
(c) Can increase in numbers to fight disease.

3) Root hair cells

- **Root Hair Cells** are found in the roots of plants.
- Their function is to **absorb water** and **mineral salts** in the soil.

Structure of root hair cells

- Long tubular extension. Vacuole with concentrated sap.



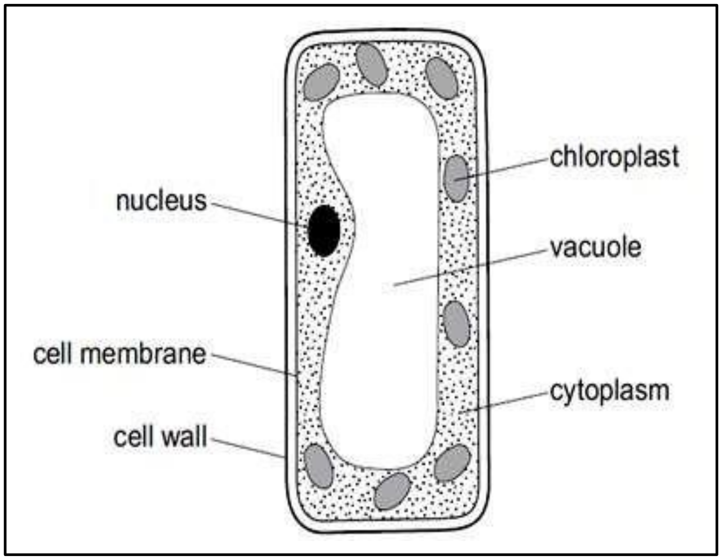
Adaptations of root hair cells to their functions

1. **Elongated** and **tubular** to **increase surface area** for absorption of water and mineral salts.
2. **Large amounts** of **mitochondria** to provide **more energy** for **active transport**.
3. **Sap vacuole lowers** its **water concentration** so that **water can be absorbed by osmosis**.

4) Palisade cell

- Found in plant leaves.
- **Function:** photosynthesis and gaseous exchange.

Structure of palisade cell



Adaptations of palisade cell

1. Packed with a light absorbing pigment called chlorophyll.
2. Tall/elongated shape increases surface area for absorbing carbon dioxide from the air inside the leaf.
3. Thin and elongated so many can be tightly packed near the top surface the leaf. This means that the palisade cell layer is close to the source of light.

5) Muscle cell

- Found in skeletal muscles
- Function: contraction.

Structure of muscle cell

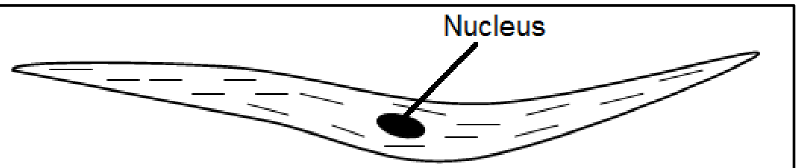


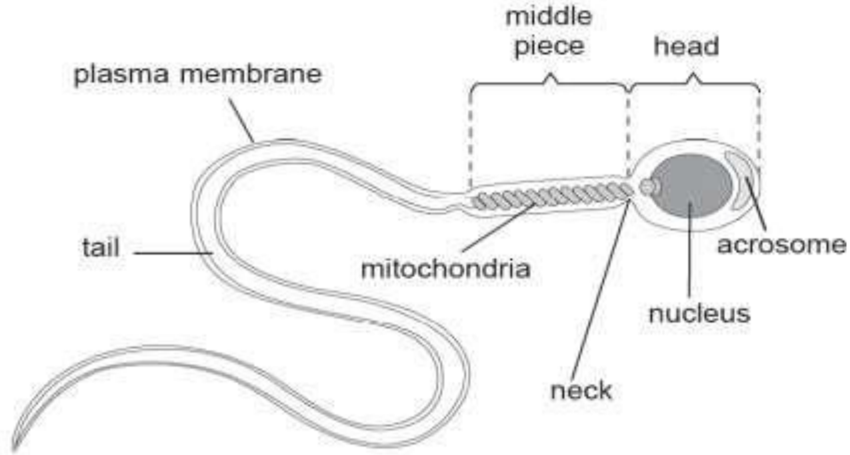
DIAGRAM: A muscle cell is long, thin, containing many fibrils and many mitochondria.

Adaptations of muscle cell

1. Many mitochondria provide lots of energy for contractions.
2. Long and thin allowing many to work side by side for greater force.
3. Long and thin for close packing to form a contractile network.
4. Each cell (or fibre) contains many smaller fibrils capable of contracting.

6) Sperm cell

- **Sperm cells** are specialised for motility and fertilisation.
- **Structure of the sperm cell**

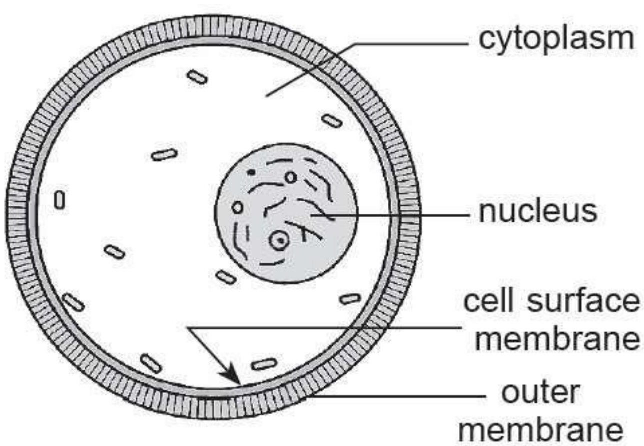


How the sperm cell is adapted to carry out its function

1. The head contains the genetic material in a haploid nucleus for fertilisation.
2. The acrosome in the head contains enzymes which dissolve the ovum membrane so that a sperm can penetrate an egg/ovum.
3. The tail enables the sperm to swim.
4. The middle piece is packed with mitochondria to release energy needed to swim and fertilise the egg.
5. Millions of sperm cells are released into vagina to increase chances of fertilisation.

7) Ovum/egg cell

- The ovum is specialised for fertilisation and food reserve.
- **Structure of the ovum**

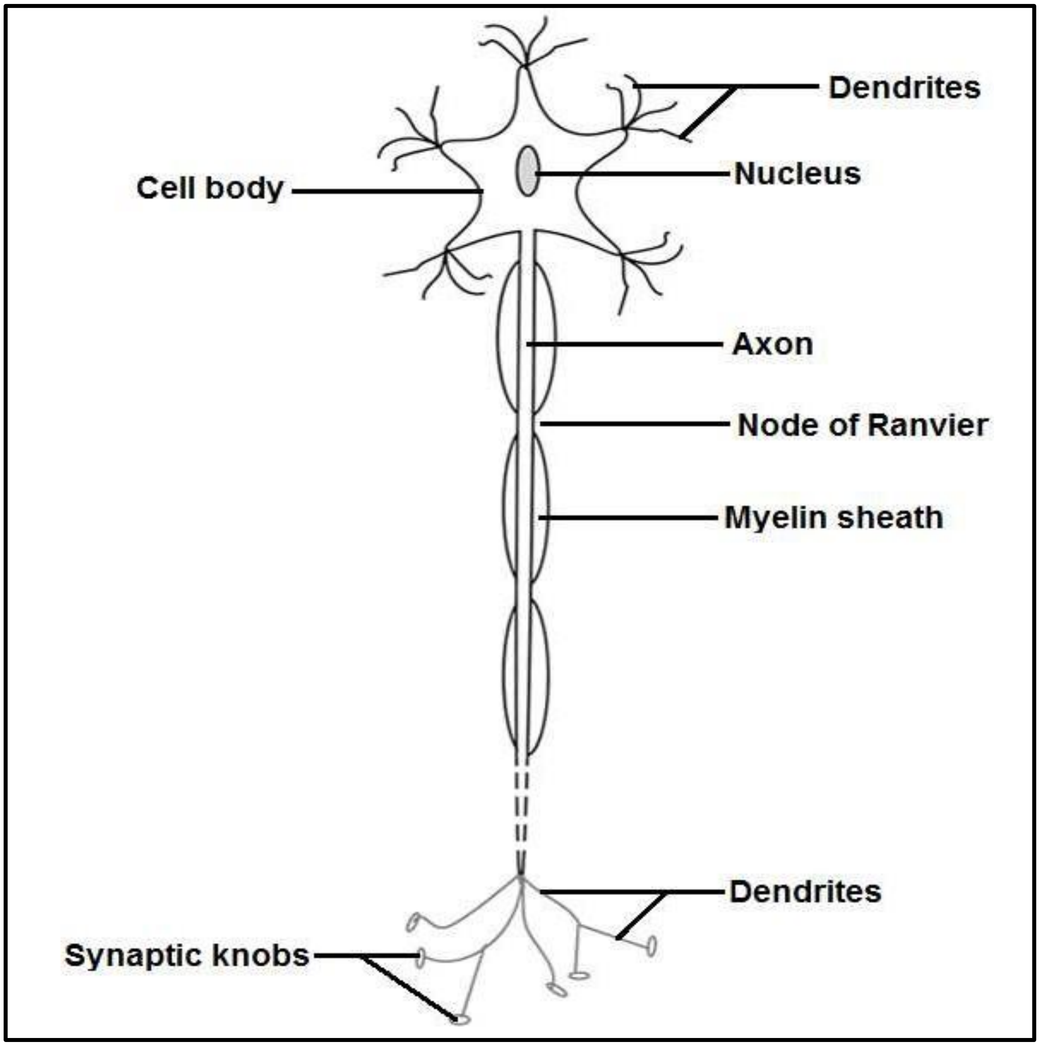


How the ovum is adapted to carry out its function

- The cytoplasm is large to contain nutrients for the growth of the early embryo.
- The haploid nucleus contains the genetic material for fertilisation.
- The cell membrane changes after fertilisation by a single sperm so that no more sperm can enter.

8) Neurons a.k.a. nerve cells

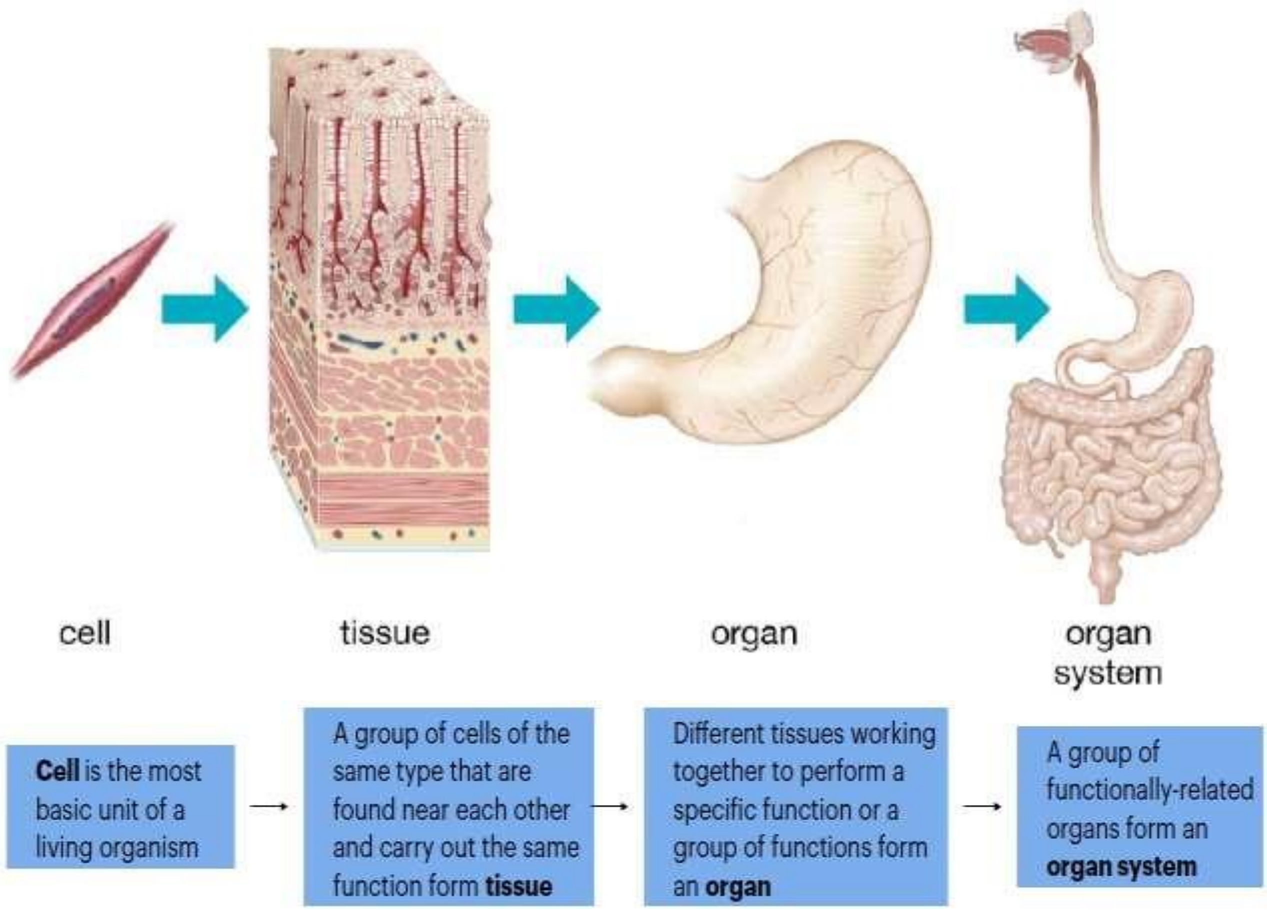
- The neurone is specialised for **transmission of impulses**.
- **Structure of a neurone**



- Neurons need to be able to rapidly transmit electrical impulses along their length. They also need to exchange signals with neighbouring neurons across the gaps, known as synapses that separate them.
- **To achieve these functions, neurones have several adaptations.**

1. Long fibre (axon) to carry impulses/messages up and down the body over long distances.
2. The axon is covered in a fatty myelin sheath, which acts as an electrical insulator. This increases the speed of transmission by forcing the impulse to jump between gaps in the sheath, known as nodes of Ranvier, rather than passing along the full length of the axon.
3. Neurones also have a lot of mitochondria, which provide the energy to synthesise neurotransmitters such as acetylcholine that pass messages across synapses.
4. Dendrites, highly branched extensions at one end of cell, provide a high surface area allowing the nerve cell to form synapses with many others.
5. Many synaptic knobs at the end of branches store neurotransmitter, ready to diffuse across the synapse as soon as an impulse arrives.

Organ System



TOPIC 5 PLANT SCIENCE
FORM 3: PLANT NUTRITION – PHOTOSYNTHESIS

KEY CONCEPT	LEARNING OBJECTIVES Learner should be able to	CONTENT (Attitudes, Knowldeg and Skills)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.5.1 Plant Nutrition	• define photosynthesis • State the word equation for photosynthesis • investigate conditions necessary for photosynthesis • explain the importance of photosynthesis in an ecosystem • investigate factors affecting the rate of photosynthesis • relate the structure of the leaf to its function in photosynthesis • investigate gaseous exchange in plants	• Photosynthesis • Word equation • Conditions necessary: chlorophyll, light, carbon dioxide • Fate of end products of photosynthesis • Importance of photosynthesis • Factors affecting the rate of photosynthesis (light intensity, • carbon dioxide • concentration and temperature) • Adaptation of the leaf to photosynthesis • Gaseous exchange in leaves	• Describing the process of photosynthesis. • Carrying out controlled experiments to show conditions necessary for photosynthesis • Discussing the fate of end products • Carrying out experiments to show the effects of these factors. • Observing the external structure of leaf. • Observing prepared slides of leaf internal structure. • Carrying out experiments to demonstrate gaseous in plants.	• Apparatus for starch testing • Potted plants • Iodine solution • Hand lenses • Microscopes • Bioviewers • Lime water / bicarbonate indicator

PHOTOSYNTHESIS

- Photosynthesis is the process by which plants convert carbon dioxide and water into glucose using light energy in the presence of chlorophyll.
- Oxygen is released as a by-product.
- Photosynthesis is summarized by the word equation:

$$\text{Carbon Dioxide + water} \xrightarrow[\text{Chlorophyll}]{\text{Light}} \text{Glucose + Oxygen}$$

Or by the chemical equation:

$$6CO_2 + 6H_2O \xrightarrow[\text{Chlorophyll}]{\text{Light}} C_6H_{12}O_6 + 6O_2$$

- The raw materials of photosynthesis are carbon dioxide, water and light energy.
- The end products of photosynthesis are glucose and oxygen.
- The carbon dioxide diffuses through the open stomata of the leaf of a plant and water is taken up through the roots.
- Chlorophyll is the green pigment that absorbs light for photosynthesis.
- Chlorophyll absorbs light energy and converts it into chemical energy for the formation of glucose/carbohydrates. Chlorophyll is found in the chloroplasts of palisade cells of the leaf.

The importance of photosynthesis

1. It reduces the amount of carbon dioxide in the air which is the main cause of global warming.
2. Produces oxygen which is used in respiration by living organisms.
3. Produces food which is a source of nutrients and energy for living organisms.
4. Energy is stored in fossil fuels through photosynthesis.
   - All the energy in fossil fuels like coal, oil and gas, came from the sun, captured through photosynthesis. Burning of fossil fuels releases energy for human activities.

The fate of the end products of photosynthesis

- Photosynthesis has two end products: glucose and oxygen.
  - ❖ Glucose is converted into:
    1. soluble sucrose for translocation through phloem from leaves to storage organs. Sucrose is then converted into insoluble starch for storage.
    2. fats and oils for storage in seeds and for making cell membranes.
    3. cellulose for making cell walls.
    4. amino acids and plant proteins when it combines with nitrates in the plant.
    5. energy during respiration.

  - ❖ Oxygen diffuses out of the leaf through stomata into the surrounding air or water.
    - The oxygen is used for respiration by plants and animals.

Try this question

What are the uses of soluble glucose in plants?

Answer

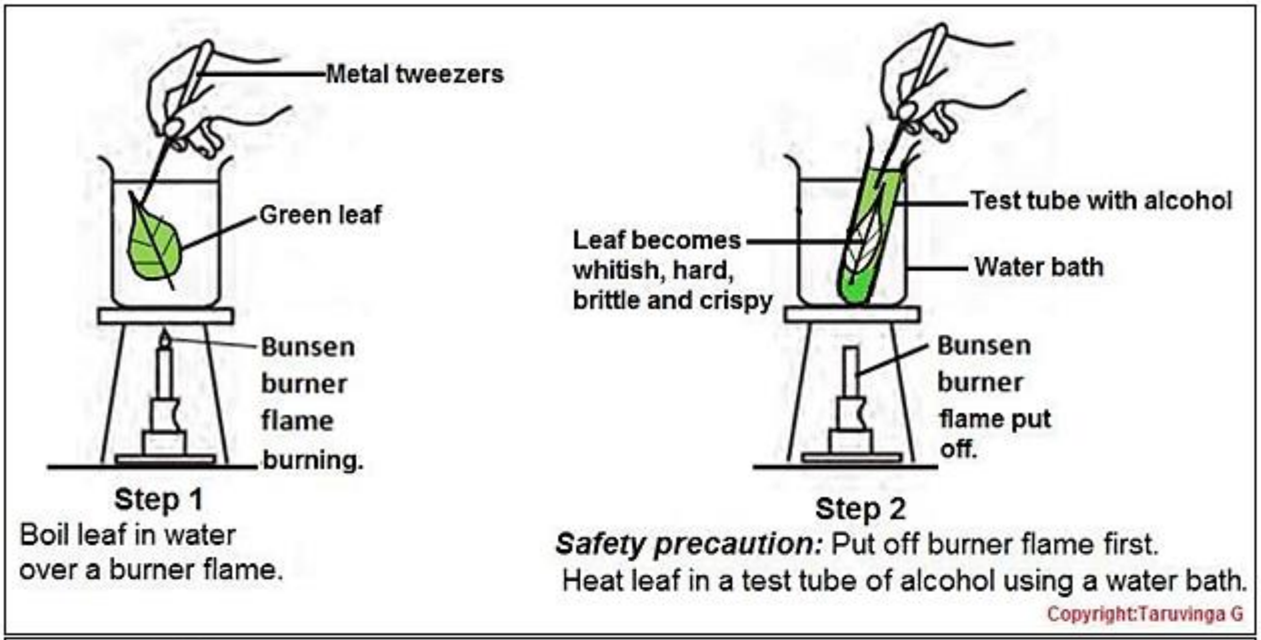
Glucose can be: converted into insoluble starch for storage, used for respiration, converted into fats and oils for storage and making cell membranes, used to produce cellulose (which strengthens cell walls) and used to produce proteins.

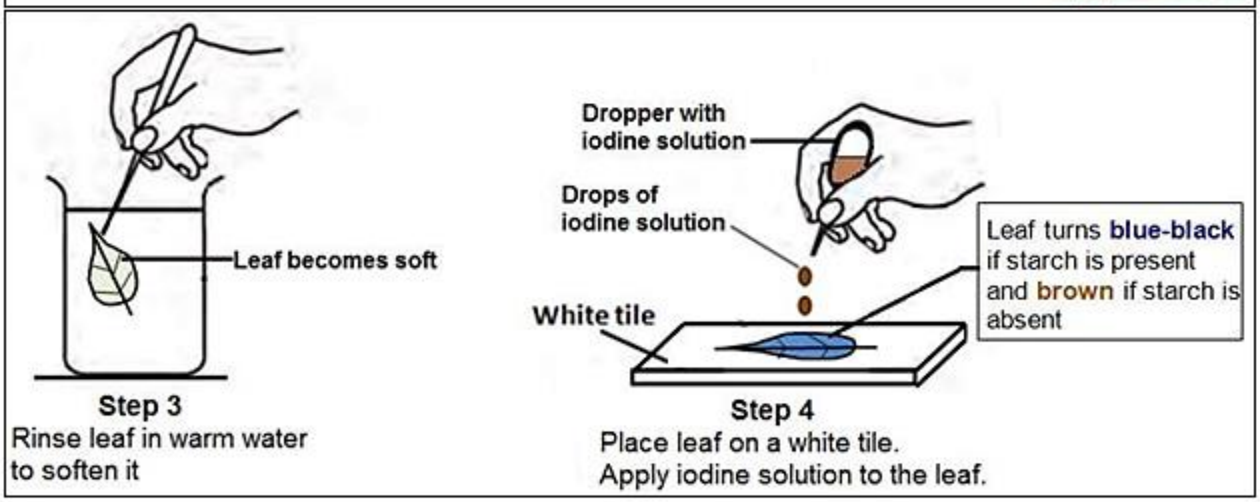
How to prove that a leaf carries out photosynthesis

- Any green part of the plant can carry out photosynthesis but the leaf is the main photosynthetic organ of the plant.
- To prove that a leaf carries out photosynthesis we need to **test it for starch**, which is one of the carbohydrates made during photosynthesis.

Testing a leaf for starch

1. Destarch leaves of a potted plant by putting it in a dark cupboard for 48 hours so that it uses up all of the stored starch.
2. To ensure the leaves are completely destarched, take some and test them for starch using the procedure shown in the diagram that follows. They should give a brown colour to show that starch is absent.
   - Why must the plant be destarched? Photosynthesis is measured by how much starch is produced so the leaf must start without starch.
3. Now place the potted plant in sunlight for a few hours to allow it to carry out photosynthesis.
4. Take some leaves form this plant and test them for starch using the steps shown in the diagrams that follows.





5. Note the colour change after applying iodine.

Observations

1. After heating the leaf in hot water it became whitish, hard, brittle and crispy.
2. The colourless alcohol became dirty green after heating the leaf in it.
3. The leaf became soft after rinsing it in warm water.
4. After applying iodine solution, the leaf turned blue-black showing that starch was present in it.

Conclusion

Therefore green leaves manufacture starch during photosynthesis.

SUMMARY: TESTING A LEAF FOR STARCH

- Destarch leaf in dark cupboard for 48 hours- removes starch.
- Put leaf in light for a few hours- photosynthesis occurs/leaf makes starch.
- Boil leaf in water - stops enzyme reactions, removes waxy cuticle/breaks down cell walls (allows easier penetration by alcohol).
- Boil leaf in ethanol - removes chlorophyll, leaf becomes whitish (to avoid masking colour changes).
- Dip in hot water - softens leaf (ethanol makes it hard, brittle,).
- Iodine added – to test for starch (blue-black colour shows presence of starch).

TRY THESE QUESTIONS

1) During the starch test on a leaf, what is the purpose of first heating the leaf in hot water? **[2]**
   - to break down the cell walls/to remove waxy cuticle;
   - to allow easier penetration by alcohol/ethanol/methylated spirits;
   - to stop enzymes reactions within the leaf.
2) What is the purpose of heating the leaf in alcohol**? [2]**
   - to remove/extract the chlorophyll;
   - to avoid masking colour changes/observations later.
3) What is the purpose of rinsing the leaf in warm water? **[1]**
   - to soften the leaf.
4) What is the purpose of applying iodine solution to the leaf at last?
   - to test for starch

9) What is the purpose of placing the leaf on a white tile? **[1]**
   - White background enables colour change to be seen easily.

10) Why is the Bunsen burner flame put off before heating the leaf in alcohol? **[1]**
   - Safety precaution: Alcohol vapour is highly flammable/inflammable**.**

11) State **four** changes that occur to the leaf after heating it in alcohol. **[4]**
   - colourless/whitish
   - hard
   - brittle
   - crispy

Factors which affect photosynthesis

- Factors which affect photosynthesis are
  - light
  - carbon dioxide (CO_2)
  - chlorophyll
  - Water.
- **Photosynthesis investigations - Principles and Starch test**

Experiments are used to find out what factors (CO_2, light, chlorophyll) are needed for photosynthesis. But first of all you need to **destarch** the plants. To be certain that they are thoroughly destarched, **test** a leaf for **starch** before you begin your investigation. Why must the plant be destarched? Photosynthesis is measured by how much starch is produced so the leaf must start without starch.

Principles of investigations

1) Investigations need controls
- **Control-** plant (or leaf) that has all substances it needs.
- **Test (**or **the experiment) -** plant/leaf that lacks one substance (light/chlorophyll/CO_2).
- **The Importance of a Control:** The control is used as a comparison so as to come up with a valid conclusion.

2) Plants must be destarched
- It is very important that the leaves you are testing should **not** have any **starch** in them at the beginning of the experiment.
- So, first of all, you must destarch the plants. Leave them in the **dark** for 48 hours. The plants use up all stores of starch in its leaves.

3) **Test the leaf for starch with Iodine solution.**

Experiments to demonstrate factors which affect photosynthesis

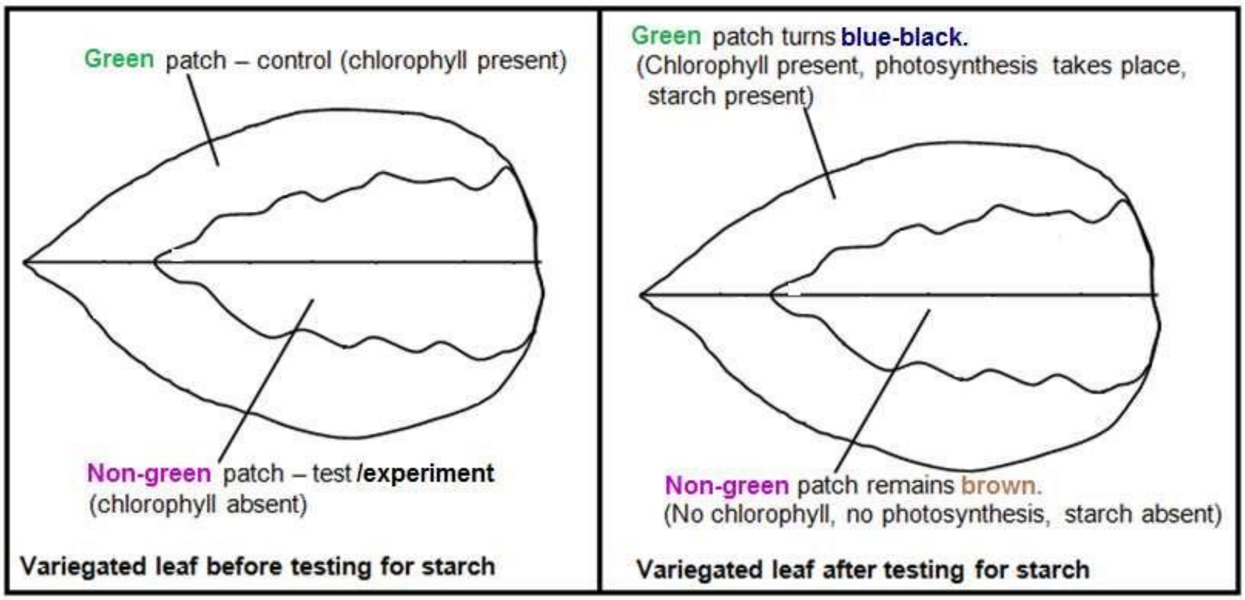
1. Chlorophyll is necessary for photosynthesis

Method

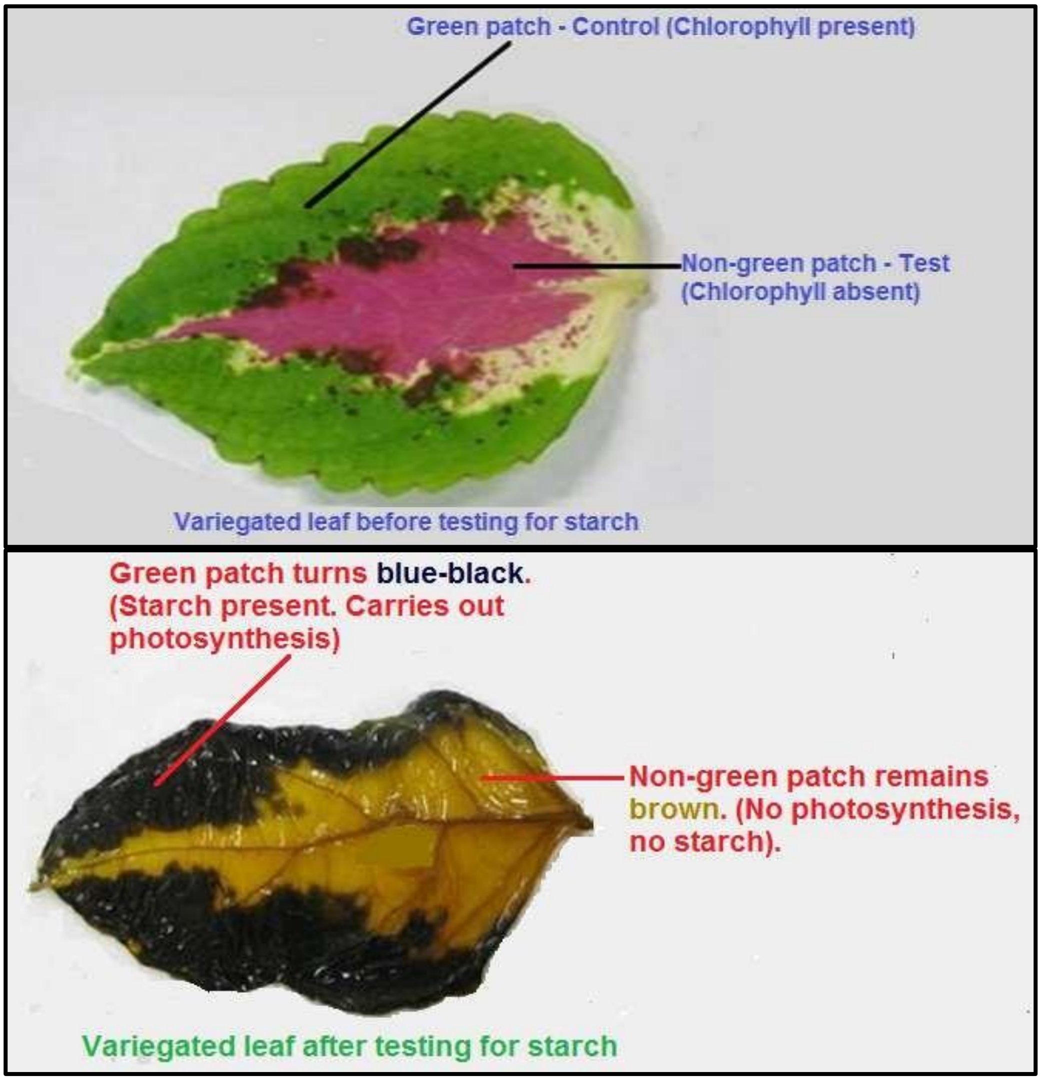
1) Take a potted plant with **variegated** leaves (leaves with green and non-green patches e.g. white/yellow).Green patch is the **control**. The non-green patch is the **test**.
2) **Destarch** the plant by keeping it in a dark cupboard for 48 hours.
3) Draw one leaf to show the white and green patches.
4) Expose the plant to the **sunlight** for about 6 hours.
5) Test one of the leaves for **starch** with iodine solution.
6) Draw the leaf again and indicate the colour changes.

Observations

- Diagram of leaf before and after testing for starch.



- Areas with previously **green** patches turn blue-black (test positive for starch).
- Areas with previously non-green patches (white/ yellow patches) remain brown (test **negative** for starch).



PICTRUE: Variegated leaf, before and after testing for starch.

Conclusion

- Photosynthesis takes place only in green patches because of the presence of **chlorophyll**.
- The pale yellow patches do not perform photosynthesis because of the absence of chlorophyll.
- Therefore, chlorophyll is necessary for photosynthesis.

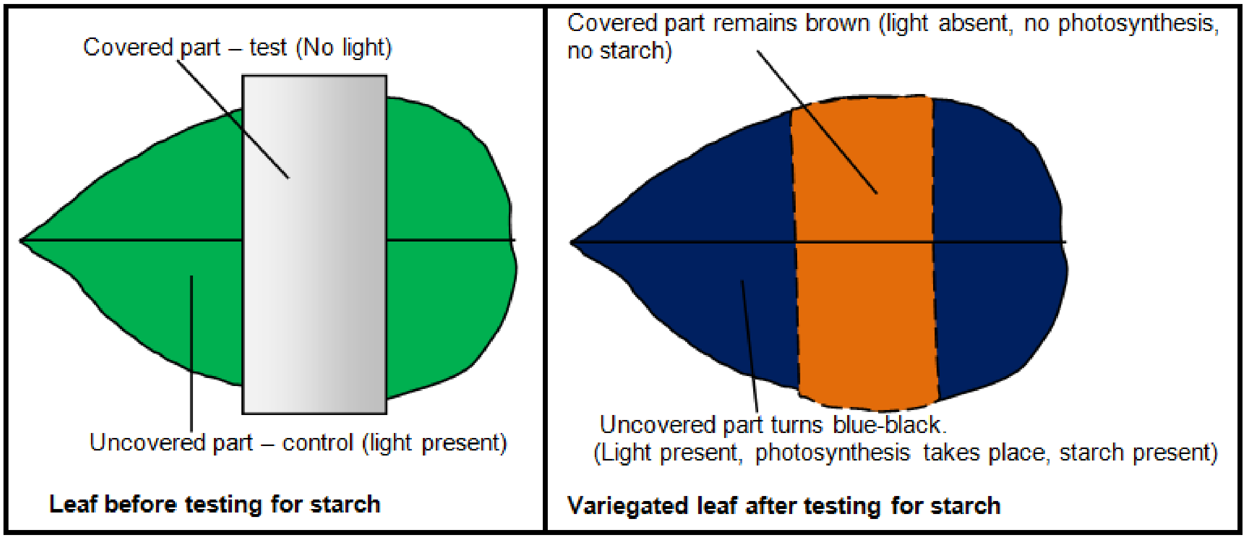
2. Light is essential for photosynthesis

Method

1) Take a potted plant.
2) **Destarch** the plant by keeping it in complete darkness for about 48 hours.
3) Test one of its leaves for starch to check that is does not contain any.
4) Cover part of the leaf with some aluminium foil to prevent light getting through. The uncovered part of the leaf is the control. The covered part is the test.
5) Place the plant in light for a few days.
6) Remove the cover and test the leaf for starch.

Observations

Only parts of the leaf that were left uncovered and received light go blue-black (positive test for starch). The covered parts which didn't receive light remain brown (negative test for starch).



Conclusion

Light is necessary for photosynthesis.

3. Carbon dioxide (CO_2) is essential for photosynthesis

Method

1) Take two destarched potted plants.
2) Cover both the plants with bell jars (or transparent plastic bags) and label them as A and B.
3) Inside Set-up A, place **$NaHCO_3$** (sodium bicarbonate). It produces CO_2. Set-up A is the control.
4) Inside Set-up B, place **soda lime** or **NaOH** (Sodium hydroxide). It absorbs CO_2. Set-up B is the test or the experiment.
5) Keep both the set-ups in the sunlight for at least 6 hours.
6) Perform the starch test on both of the plants.

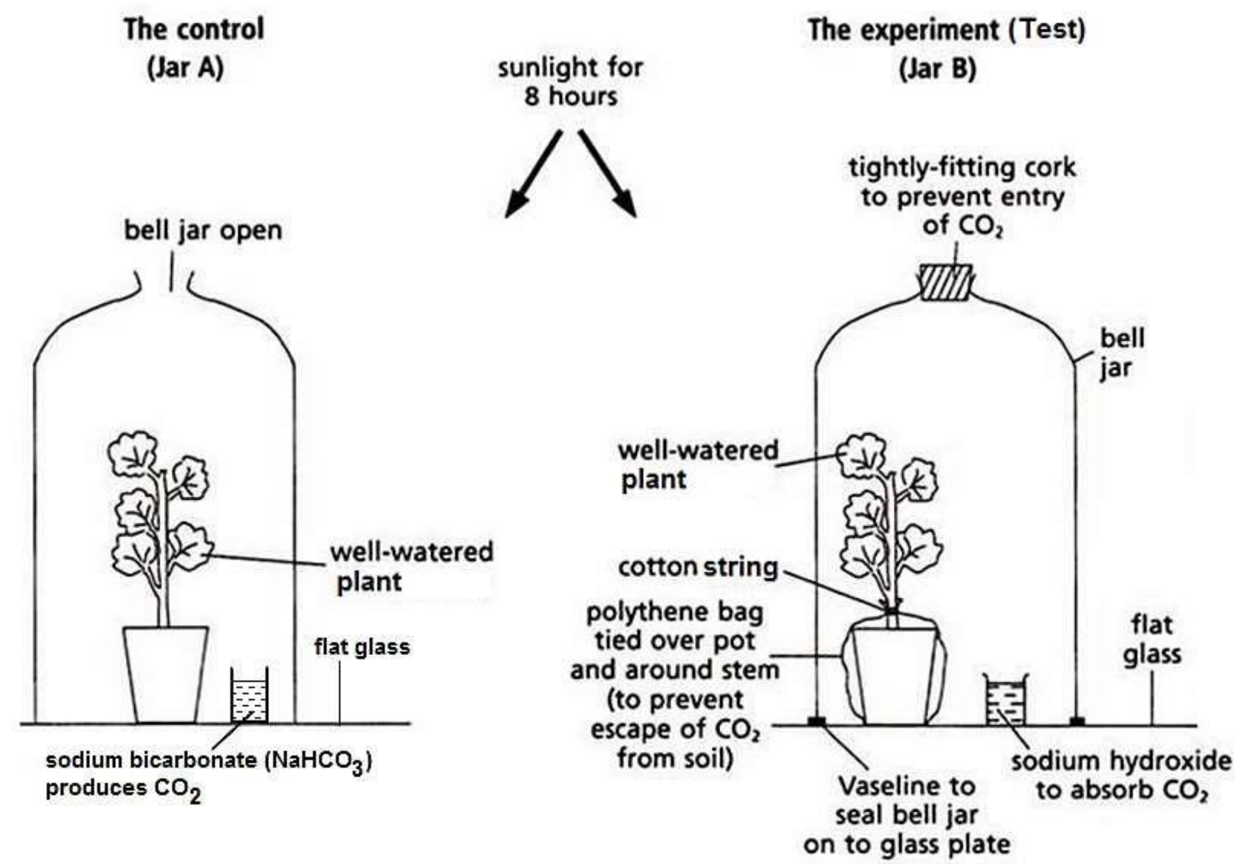


DIAGRAM: Experiment to show carbon dioxide is necessary for photosynthesis

Observations
Leaf from the plant in which **$NaHCO_3$** has been placed turns blue-black (gives a **positive** test for starch).
Leaf from the plant in which **NaOH** has been placed remains brown (gives a **negative** test for starch).

Conclusion
Leaf that gets CO_2 gives a positive test for starch and leaf that does not get CO_2 gives a negative test for starch. Therefore it means CO_2 is essential for photosynthesis.

Summary: Experiments to demonstrate factors which affect photosynthesis

Describe an experiment to investigate the necessity of chlorophyll using appropriate controls.
- Take a destarched, variegated plant.
- Place the plant in sunlight for about 6 hours.
- Draw one leaf to show the white and green parts.
- Test the variegated leaf for starch using a few drops of iodine solution.
- Only the green parts of the leaf go blue-black, as these parts of the leaf contains chlorophyll.

Describe an experiment to investigate the necessity of light using appropriate controls.
- Take a destarched plant.
- Cover part of the leaf with some aluminium foil to prevent light getting through.
- Leave the plant in the light for a few hours.
- Test the leaf for starch using a few drops of iodine solution.
- Only the parts of the test leaf that were left uncovered go blue-black, as the covered parts didn't receive light.

Describe an experiment to investigate the necessity of carbon dioxide using appropriate controls.

- Take a destarched plant.
- Enclose it in a plastic bag with soda lime that absorbs carbon dioxide.
- Leave the plant in the light for a few hours.
- Test a leaf for starch using a few drops of iodine solution.
- The leaf should show a negative result for the starch test.
- A control experiment should be set up in exactly the same way but without soda lime.

Factors that affect the RATE of photosynthesis are

- light intensity
- carbon dioxide concentration
- temperature

Describe an experiment to measure the effect of light intensity on the rate of photosynthesis.

- Cut a piece of pondweed about 5cm in length.
- Put a paperclip on the pondweed to stop it floating to the surface.
- Put the lamp close to the plant and measure the distance between the plant and the lamp.
- Count the number of bubbles released over 5 minutes. Repeat several times and calculate the average.
- Repeat this procedure with the lamp at different distances from the plant.
- The number of bubbles should decrease as the distance between the lamp and the plant increases.

Describe an experiment to measure the effect of temperature on the rate of photosynthesis.

- Cut a piece of pondweed about 5cm in length.
- Put a paperclip on the pondweed to stop it floating to the surface.
- Set up the apparatus with water baths at different temperatures.
- Count the number of bubbles released over 5 minutes.
- Remember that factors like light intensity and carbon dioxide concentration must be kept constant.
- You should find that the increasing temperature of pondweed increases the rate of photosynthesis.
- However at higher temperatures the rate of photosynthesis will decrease.

Describe an experiment to measure the effect of carbon dioxide on the rate of photosynthesis.

- Cut a piece of pond weed about 5cm in length.
- Put a paperclip on the pond weed to stop it floating to the surface.
- Count the number of bubbles released over 5 minutes.
- Adding sodium hydrogen carbonate to water increases the concentration of carbon dioxide as it dissolves in water.
- Other factors like light intensity and temperature must stay constant.
- You should find that the plant produces more bubbles of gas as the carbon dioxide concentration increases.

LEAF STRUCTURE

External structure of a leaf

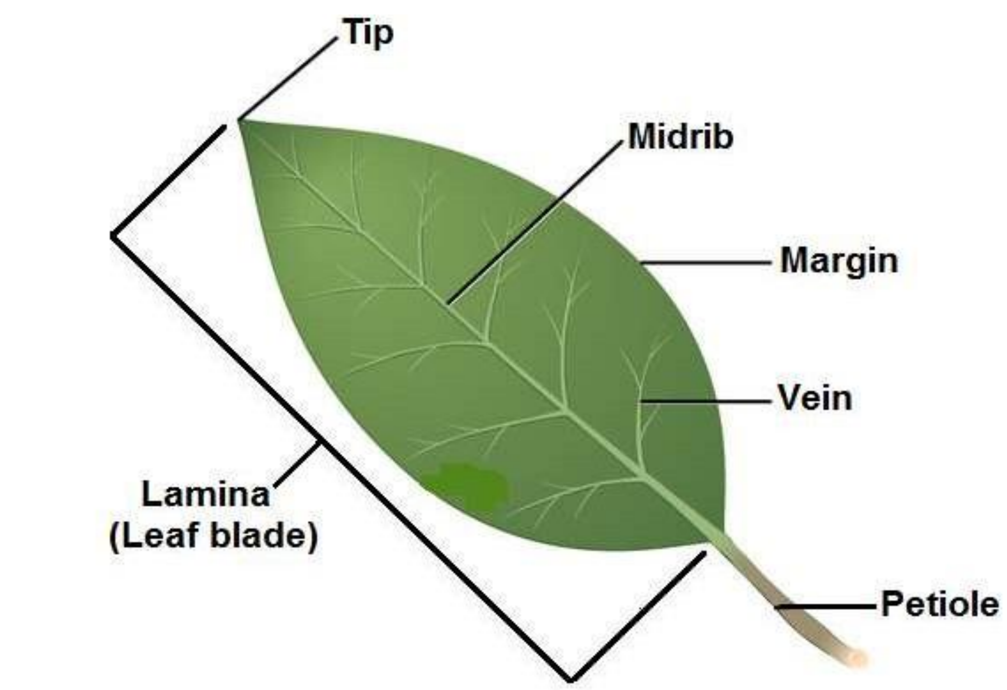


DIAGRAM: External structure of a dicotyledonous leaf

Name the different external parts of the leaf.

- margin
- lamina
- petiole
- veins
- mid rib
- tip.

What is the function of the lamina?

- The lamina or lead blade has a large surface area to maximize absorption of sunlight. It also allows for rapid diffusion of carbon dioxide into the leaves (large surface area).

What is the function of the petiole?

- The petiole positions the lamina for maximum absorption of sunlight and gaseous exchange.

What is the function of the veins?

- The veins allow transport of water and mineral salts to the cells of the lamina. They also transport manufactured food from the leaves to the other parts of the plant.

How are the leaves USUALLY arranged?

- They are usually arranged in a regular pattern around the stem (paired/alternate arrangement).

Why are they arranged in this manner?

- This is to ensure that the leaves are not blocking one another from sunlight and that each leaf receives optimum amount of light.

Internal structure of a leaf

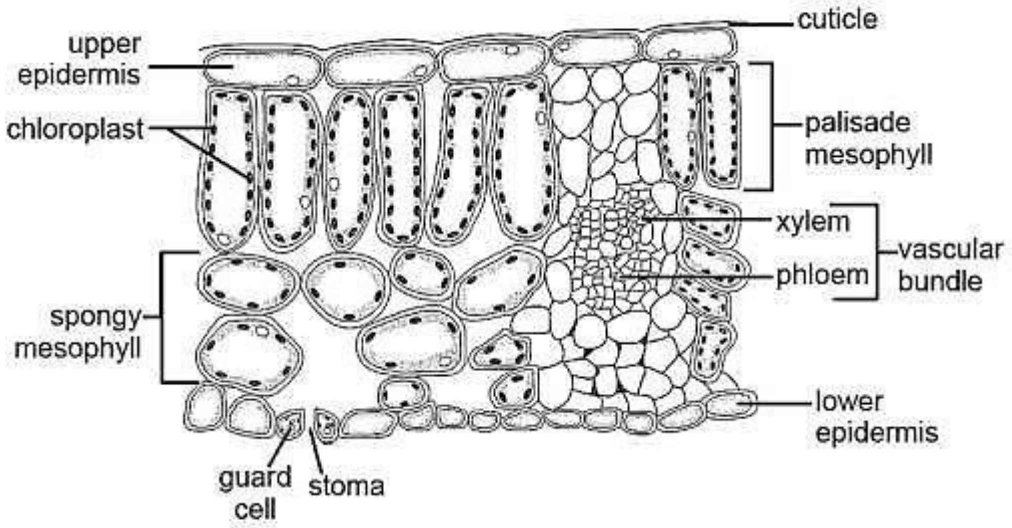


DIAGRAM: Cross section of a dicotyledonous leaf

❖ Name the different parts of the internal structure of the leaf.
- cuticle
- upper epidermis
- palisade mesophyll
- spongy mesophyll
- vascular bundle (xylem and phloem)
- lower epidermis
- guard cells

❖ Describe the cuticle and its function.
Waxy layer above the epidermis which prevents excessive water loss. It is transparent to allow sunlight to penetrate to the mesophyll.

❖ Describe the upper epidermis and its function.
Single layer of closely packed cells which have no chloroplasts.

❖ Describe the palisade mesophyll and its function.
A few layers of closely packed cells which are long, cylindrical and contain numerous chloroplasts for maximum absorption of sunlight.

❖ Describe the spongy mesophyll and its function.
Irregularly shaped cells with numerous large intercellular air spaces. This allows for rapid diffusion of gases inside the leaf.

❖ Describe the vascular bundle and its function.
Contains xylem and phloem, allowing for the transport of water and food minerals respectively within the plant.

❖ Describe the lower epidermis and its function.
Single layer of closely packed cells. Many minute openings called stomata found here. When do stomata open and close?
Stomata open in the light and close in the dark.

❖ How do the stomata open in the day?
In the day, water from the adjacent epidermal cells enters the guard cells. Guard cells swell and become turgid. This causes the guard cells to become curve and pull the stomata open.

❖ How do stomata close during the night?

In the night, K^+ ions diffuse out of the guard cells. Water potential in guard cells increases leading to the exit of water by osmosis. Guard cells then become flaccid and stoma closes.

❖ How does carbon dioxide enter the leaf?

CO_2 diffuses into the leaf via the stomata. CO_2 dissolves into the film of water surrounding the mesophyll cells and diffuses into cells.

❖ How does water enter the leaf?

Water is brought to the leaves via the xylem vessels in the veins. Water leaves the veins and moves from cell to cell in the meanwhile by osmosis.

How the leaf is adapted for photosynthesis

1) Broad so large surface area is exposed for light absorption.
2) Thin for short distance of diffusion and light absorption.
3) Chloroplasts in palisade cells absorb light.
4) No chloroplasts in the epidermal cells to allow sunlight to penetrate the mesophyll layer.
5) Waxy cuticle reduces water loss by evaporation.
6) Network of xylem that brings water to the leaf cells.
7) Network of phloem that carries away the sugars produced in photosynthesis.
8) Transparent waxy cuticle and epidermis to allow light to enter.
9) Numerous stomata for gaseous exchange.
10) Stomata that open and close in response to light intensity.
11) Guard cells that control the opening and closing of stomata.
12) Leaf arrangement on plant stem that minimizes overlapping and shadowing of one leaf by another.
13) Long narrow upper mesophyll cells packed with chloroplasts that collect sunlight.
14) Many air spaces in lower mesophyll layer to allow diffusion of carbon dioxide and oxygen.
15) Tall/elongated palisade cells increase surface area for absorbing carbon dioxide.
16) Thin and elongated palisade cells so many can be tightly packed near the top surface of the leaf. This means that the palisade cell layer is close to the source of light.

TRY THESE QUESTIONS

1) What increases light absorption in plants?

– large surface area.
– **palisade mesophyll** is composed of tall thin cells.
– **waxy cuticle** is transparent to allow light to enter.

2) What increases gas exchange in plants?

- **spongy mesophyll** has air spaces (inter-cellular spaces) to allow gases to move easily.
- epidermis of leaf perforated by **stomata.**
- **guard cells** change shape in order to control the size of the pore.

FORM 3: PLANT MINERAL NUTRITION

KEY CONCEPT	LEARNING OBJECTIVES Learner should be able to	CONTENT (Attitudes, Knowldeg and Skills)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.5.1 Plant Mineral Nutritioı	• describe the functions of nitrogen phosphorus and potassium on plant growth • describe the effects of deficiencies of nitrogen, phosphorus and potassium	• Mineral Nutrients - Nitrogen (N) – protein synthesis and • chlorophyll formation - Phosphorous (P) – synthesis of ATP and - nucleic acids - Potassium (K) – ion and osmotic balance • Defiency signs - N – stunted growth and chlorosis - P – Stunted root growth and purplish leaf colouring - K – yellow and brown leaf margins and premature death (poor flowering and fruiting)	• Discussing the functions of nitrogen, phosphorus and potassium on plant growth. • Carrying out culture experiments	• ICT tools • Braille software/. Jaws • Culture solutions • Plants

MINERAL NUTRITION IN PLANTS

- Plants need mineral elements for healthy growth and development.
- Mineral elements needed by plants in large quantities are nitrogen (N), Phosphorus (P) and Potassium (K).
- Mineral elements are found as **ions** in soil **salts** dissolved in water from where plant roots absorb them.
  - Nitrogen (N) is absorbed as nitrate ions ($NO_3^{\ -}$) and sometimes as ammonium ions (NH_4^+).
  - Phosphorus (P) is absorbed as phosphate ions (PO_3^{2-}).
  - Potassium (K) is absorbed as potassium ions (K^+).
- **The sources of mineral elements in the soil** are:
  - **Rocks:** the mineral salts in rocks are dissolved by rainwater and washed into soil.
  - **Natural fertilizers:** decay of animal faeces and dead plants and animals recycle mineral nutrients in the soil.
  - **Artificial fertilizers:** these are added to the soil by farmers and contain ready-made mixtures of mineral elements needed by plants:
    - nitrogen is supplied by ammonium nitrate, ammonium sulphate and compound fertilizers.
    - phosphorus is supplied by ammonium phosphate fertilizer.
    - potassium is supplied by potassium chloride and potassium sulphate fertilizers.

Functions of Mineral Nutrients in plants

- ❖ **Nitrogen (N)** – protein synthesis and chlorophyll formation.
- ❖ **Phosphorous (P)** – synthesis of ATP and nucleic acids.
- ❖ **Potassium (K)** – ion and osmotic balance. Flower and fruit. Improves disease resistance.

Mineral deficiency

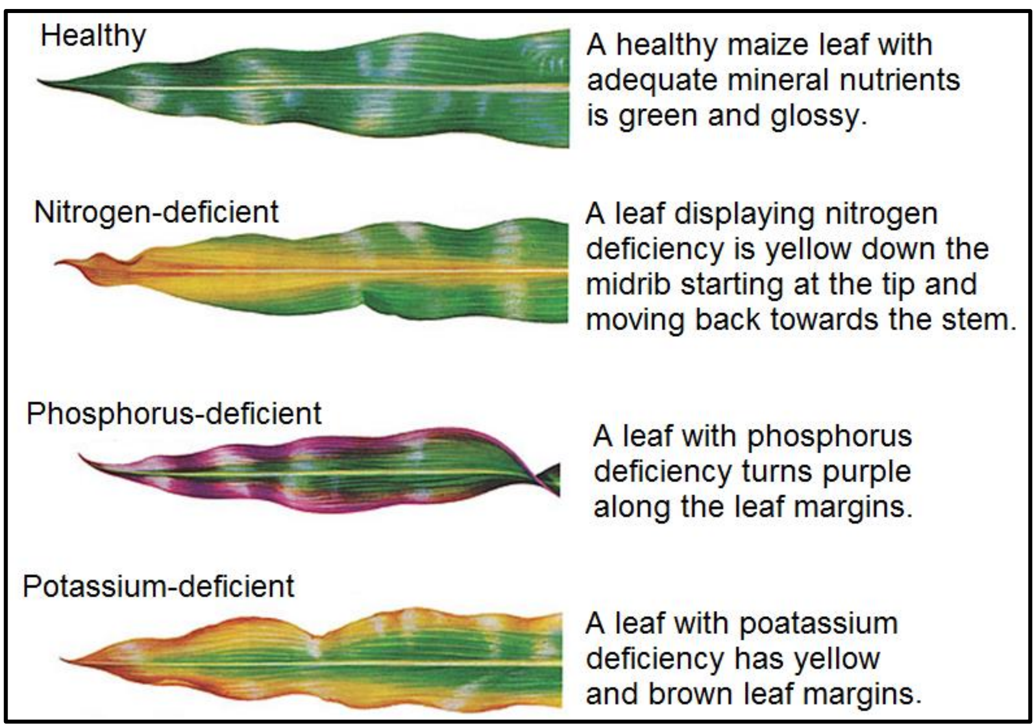
- If a plant does not get enough of a mineral element it is said to have a mineral deficiency.

Deficiency signs and symptoms in plants

- Nitrogen (N) – stunted growth and chlorosis (yellowing of leaves).
- Phosphorus (P) – Stunted root growth and purplish leaf colouring. Necrosis (Plant parts turn brown and die).
- Potassium (K) – yellow and brown leaf margins and premature death, poor flowering and fruiting.

Signs that a plant has adequate nutrients

- A plant with **enough nitrogen** has good leaf growth, large dark green leaves and a well-developed shoot system.
- A plant with **enough Phosphorus** has large healthy roots, shows normal growth and normal seed development.
- A plant with **enough Potassium** will have healthy flowers and good fruit setting.



Picture: NPK nutrient deficiency symptoms.

WATER CULTURE EXPERIMENT

- The **water culture experiment** is also known as **soil-less culture experiment** or **hydroponics**. It is used to find out whether mineral nutrients such as nitrogen (N), phosphorus (P) and potassium (K) are essential for plant growth.

AIM: To investigate the effects of N, P and K on plant growth.

Materials: Five jars, germinating bean seeds, water culture solutions, opaque covers (e.g. black paper or aluminium foil).

Precautions

- Before setting up the experiment, all apparatus are to be sterilized to ensure that the apparatus is free from micro-organisms which might interfere with the growth of the seedlings.
- The outside of the gas jars are to be covered completely with opaque covers (e.g. black paper or aluminium foil) to prevent light from entering the jars. This prevents algae from growing around the roots and hindering their normal functioning. This also prevents the algae from using up the nutrients intended for the plant.
- The gas jars are to be placed in such a way that they receive enough sunlight but be very careful not to place the gas jars in direct sunlight as it may cause the leaves of the seedlings to scorch and heat up the culture solutions.
- Keep the cotton wool around the seedling dry to prevent the stem from rotting.
- Renew the culture solution every 2 weeks to avoid depletion of the nutrients.
- Aerate the solution by blowing air through the delivery tube to supply oxygen for root respiration.

Method

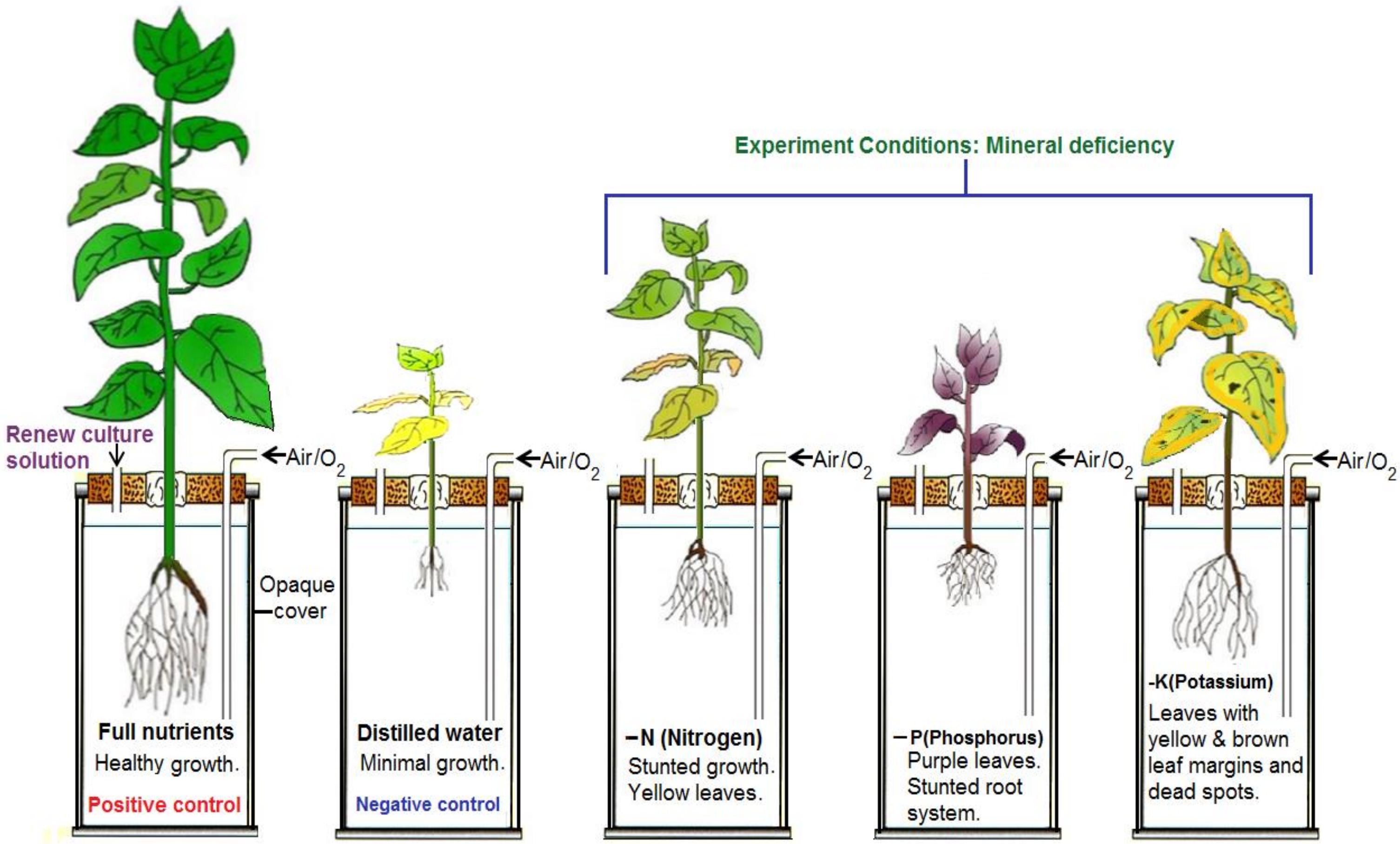
1. One plant seedling is grown in a jar with all the required nutrients / minerals being present. This is the positive control.
2. Another seedling is grown in distilled water without the required minerals. This is the negative control.
   - The controls are used as comparisons to help come up with a valid conclusion.
3. Other seedlings are grown in different culture solutions each lacking only one mineral which is being investigated.
   - To investigate whether Nitrogen (N) is really needed for plant growth, omit nitrates in the culture solution.
   - To investigate whether Phosphorus is really needed for plant growth, omit phosphates in the culture solution.
   - To investigate whether Potassium (K) is really needed for plant growth, omit it in the culture solution.
4. Examine the seedlings daily for about four weeks and take note of the following:
   - change in root length
   - change in length of shoot
   - colour of leaves
   - number of leaves.

Results

1. Initially during the first few days the seedlings in all the culture experiments appeared healthy because they were using the nutrients stored inside their seeds for growth.
2. After the stored nutrients were used then the seedlings started to show the effects of mineral deficiency.
3. The first jar had all the mineral nutrients available and the seedling showed healthy growth, with many large dark green leaves and well developed root and shoot systems. This was the positive control. It gave positive results.
4. The second jar with distilled water lacked all mineral nutrients and therefore the

seedlings had minimal growth. The leaves were very few, small and showed many signs of nutrient deficiency such yellow, brown and purple colouration and dead spots. The root and shoot systems had stunted growth. This was the negative control. It gave negative results.

5. The third jar had no nitrogen and the seedling had stunted growth and yellowish leaves. All the leaves had signs of chlorosis. The lower leaves were yellow and dead. The upper leaves were pale green.
6. The fourth jar had no phosphorus and the seedling had stunted roots and purple leaves.
7. The fifth jar had no potassium and the seedling had yellow/brown leaf margins and dead spots.



– means minus.

DIAGRAMS: Culture experiments to investigate the deficiency of N, P and K.

Conclusion

1. The Deficiency effects of nitrogen, phosphorus and potassium are as follows:
   - Nitrogen (N) – stunted growth and chlorosis (yellowing of leaves).
   - Phosphorus (P) – Stunted root growth and purplish leaf colouring.
   - Potassium (K) – yellow and brown leaf margins and dead spots
2. Mineral elements such as N. P and K are needed for healthy plant growth.

FORM 3: PLANT PRODUCTIVITY

KEY CONCEPT	LEARNING OBJECTIVES Learner should be able to	CONTENT (Attitudes, Knowldeg and Skills)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.5.2 **Plant Productivity**	• define biomass • define productivity	• Organic content • Increase in biomass overtime	• Carrying out experiments to measure biomass • Discussing productivity in terms of increase in biomass .	• Balances • Ovens • Print media
	• explain factors affecting productivity • define pest	• Factors affecting productivity - Light - Mineral salts - Temperature - Water availability - Pests and diseases	• Discussing factors affecting productivity	
	• identify types of plant pests and diseases • explain how pests and diseases affect productivity	• Pest as an organism which reduces productivity	• Observing plants affected by various pests and diseases • Classifying pests • Discussing effects of plant pests and diseases on productivity.	• School grounds and neighboring fields • Preserved plant pests and affected plants • Resource person
	• describe methods of controlling pests and diseases • outline the advantages and disadvantages of each method	• Pests: - Tissue – eating - Sap-sucking pests • Diseases: - Bacterial wilt - Fungal rust - Reduction of yields • Management, cultural, chemical and biological	• Discussing methods of controlling plant pests and diseases with specific reference to maize, cotton and tobacco	

PRODUCTIVITY

- Productivity is the increase in biomass over time.
- Productivity is the amount of organic matter or biomass produced by a plant per unit time.
- Biomass is the dry mass of an organism e.g. a plant.
- A plant's biomass increases as it grows.
- The yields we get from crop plants are part of the plant's productivity.
- Some plants are more productive than others, that is, they carry out photosynthesis more efficiently than others and produce more yields.
- To ensure that plants are as productive as possible:
  1. we should maximise the following factors:
     a) Light e.g. in green houses
     b) Water availability
     c) Mineral salts/mineral elements (NPK as major elements).
     d) Temperature around $25^{O}C$ e.g. in green houses.
  2. we should also control plant pests and diseases.

FACTORS AFFECTING PLANT PRODUCTIVITY

As listed previously are:

1) Light

2) Water availability
3) Mineral salts
4) Temperature
5) Plant pests and diseases.

NB. All factors which affect photosynthesis will also affect plant productivity (Factors 1 to 4 above).

TRY THESE QUESTIONS

1. Using your knowledge of photosynthesis and mineral nutrition describe how the following factors affect productivity:
   (a) Light intensity
   (b) Temperature
   (c) Water availability.
   (d) Minerals salts (NPK).
2. How do plant pests and diseases reduce productivity and hence crop yields?

EXPERIMENT TO MEASURE THE BIOMASS OF A PLANT

1. Plant seeds of a fast growing plant e.g. bean seed. Apply the NPK nutrients as recommended for the bean variety. Apply enough water and pick off pests by hand.
2. One week after emergency remove some of the plants (say 10) from the soil and wash off any loose soil. This is the first harvest at time t_1= 1week.
3. Blot the plants removing any free surface moisture.
4. Dry the plants in an oven set to low heat (70° C) overnight or until the mass is constant.
5. Let the plants cool in a dry environment (a Ziploc bag will keep moisture out) - in a humid environment the plant tissue will take up water. Once the plants have cooled weigh them on a milligrams electronic balance.
6. Divide the total dry mass by number of plants to get the average dry mass 1 (m_1) at first harvest (time, t_1).
7. After some time (t_2), say 3 weeks, do the second harvest and repeat procedures 3 to 6 to get the average dry mass 2 (m_2).
8. Calculate the time taken using the formula: Time = $t_2 - t_1$.
9. Calculate the increase in biomass using the formula: Increase in biomass = m_2-m_1.

Units => g

10. Calculate productivity using the formula:

Plant productivity = (Increase in biomass)/(time)

Units => g/week

11. Stages 3 to 6 can be repeated until time of harvest so as to get the overall increase in biomass.
12. This experiment can be used to compare the productivity of two or more sets of plants grown under different conductions e.g. well watered plants supplied with NPK versus well water plants deprived of NPK.

Typical results of bean seedlings

Time/weeks	1	3
Dry mass of 10 seedlings/g	12.65	50.85
Average dry mass/g	1.265	5.085

1. Show how the average dry masses were obtained.
2. Calculate the increase in biomass of the bean seeds over the 3 week period.
3. Calculate the productivity of the bean seeds over the three week period.
4. Explain why the plants are well watered and supplied with all nutrients?

ANSWERS

1. Show how the average dry masses, m_1 and m_2, were obtained.

$m_1 = \frac{12.65\ g}{10}$

$= 1.265\ g$

$m_2 = \frac{50.85g}{10}$

$= 5.085\ g$

2. Calculate the increase in biomass of the bean seeds over the 3 week period.

Increase in biomass = $m_2 - m_1$
= 5.085 g – 1.265 g
= 3.82 g

3. Calculate the productivity of the bean seeds over the three week period.

Plant productivity = (Increase in biomass)/(time)

= 3.82 g / 3 - 1 weeks

$= \frac{3.82\ g}{2\ weeks}$

= 1.91 g/week

4. Explain why the plants are well watered and supplied with all nutrients?

To ensure maximum growth

5. Why were the plants dried in an oven?

To remove moisture to get the dry mass/dry weight which is the biomass

6. How do you tell that all the moisture in the plant has been removed?

Dry the plants in an oven until the mass (measured by an electronic balance) is constant.

PLANT PESTS AND DISEASES

- A plant pest is an organism which reduces plant productivity.
- Plant pests cause damage or harm to plants and also act as vectors or pathogens in plants.
- Vectors spread diseases among plants.
- Pathogens such as bacteria, fungi and viruses cause various plant diseases.

Examples of plant diseases:

1. **Bacterial wilt**
   - This is a disease caused by bacteria affecting plants such as tomatoes, potatoes, tobacco and cucumbers.
   - When a plant is infected with a bacterial wilt, it starts showing browning of individual leaves spreading to the rest of the plant.
   - The infection builds around the xylem of the plant resulting in blockage of water movement.
   - Slimy sticky ooze where the plant is cut.

Control of bacterial wilt

- Once a plant is affected, there is no way of stopping the spread of the disease.
- Pull off infected plants as soon as possible and practice crop rotation.
- Grow and propagate from pathogen-free plant seeds and material.

2. **Fungal rust**
   - This is a disease caused by microscopic fungi which affect leaves and stems of a plant.
   - The disease is usually shown by white spots on the undersides of leaves and on the stems.
   - After a while the spots become covered with rust-coloured spore masses.
   - Later on, the different sized leave pustules on the surface turn yellow or orange.
   - Cereal crops like maize are largely affected by the fungal rust disease.
   - Severe infestation will result in the deformation and yellowing of leaves and eventually leaf drop.

Control and treatment of fungal rust

- Choose resistant varieties when possible.
- Pick off and destroy the infected leaves.
- Weed and rake fallen debris to keep clean under the plants frequently.
- Avoid overhead sprinkling where possible.
- Practice crop rotation.
- Use Dithane M45.

PLANT PESTS

- There are different types of plant pests and these affect plants in different ways.
- Pests feed on plants in different ways and this enables farmers to identify them and the different types of diseases that they cause.

Note that not all insects are pests.

An insect is called a pest based on its potential to cause harm to human needs, plants or animals.

Identifying the major types of plant pests and diseases

The **two** major categories of plant pests are **tissue eating pests** and **sap-sucking pests**.

1. Tissue-eating (biting and chewing)

- Tissue eating pests (a.k.a. chewing and biting pests) have biting jaws or chewing and biting mouthparts.
- These pests cut edges of plant organs such as leaves, flowers, stems and roots.
- They also make holes and gaps on plants parts such as leaves and tender stems, seeds and fruits.

Examples of Tissue-eating pests

a) Cutworms
b) Locusts
c) Grasshoppers
d) Caterpillars
e) Beetles
f) weevils
g) Crickets
h) Millipedes
i) Snails
j) Rodents e.g. moles, rats, mice.

Control of tissue-eating pests

- Different pesticides are used to control pests.
- Pesticides are chemicals used to kill pests.
- Tissue-eating pests can be controlled with stomach poisons and contact poisons.
- Apply the pesticides both at the top and the lower parts of the foliage or leaf.

2. Sap-sucking pests (piercing and sucking pests)

- Sap sucking pests (a.k.a. piercing and sucking pests) have hollow needle mouthparts or piercing and sucking mouthparts which they use for piercing into phloem of leaves or stems and sucking the juice or sap.
- Young seedlings or plants are the most affected by sap-sucking pests.

Examples of sap-sucking pests

a) Red spider mites
b) Aphids
c) Leafhoppers
d) Plant bugs

The symptoms of sap-sucking pests are spots on leaves and stems.

Control of sap-sucking pests

- These types of pests can be controlled with the use of contact or systematic pesticides.
- Systematic pesticides are those that may be absorbed through plant's roots, bark or leaves and with the help of the sap translocated throughout the plant.

Effects of plant pests and diseases

Pests and diseases have a huge impact on crop productivity.

- They reduce yields.
- They affect food crops, causing significant losses to farmers and
- Threatens the food security of a country.
- Huge losses of pastures.

Methods of pest control

Three types of pest control methods are:
(a) Cultural/management control
(b) Biological control
(c) Chemical control

Cultural control (a.k.a. management control)

- Cultural/management control is the use of good farming practices or farming precautions related to growing crops to control pests.
- These farming practices include the growing of health seeds, planting resistant varieties, removing diseased plants, crop rotation, early planting, good sanitation and clean crop environment, removal of crop residue, weeding and handpicking large pests and stamping them underfoot.
- Crop rotation and removal of crop residues help in breaking life cycles of pests and other diseases in the soil.
- Weeds should be removed as they may harbour pests and diseases.
- The cultural methods are mostly used in the growing of maize, cotton and tobacco.

Biological control

- Biological control is the use of a natural enemy of a pest such as a predator or disease to control pest population.
- The main challenge with introducing a predator is that it may end up feeding on the desirable plant.
- Examples of the use of biological control.
  1. Ladybirds introduced into a garden feed on aphids.
  2. Poultry/birds introduced into a field to feed on caterpillars, locusts, grasshoppers etc.

Chemical control

- Chemical control is the use of chemicals to control plant pests and diseases.
- Pesticides are poisonous chemicals used to control/kill plant pests.
- Pesticides are grouped into three main categories depending on the purpose they are usually applied for.
  1. Fungicides – applied to kill fungi.
  2. Herbicides – used against weeds.
  3. Insecticides – used to kill harmful insects.

- Chemical controls are available in a variety of forms, including spray, powders, concentrates and granules.
- All pesticides are dangerous or toxic and should be used with great care to avoid harm.
- There are four main categories of levels of toxicity and these are indicated by triangles of various colours.
- The triangle colours commonly used and their levels of toxicity from the least to the most dangerous are as follows:

i. Green triangle

Green Triangle

- Pesticides with a green triangle label are **less toxic**. These may be used in the home. However, these should be used with **caution** to avoid harm.

ii. Amber triangle

Amber Triangle

- Amber colour label means that the pesticide is **poisonous**.

iii. Red triangle

Red Triangle

- The red triangular colour means that the pesticide is **very poisonous** and very dangerous.

iv. Purple triangle

Purple Triangle

- This is an **extremely poisonous** pesticide. Extra caution should be exercised to prevent harm.

SAFE USE OF AGRO-CHEMICALS

Chemicals used in agriculture are called agro-chemicals. Greatest care in handling the chemicals should be observed because the chemicals are poisonous/toxic/dangerous. Examples of agro-chemicals are herbicides, pesticides and fungicides. The safe use of agro-chemicals is described under the following headings:

1. **Warning symbols**,
   (a) Purple triangle on container; means extremely poisonous.
   (b) Red triangle; means Very poisonous.
   (c) Amber triangle; means poisonous.
   (d) Green triangle; means less toxic but must be used with caution. May be used in the home.

2. **Protective clothing**,

To prevent skin contact with chemicals wear overalls, gumboots, gloves, goggles, and face mask.

3. **Hygiene,**
   a) Wash hands/body/clothes after handling chemicals to avoid contamination.
   b) Do not eat, smoke or drink when working with chemicals.
   c) Do not inhale/breath vapour from chemicals.
   d) Do not spray against the wind to prevent inhaling chemical vapour.
   e) Choose a calm day for spraying when there is no wind as chemicals may be blown away.
   f) Crops/vegetables to be eaten should not be sprayed later than 3 weeks before the date of harvest.

4. **Storage**
   a) Store chemicals under heavy lock and key.
   b) Keep chemicals out of reach of children.
   c) Store chemicals separately from food.

5. **Proper disposal**

a) Burn or bury empty containers.
b) Never throw empty containers in water to prevent contamination /pollution of water.
c) Expired chemicals/empty containers can be recycled.

6. **Instructions**
Read instructions on the container carefully and follow them closely.

COMPARING THE DIFFERENT PESTS CONTROL METHODS

- Considering the advantages and disadvantages of different pests control methods enables farmers to determine the most effective method to use in their gardens.

Advantages of chemical control method

1. Provides a quick and efficient means of killing pests.
2. Pesticides can be used over a considerable large area.
3. May be used rapidly implemented as needed.

Disadvantages of chemical control method

1. Chemicals are expensive.
2. Bioaccumulation (bio-magnification) – increase in the concentration of chemicals along a food chain that finally poisons the tertiary consumers.
3. Non-specific: may harm some beneficial organisms e.g. pollinators, decomposers and pest predators.
4. Human and animal exposure to these pesticides can be very harmful.
5. Contribute to the damage of the ozone layer and harm other organisms living within the environment.
6. Rapid application of chemicals may result in pests becoming resistant.

Pesticide resistance

- This is a change in the sensitivity of a pest population to a pesticide, resulting in the failure of a correct application of the pesticide to control the pest.
- This may be as a result of applying a similar pesticide over and over again with the same mode or doses.
- Pesticide resistance can be managed by reducing and alternating chemical application with different modes.

Advantages of using biological control instead of pesticides (chemical control)

1. Lasts longer / no need to reapply
2. Specific / no harm to other species e.g. pollinators and decomposers.
3. Safe and environmental friendly since predator is part of the environment or food chain.
4. No bioaccumulation / no bio-magnification /no bio-amplification
5. No resistance;
6. No harm/effect on food chains.
7. Cheap after start up.

Disadvantages of using biological control

1. The natural enemies of the pest may move away especially when using a predator.

2. Predator may end up feeding on the desirable plant.
3. Invasive predators of pests (foreign predators imported from other countries) may outcompete and drive out native predators.

Advantages of cultural pests control methods

1. Safe and environmentally friendly since some practices are actually part of the natural environment.
2. Low cost and readily available.
3. Natural pest control is very targeted and therefore an effective way to control particular pests

Disadvantages of cultural control methods

1. They are relatively slow and may need to be repeated more often.

FORM 3: TRANSPORT IN PLANTS

KEY CONCEPT	LEARNING OBJECTIVES Learner should be able to	CONTENT (Attitudes, Knowldeg and Skills)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.5.3 Transport	• describe the general structure of a plant	• External Plant structure	• Educational tour to observe the general structure of plants	• Hand lenses • Maize and bean plants
	• distinguish between monocotyledonous and dicotyledonous plants	• Stem , leaves, flowers ,seeds and roots	• Comparing the structures of dicotyledonous and monocotyledonous plants	• Maize and bean seeds
	• identify parts of the internal structure of a young dicotyledonous stem and root	• Plant stem and root internal structure • Stem: Epidermis, cortex and vascular tissue and pith	• Observing prepared slides showing internal structures of stem and root	• Microscopes • Bio viewers and bio sets
	• compare internal structure of dicotyledonous stem and root	• Root: Epidermis, cortex, vascular tissue and root hairs • Xylem – support and transport • Phloem – transport	• Making labelled drawings of observed structures - Discussing the functions of xylem and phloem	• Prepared slides
	• state the functions of xylem and phloem vessels • define transpiration	• Transpiration • Functions of transpiration - Water and ion movement in xylem - Cooling effect	• Discussing the concept and functions of transpiration	• Dyes • Celery plants or spinach
	• describe the functions of transpiration • investigate factors affecting rate of transpiration	• Factors affecting rate of transpiration - Surface area - Stomata distribution - Temperature - Wind - Humidity - Light intensity	• Carrying out experiments to investigate factors affecting the rate of transpiration. • Demonstrating • transpiration	• Microscope slides • Bioviewers • Leaves • Vaseline • Transparent plastic bags
	• describe adaptations of leaves to minimize water loss • describe how wilting occurs • define translocation • describe the effect of ring barking	• Adaptations - Reduction of surface area - Thickness of cuticle - Distribution of stomata - Presence of hairs • Wilting and its significance • Translocation • effects to plants and the environment	• Observing leaves. • Carrying out experiments to investigate the distribution and role of stomata in water loss. • Observing wilting plants. • Explaining wilting and its significance. • Discussing translocation • Explaining the process of bark ringing. • Explaining the effects of ring barking	• Potted plants • Print media • Educational touring

External Plant Structure

- Flowering plants, aka angiosperms, are the most abundant plants on earth.
- The external structure of a flowering plant is as follows.

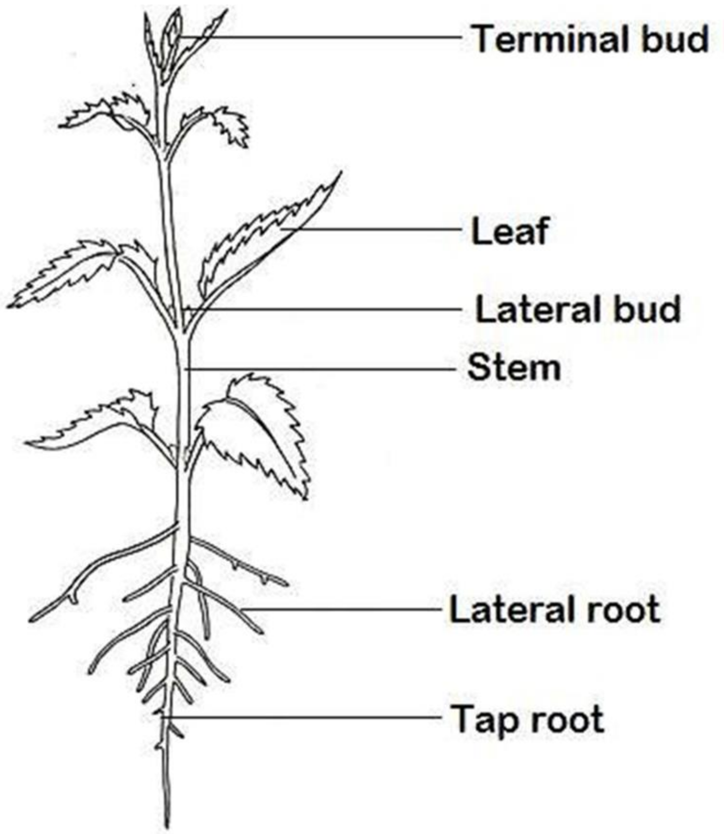


DIAGRAM: External plant structure.

Functions of the external parts of a plant

- ❖ **The terminal bud**
  - continues growth in height,
  - produces new leaves and flowers.
- ❖ **Flowers**
  - Structures for sexual reproduction.
  - Produce fruit and seeds.
- ❖ **The leaf**
  - produces carbohydrates during photosynthesis.
  - absorbs sunlight for photosynthesis.
  - evaporates water during transpiration to cool the plant.
- ❖ **The lateral bud**
  - makes branches,
  - produces new leaves flowers.
- ❖ **The stem**
  - carries water to the leaves,
  - conducts food to the roots,
  - spaces out the leaves,
  - If the stem is green it may also absorb sunlight and produce carbohydrates by photosynthesis.
- ❖ **The root**
  - absorbs mineral salts,

- absorbs water,
- anchors the plant in the soil.

Monocots and dicots

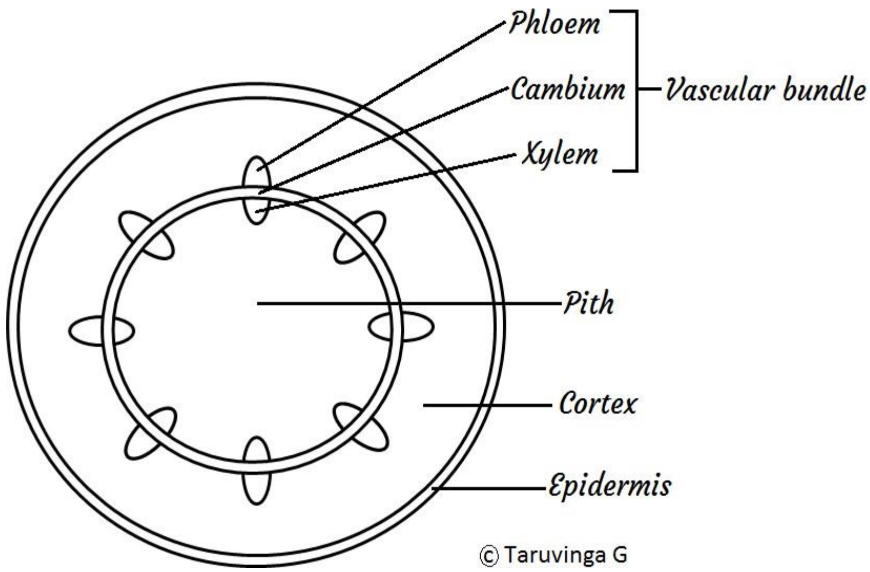
- The two major groups of flowering plants are the monocotyledonous plants (aka monocots) and the dicotyledonous plants (aka dicots).
- Basically monocots have one cotyledon whereas dicots have two cotyledons.
- The table that follows summarises all the differences between monocots and dicots.

TABLE: Differences between Monocots and Dicots.

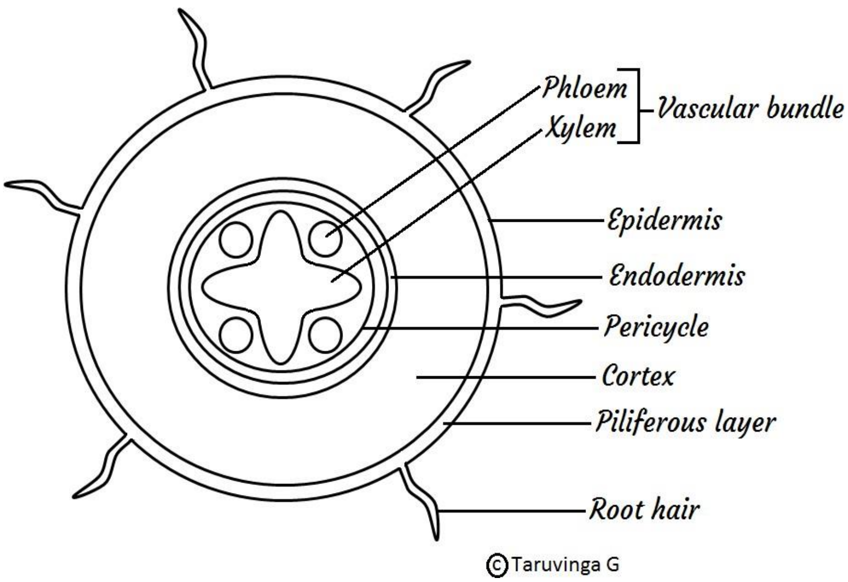
	Monocot	Dicot
Embryos (Seeds)	 One cotyledon	 Two cotyledons
Leaf venation	 Paralell venation (Veins usually parallel)	 Net venation (Veins usually netlike)
Stems	 Vascular tissue scattered	 Vascular tissue usually arranged in ring
Roots	 Root system usually fibrous (no main root)	 Taproot (main root) usually present
Pollen	 Pollen grain with one opening	 Pollen grain with three openings
Flowers	 Floral organs usually in multiples of three	 Floral organs usually in multiples of four or five
Examples	Maize and most grasses	Bean & groundnut plants

Internal structure of a young dicotyledonous stem

TS – Dicot stem



Internal structure of a young dicotyledonous root **TS – Dicot root**



- **Phloem** and **xylem** make up the **vascular bundle,** aka **vascular tissue.**
- **The function of phloem in a plant** is to transport sugars from leaves to all parts of the plant both up and down the plant.
- **The functions of xylem in a plant are:**
  a) To transport water up the plant.
  b) To transport dissolved minerals.
  c) For structural support: Xylem makes stems strong.

Try these questions

The diagrams represent transverse sections through a root and a stem.

(a) Say which one represents the stem and which one represents the root, giving your reasons.

(b) Name the parts of these organs represented by the letters A to J.

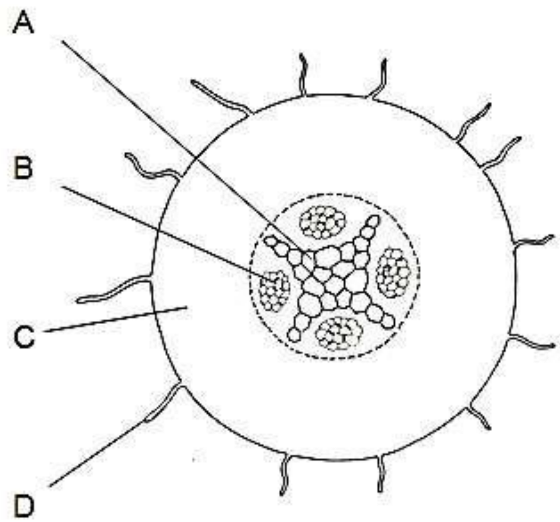


Figure 1

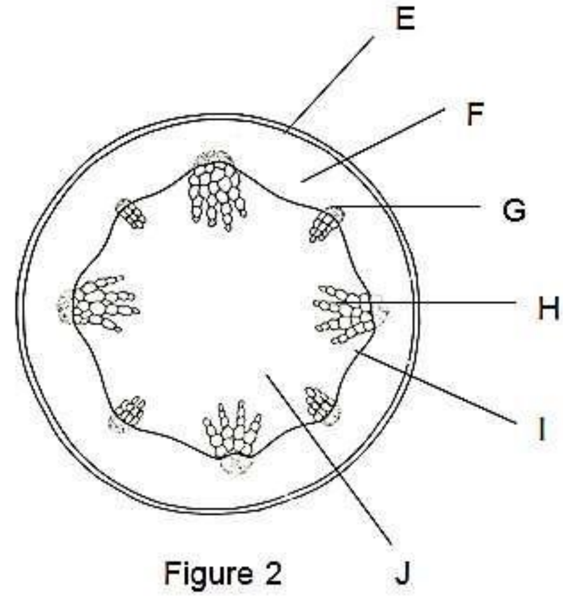


Figure 2

Answers

(a) Figure 1 represents a transverse section through a **root**.
Reason: The presence of root hairs and the central position of the vascular tissue (xylem and phloem) are the features which identify this structure as a root.
Figure 2 represents a transverse section through a **stem**.
Reason: The diagnostic features are the distinct epidermis, the central pith and the distribution of vascular bundles round the periphery.

(b) Figure 1:
A – xylem (or vessels), B – phloem, C – cortex, D – root hair,

Figure 2:
E – epidermis, F – cortex, G – phloem,
H – xylem (or vessels), I – cambium,
J – pith.

Identifying xylem vessels using the red dye eosin.

- Dip the cut end of the stem of celery or spinach plant in water dyed with the red dye eosin.
- After 30 minutes remove the stem and cut thin sections.
- Mount the thin sections on a slide and observe them under a microscope.

Observation
- The xylem vessels are stained red by the dye.
- The phloem vessels are not stained.

Conclusion
- The xylem absorbed the water and the dissolved dye and transported them up the plant.
- Therefore, xylem transport water and dissolved mineral salts.
- The phloem was not affected because it does not transport water.

TRANSPIRATION

- Transpiration is the process by which plants lose water in the form of water vapour into the atmosphere.
- Water is lost through the stomata, cuticle and lenticels.

Stomatal transpiration:

- Stomatal transpiration is loss of water through the stomata of leaves.
- This accounts for 80-90% of the total transpiration in plants.
- Stomata are small pores found on the leaves.

Cuticular transpiration:

- Cuticular transpiration is loss of water through the cuticle.
- The cuticle is found on leaf surfaces, and a little water is lost through it.
- Plants with thick cuticles minimize water loss through the cuticle.

Lenticular transpiration

- Lenticular transpiration is loss of water through lenticels.
- These are found on stems of woody plants.
- Water lost through the stomata and cuticle by evaporation leads to evaporation of water from surfaces of mesophyll cells.
- The mesophyll cells draw water from the xylem vessels by osmosis.
- The xylem in the leaf is continuous with xylem in the stem and root.

Functions of Transpiration

1) Cooling of the plant in hot weather.
2) Water and ion transport through xylem.

Demonstrating transpiration using a leafy twig and leafless twig tied in plastic bags, and testing the vapour produced using cobalt chloride paper or anhydrous copper (II) sulphate.

- A transparent plastic bag is tied over a leafy twig X to enclose its leaves.
- Another transparent plastic bag is tied over a similar but leafless twig Y which has had its leaves pruned. This is the control.
- After a few hours water vapour produced by transpiration of the leaves of twig X is seen condensing on the inside surface of the plastic bag.
- The condensed liquid turns blue cobalt chloride paper to pink, confirming that it's indeed water that has been produced. The liquid is also tested with white anhydrous copper sulphate and it turns it to blue further confirming that the liquid is water.
- The leafless twig Y which is the control shows no water vapour condensing inside the plastic bag.
- This indicates that the water vapour has come from the leaves.

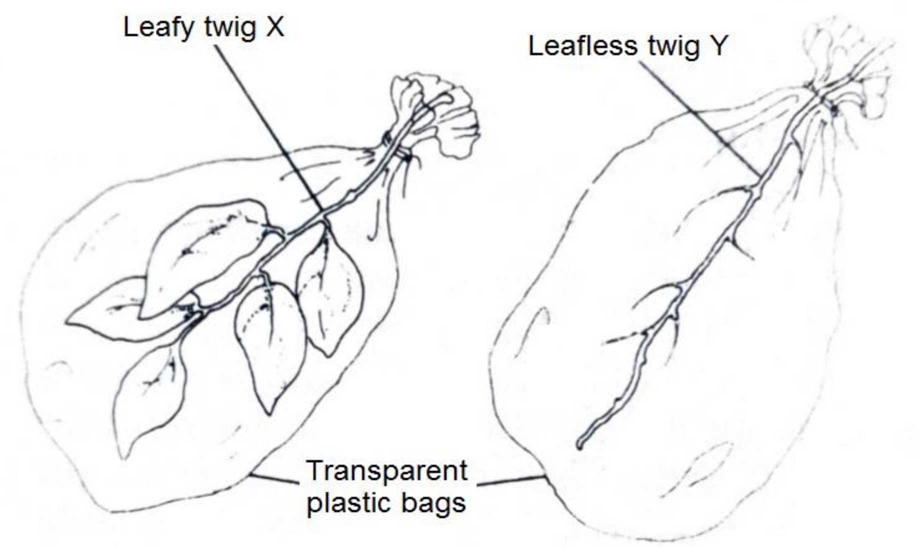


DIAGRAM: Demonstrating transpiration.

Demonstrating transpiration by measuring the initial and final masses of a leafy plant.

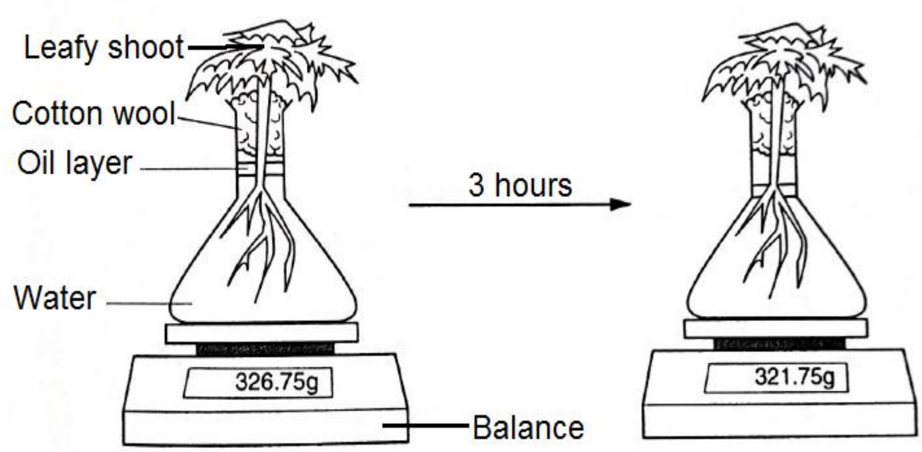


Diagram: Demonstrating transpiration by loss in mass of a shoot.

- A leafy plant in a flask of water is left for three hours in a warm and windy place in bright sunshine.
- All the other parts of the plant are prevented from water loss except the leaves. The oil layer above the water prevents loss of water by evaporation from the water surface. The cotton wool holds the plant in place.
- Observation: After about three hours the balance records a decrease in mass. The water level in the flask also falls.
- Conclusion: The decrease in mass and water level in the flask shows that some of the water absorbed by the plant has evaporated through the leaves during transpiration.
- A potted plant can also be used in this investigation; the pot (vase) and soil are sealed off in a polythene / plastic bag to prevent water loss from the soil in the vase.

FACTORS AFFECTING TRANSPIRATION

- The factors that affect transpiration are grouped into **two**, that is, **environmental** and **structural**.

Environmental factors

Temperature

- High temperature increases the internal temperature of the leaf, which in turn increases kinetic energy of water molecules which increases evaporation.
- High temperatures dry the air around the leaf surface maintaining a high concentration gradient. More water vapour is therefore lost from the leaf to the air.

Humidity

- Humidity is the amount of water vapour in the air / atmosphere.
- The higher the humidity of the air around the leaf, the lower the rate of transpiration.
- The difference in humidity between the inside of the leaf and the outside is called the saturation deficit.
- In a dry atmosphere, the saturation deficit is very high.
- At such times, transpiration rate is high.

Wind

- Wind carries away water vapour as fast as it diffuses out of the leaves.
- This prevents the air around the leaves from becoming saturated with vapour.
- On a windy day, the rate of transpiration is therefore high.

Light Intensity

- When light intensity is high; more stomata open hence high rate of transpiration.

Atmospheric Pressure

- The lower the atmospheric pressure the higher the kinetic energy of water molecules hence more evaporation.
- Most of the plants at higher altitudes where atmospheric pressure is very low have adaptations to prevent excessive water-loss.

Availability of Water

- The more water there is in the soil, the more is absorbed by the plant and hence a lot of water is lost by transpiration.

Structural Factors

Cuticle

- Plants growing in arid or semi-arid areas have leaves covered with a thick waxy cuticle.

Stomata

- The more the stomata, the higher the rate of transpiration.
- Xerophytes have fewer stomata which reduce water-loss.
- Some have sunken stomata which reduces the rate of transpiration as the water vapour accumulates in the pits.
- Others have stomata on the lower leaf surface hence reducing the rate of water loss.
- Some plants have reversed stomatal rhythm whereby stomata close during the day and open at night.
- This helps to reduce water loss by transpiration.

Leaf size and shape

- Plants in wet areas have large surface area for transpiration.
- Xerophytes have small narrow leaves to reduce water loss.

Investigating the role and distribution of stomata

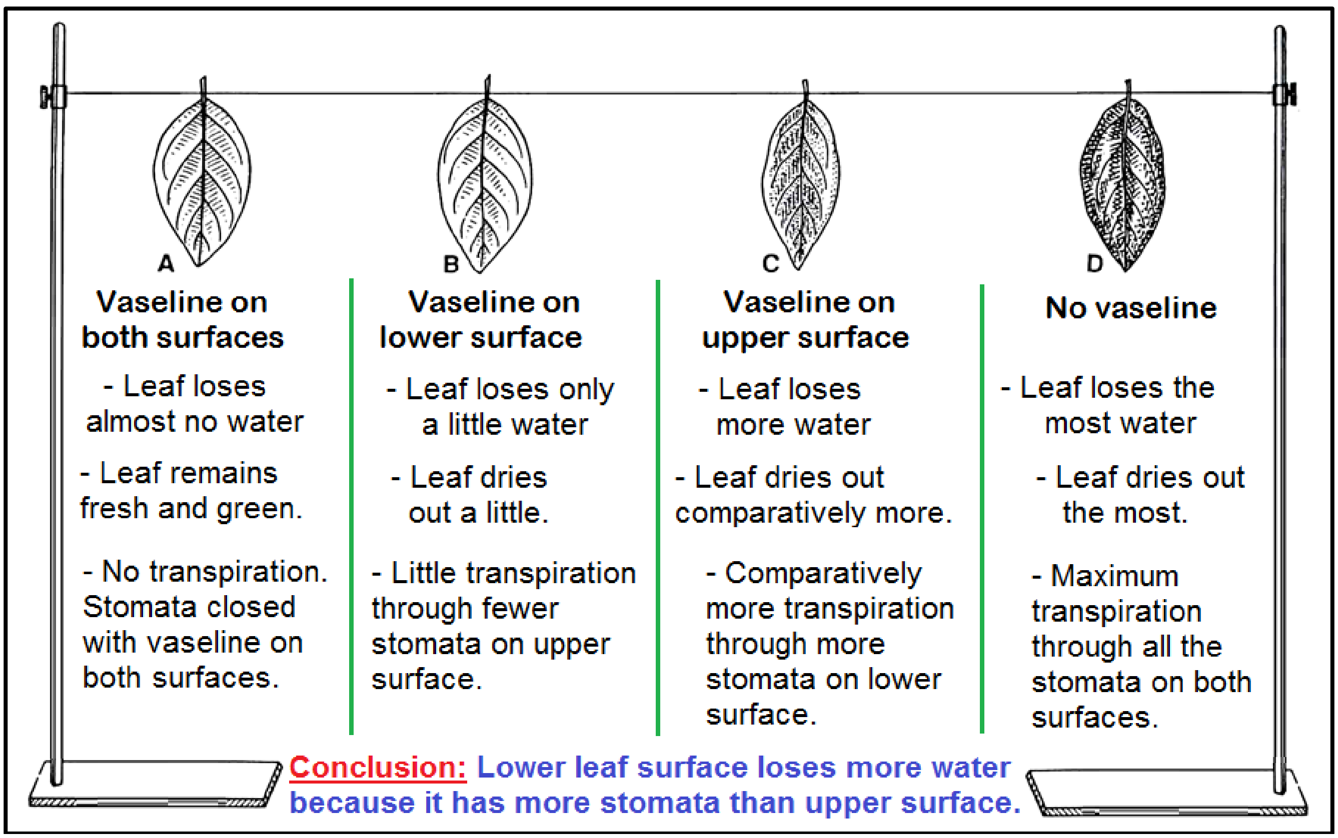
Experiment to compare transpiration rates from upper and lower leaf surfaces

Aim: Comparing transpiration rates through stomata from upper and lower leaf surfaces.

Materials required: four leaves, Vaseline, string.

Procedure:

1) The four fresh leaves are taken and labelled as A, B, C and D.
2) Leaf **A** is coated with Vaseline on both surfaces.
3) The lower surface of leaf **B** is coated with Vaseline. On the contrary, the upper surface of leaf **C** is coated with Vaseline.
4) The leaf **D** kept uncoated.
5) Vaseline closes the stomata and therefore prevents the loss of water through the stomata.
6) The leaves are tied in a similar sequence on the string. The setup is then placed in the sunlight for about one or two days.



ANOTATED DIAGRAM: Comparing transpiration rates from upper and lower leaf surfaces.

Observation:

The following observation has been found:

1) Leaf **A** is fully coated with vaseline and remains fresh and green.
2) Leaf **B** dried out a little.
3) Leaf **C** dried out comparatively more.
4) Leaf **D** is dried out the most.

Conclusion:

1) No transpiration has occurred through leaf **A** as stomata were closed on both surfaces due to the application of vaseline
2) Leaf **B** loses a little water by transpiration as its upper surface with less stomata is exposed for transpiration.
3) Leaf **C** shows comparatively more transpiration as its lower surface with more stomata is exposed for transpiration.
4) Leaf **D** shows the maximum rate of transpiration as stomata on both sides are exposed and transpiring.

- This experiment shows that the **lower surface of the leaf loses more water vapour** by transpiration than the upper surface. This is because **there are more stomata on the lower surface of the leaf**.

POTOMETER

- The potometer can be used to determine transpiration in different environmental conditions.

A potometer actually measures **water uptake by the plant**.

- This can be used as an **estimate** of **water loss through transpiration**.
- With the assumption that **water uptake** = **water loss through transpiration**.
- Assumed that the detached branch is representative of the plant as a whole.
- Problems with this,
  - some **water** may be **used** for **turgidity** of the **plant cells**.
  - some **water** will be **used** in **photosynthesis**.

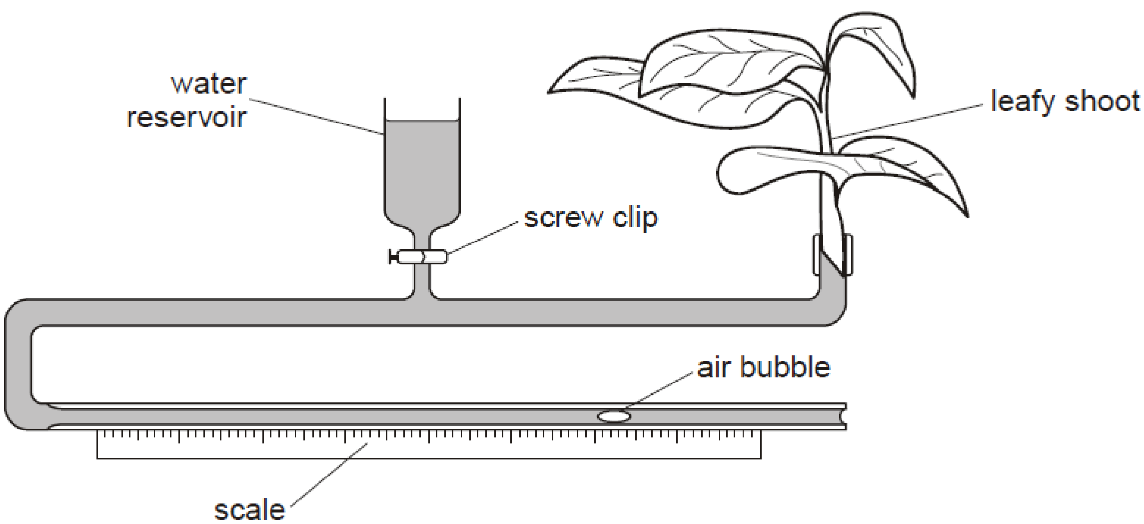


DIAGRAM: A potometer

How to setup a potometer

1. cut (healthy) shoot under water (to stop air entering xylem vessels);
2. cut shoot at a slant (to increase surface area);
3. check apparatus is full of water / is air bubble free / no air locks;
4. insert shoot into apparatus under water / AW;
5. remove potometer from water and ensure, airtight / watertight, joints around shoot;
6. dry leaves / AW;
7. keep, condition(s) / named condition(s), constant;
8. allow time for shoot to acclimatise / AW;

9. shut screw clip;
10. keep ruler fixed and record position of air bubble on scale;
11. start timing and, measure / calculate, distance moved per unit time

ADAPTATIONS OF LEAVES TO MINIMISE WATER LOSS BY TRANSPIRATION

1. **Small, spiny / needled shaped leaves** to reduce surface area exposed for evaporation, e.g. conifer needles and leaves replaced by thorns in cactus.
2. **Thick, waxy cuticle** to reduce cuticular transpiration / stops uncontrolled evaporation through leaf cells.
3. **Reduced number and size of stomata** (low stomata density) – fewer gaps in leaves and therefore less evaporation.
4. **Sunken stomata** (stomata in grooves) to trap humid air around stomata which decreases water loss by transpiration, e.g. marram grass.
5. **Stomatal hairs** (Stomata surrounded by hairs) to maintain humid air around stomata which decreases water loss by transpiration.
6. **Hairy leaves** trap cushion of moist air close to leaves which cools leaves and reduce evaporation.
7. **More stomata on lower surface of leaf** - more humid air on lower surface, so less evaporation, e.g. most dicots.
8. **Rolled leaves in hot weather** maintain humid air around stomata which reduces water loss by transpiration.
9. **Shiny cuticle (Highly reflective cuticle)** reflects away incident heat waves. This reduces leaf temperature, hence reduces water loss.
10. **Succulent leaves and stems** to store water, e.g. cactus.
11. **Shedding leaves in dry/cold season** - reduce water loss at certain times of year, e.g. deciduous plants.

WILTING

- The term wilting describes the drooping of leaves and stems of herbaceous plants after considerable amounts of water have been lost through transpiration.
- It is observed in dry weather.
- This is when the amount of water lost through transpiration exceeds the amount absorbed through the roots.
- Individual cells lose turgor and become plasmolysed and the leaves and stems droop.
- The condition is corrected at night when absorption of water by the roots continues while transpiration is absent.
- Eventually, wilting plants may die if the soil water is not increased through rainfall or watering.

TRANSLOCATION

- Transport of soluble organic products of photosynthesis within a plant is called translocation.
- It occurs through phloem in sieve tubes.
- Substances translocated include glucose, amino acids, and vitamins.
- These are translocated to the growing regions like stem, root apex, storage organs e.g. tubers, bulbs and secretory organs such as nectar glands.

RING BARKING

- Ring barking is cutting and removing a complete ring of bark from the stem of a tree.
- This removes the phloem since they are on the outside and leaves the xylem intact.

Effects of ring barking

- After a week there is a swelling above the ring, reduced growth below the ring and the leaves are unaffected.
- This is evidence that sugars are transported downwards in the phloem:
  - sugars cannot pass the cut
  - decrease water potential;
  - water moves into cells;
  - damage triggers increased cell division;
  - to produce cells to store sugars;

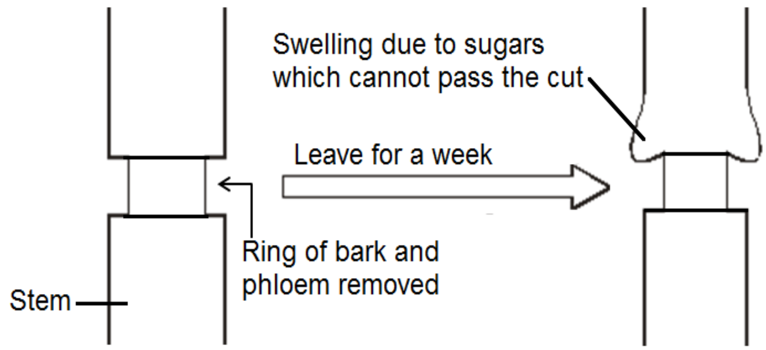


DIAGRAM: Ring barking

Try this question:
Suggest explanations for the following observation: Fruit growth is suppressed if a ring of bark between the fruit and mature leaves is removed. [3]

Answer
Removal of bark removes phloem;
responsible for transport of sugars to fruit;
to enable fruit development/formation of food store in fruit;

FORM 4: REPRODUCTION IN PLANTS
FORM4: ASEXUAL REPRODUCTION IN PLANTS – VEGETATIVE REPRODUCTION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (ATTITUDES, SKILLS AND KNOWLEDGE)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.15.1 Reproduction * Asexual Reproduction In Plants	• define asexual reproduction • describe natural and artificial methods of vegetative reproduction in plants	• Asexual reproduction in plants • Vegetative reproduction • Natural methods - Rhizomes, tubers, suckers, runners bulbs • Artificial methods - Cuttings, grafting	• Discussing asexual reproduction in plants. • Observing vegetative structures in plants. • Demonstrating various methods of vegetative reproduction.	• Plant vegetative structures such as: stems, tubers, leaves, bulbs • Gardens
	• state advantages and disadvantages of vegetative reproduction	- Resistance to diseases and pests - genetic variation - survival of offspring - rate of propagation	• Discussing advantages and disadvantages of vegetative reproduction	• ICT tools • Braille software/Jaws • Grass • Irish potatoes • Bananas • Sugar cane • Sweet-potato stems • Onion bulbs • Strawberry plants

REPRODUCTION

- **Reproduction** is the process that makes more of the same kind of an organism.

- Reproduction is the production of offspring through a **sexual** or **asexual** process.

Types of reproduction

- There are 2 types of reproduction: **asexual** and **sexual**. Sexual reproduction occurs through fertilisation of gametes of two parents whereas asexual reproduction occurs by mitosis of vegetative structures of a single parent.

1. ASEXUAL REPRODUCTION:

- Is the production of ***genetically identical offspring*** from one parent.
- Is formation of a new organism, ***without involvement of gametes or fertilisation***.
- Asexual reproduction involve **mitosis,** a process by which a cell divides into two daughter cells, each of which has the same number of chromosomes as the original (parent) cell. The two cells further divide into four and so on.

Examples of asexual reproduction

- Binary fission in bacteria
- Spores in fungi
- Budding in yeast
- Vegetative reproduction/propagation in plants.

VEGETATIVE REPRODUCTION/PROPAGATION IN PLANTS

- Vegetative reproduction is the production of new and identical plants from **vegetative structures** (outgrowths) or **fragments** of a single parent plant.
- Through asexual reproduction, a plant produces clones (genetically identical offshoots) of itself, which then develop into independent plants.
- Some plants, however, reproduce sexually by pollination, fertilisation and seeds.
- Examples of plant vegetative structures (outgrowths) are rhizomes, tubers, cuttings, suckers, runners and bulbs.
- Vegetative reproduction in plants can either be **natural** or **artificial**.
- Natural vegetative reproduction occurs by means of rhizomes, tubers, suckers, runners, tubers, etc.
- Artificial vegetative reproduction occurs by means of cuttings, grafting, etc.

NATURAL VEGETATIVE REPRODUCTION

- **DEFINITION:** Natural vegetative reproduction is the natural propagation of a new plant from vegetative structures without human intervention.
- **OCCURRENCE:** Natural vegetative propagation occurs naturally in plants.
- **EXAMPLES:** Natural vegetative propagation occurs by means of rhizomes, tubers, suckers, runners, bulbs, tubers, etc.

a) Rhizomes

- A rhizome is an underground stem that grows horizontal to the ground.
- Grasses reproduce by means of rhizomes

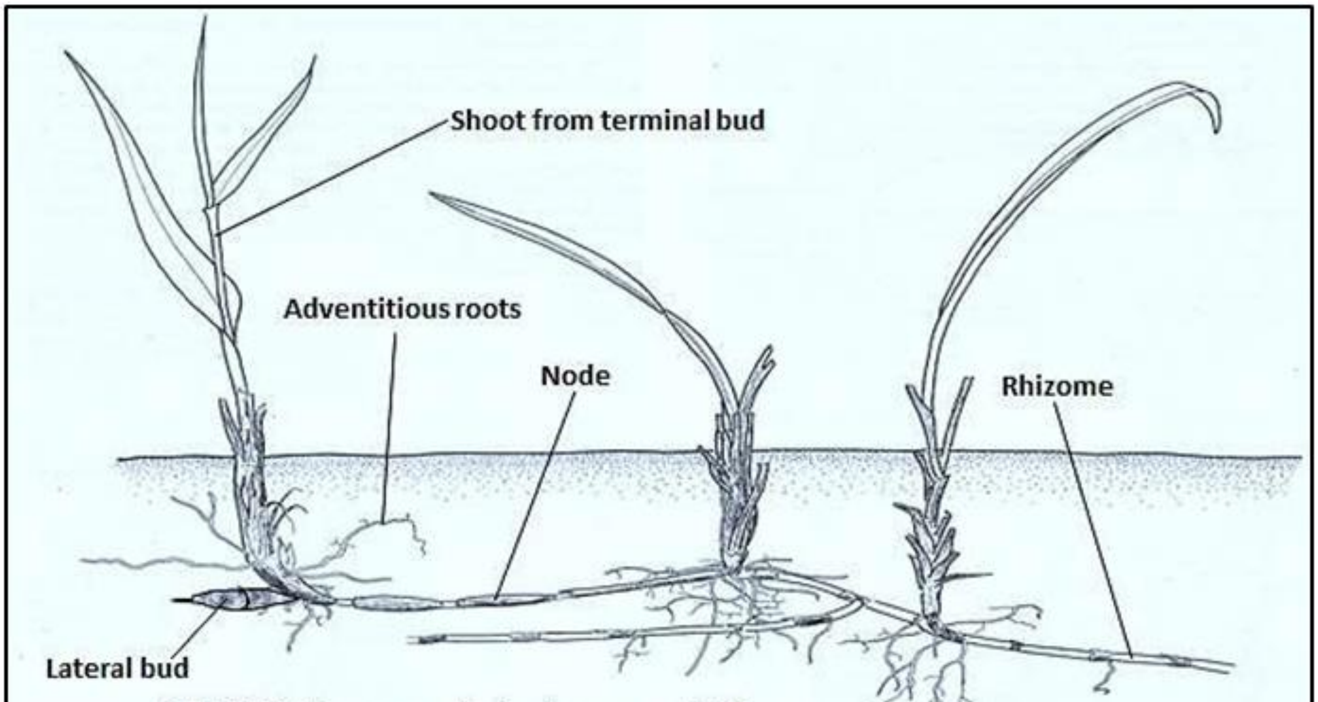


DIAGRAM: Grass reproducing by means of Rhizomes

- A rhizome produces new plants at each bud on the nodes along its length.
- A rhizome stores food which enables the plant to grow the next season and perennate (survive from year to year).

b) Suckers

- A sucker is a shoot that grows from an underground root or stem.
- **Banana plants** reproduce by means of suckers. New stems grow from the base of old ones, forming new plants.

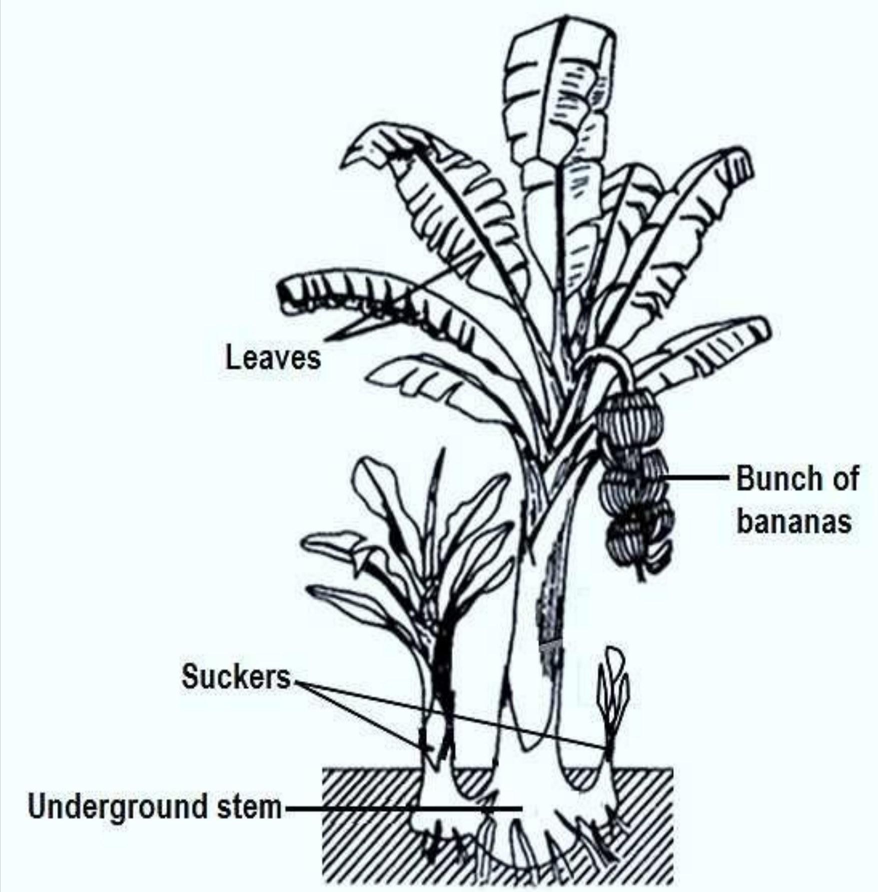


DIAGRAM: Banana plant reproducing by means of suckers

c) Tubers

- A tuber is a swollen underground stem or root.
- The tuber is swollen with stored food in the form of starch.

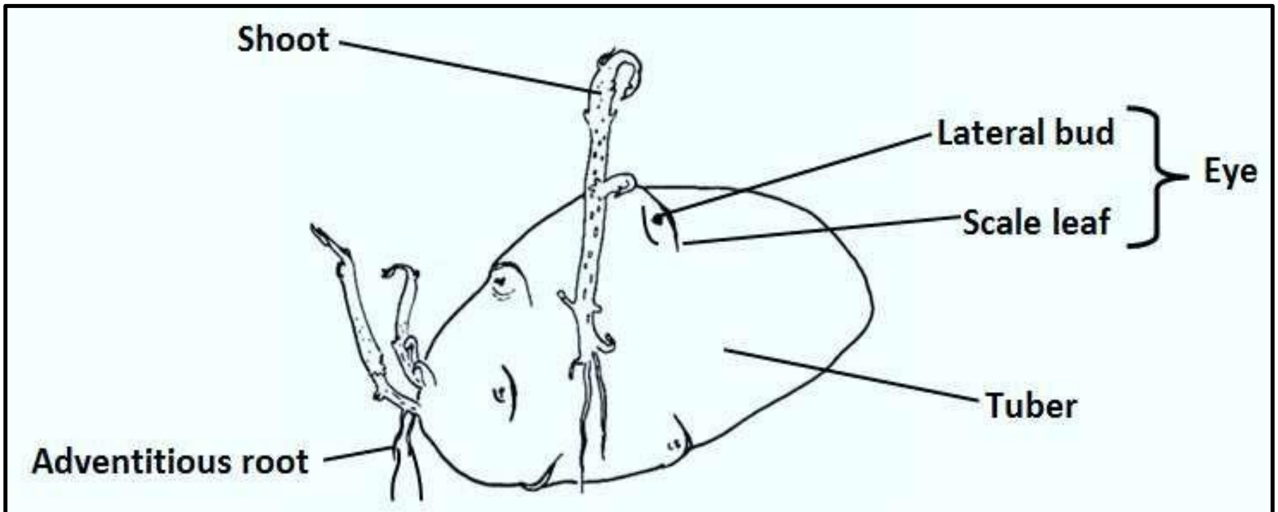


DIAGRAM: Irish potato sprouting

- **Irish potatoes** reproduce by means of tubers.
- The parent tubers are allowed to produce short shoots from their eyes or lateral buds and the planted to produce new potato plants.

d) Runners

- Runners aka stolons.
- Runners are thin, long horizontal stems that spread above the ground from the main plant.
- Entirely new plants develop from nodes located at intervals on the runners; each node gives rise to new roots and shoots.
- Strawberry plants and runner grass are examples of plants that propagate by means of runners.

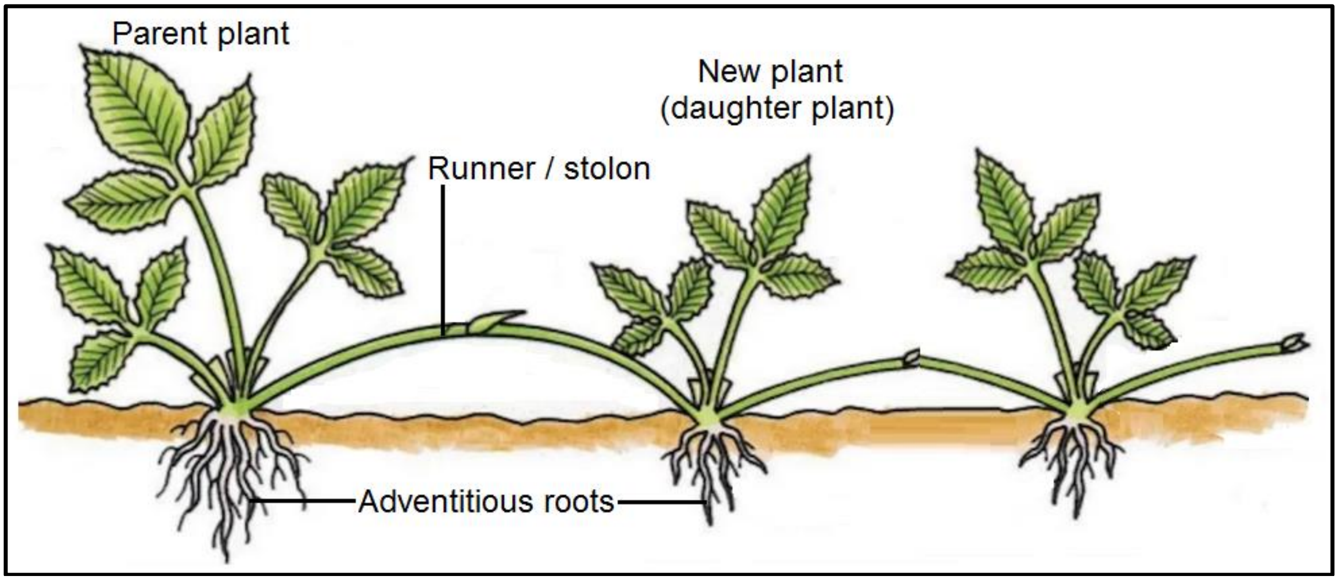


DIAGRAM: Strawberry plant propagating by means of runners.

PICTURE: Strawberry runners branching off parent plant.

e) Bulbs

- Bulbs are spherical underground storage structures.
- Bulbs are stems or buds that are enclosed with fleshy layered leaves.
- The parent bulb produces smaller buds called lateral buds.
- As the mature plant gets to the end of its life, the lateral buds develop into smaller bulbs that are attached to the base of the parent bulb.
- The smaller bulbs then develop into new plants during the new growing season.
- The new bulbs can also be physically separated and planted.
- Examples of plants that reproduce by means of bulbs include onions, tulips and daffodils.
- The propagation by bulbs can be summarised as follows:

Parent bulb $\xrightarrow{\text{produces}}$ Lateral bulbs $\xrightarrow{\text{which develop into}}$ Smaller daughter bulbs $\xrightarrow{\text{which develop into}}$ New plants

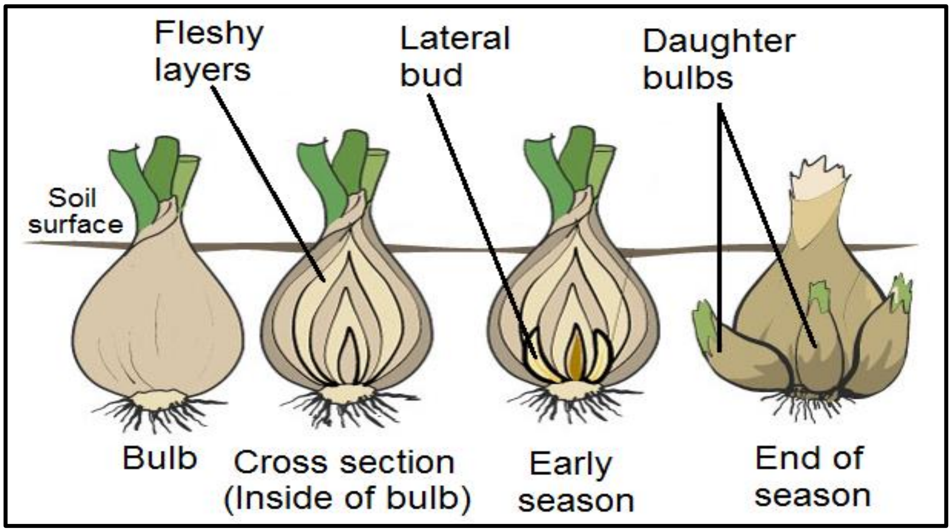


DIAGRAM: Vegetative propagation by bulbs.

ARTIFICIAL METHODS OF VEGETATIVE REPRODUCTION

- **DEFINITION:** Artificial vegetative reproduction is the artificial propagation of plants under the influence of man.
- **OCCURRENCE:** Artificial vegetative reproduction occurs artificially in plants.
- **EXAMPLES:** Artificial vegetative reproduction occurs by means of cuttings, grafting etc.

(a) Cuttings

- Cuttings: this is simply where a stem is cut from a plant and replanted.
- Examples of plants that can reproduce by cuttings are **sugar cane** and **sweet potato**.
- The cutting part buried underground quickly grows roots from nodes to become a new plant.

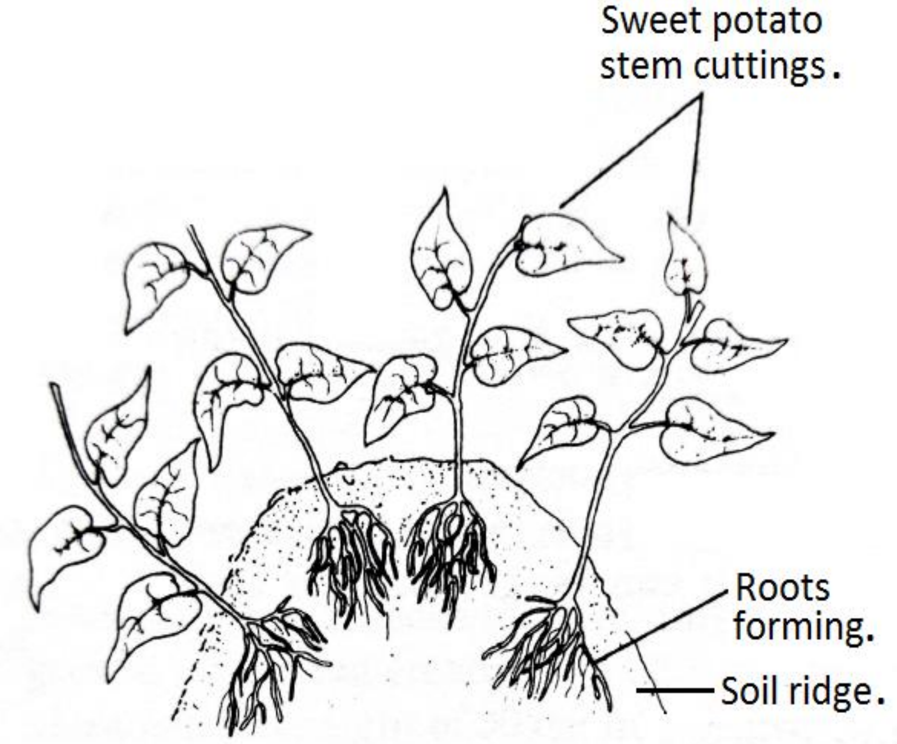


DIAGRAM: Sweet potato stem cuttings.

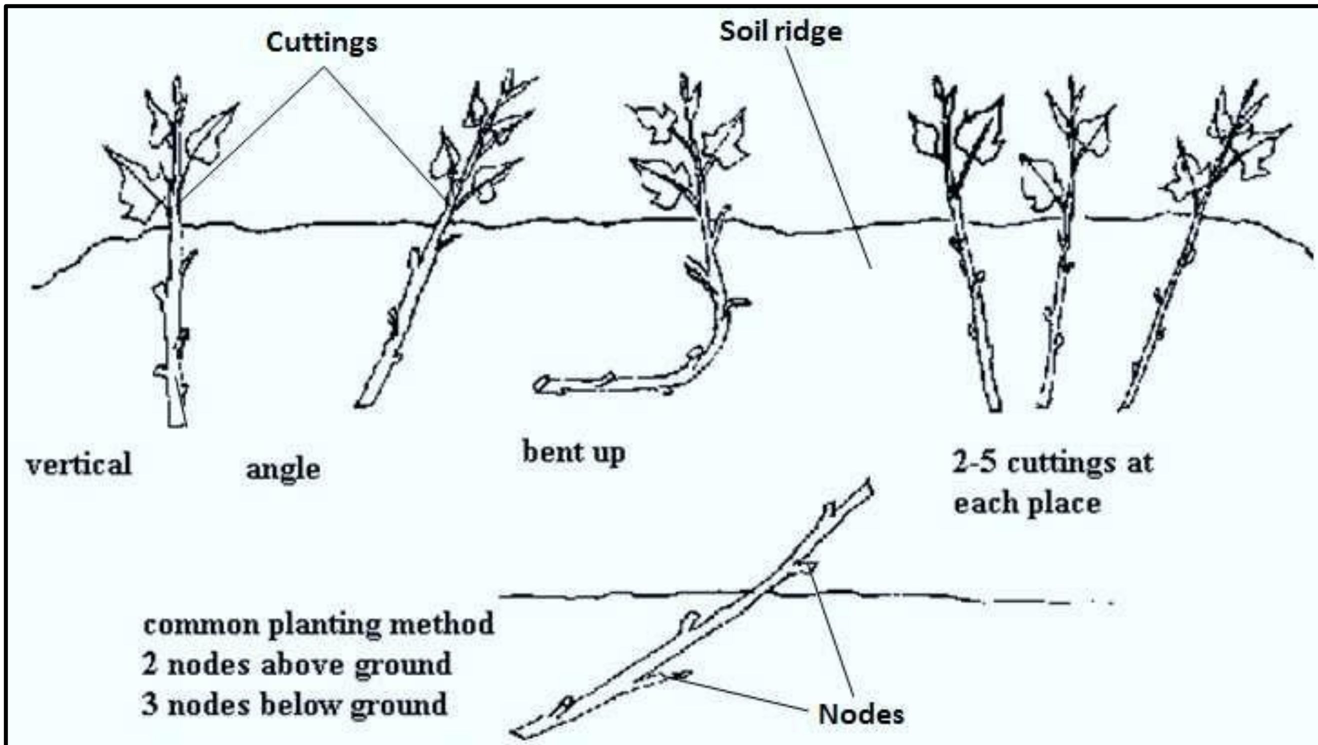


DIAGRAM: Common methods of planting sweet potato cuttings.

(b) Grafting

- Grafting is insertion of **shoot** or **bud** onto a related plant so they grow as one plant.
- Grafting joins plant parts from two related plants so they will heal and grow as one plant.
- Two plant portions, **rootstock** and **scion**, of the same variety are joined.

- The **lower part** of the combined plant called the rootstock or stock provides the root system.
- **Definition:** The **stock** or **rootstock** is a **rooted stem** into which a **scion** or a **bud** is grafted.
- The **upper part** of the combined plant called the scion provides the shoot system.
- **Definition:** The **scion** is a **piece** of **young stem** or **bud** which is inserted into a root stock.
- Basically there are **two major types of grafting** namely **stem grafting** and **bud grafting**.

(i) Stem grafting

- This is when a **shoot** of a selected desired plant (known as the **scion**) is grafted onto the **stem** of another plant (known as the **stock**), so that the cambia of the two combine and grow as one plant.
- **The following are the stages in stem grafting:**
  Both scion and stock are cut at a slanting angle, placed in close contact facing each other, and are then held together by binding around with tape.

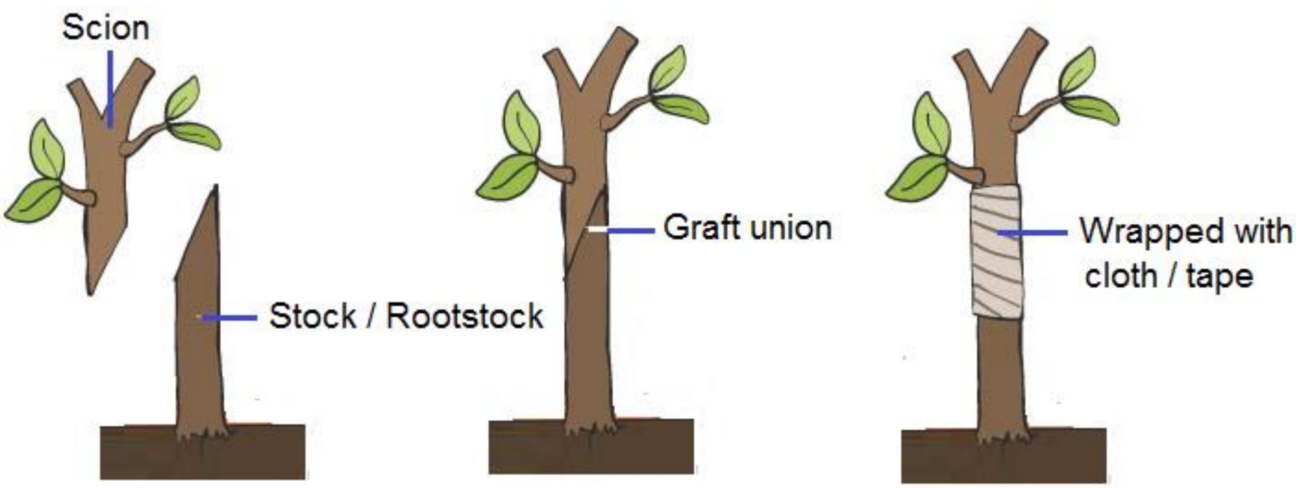
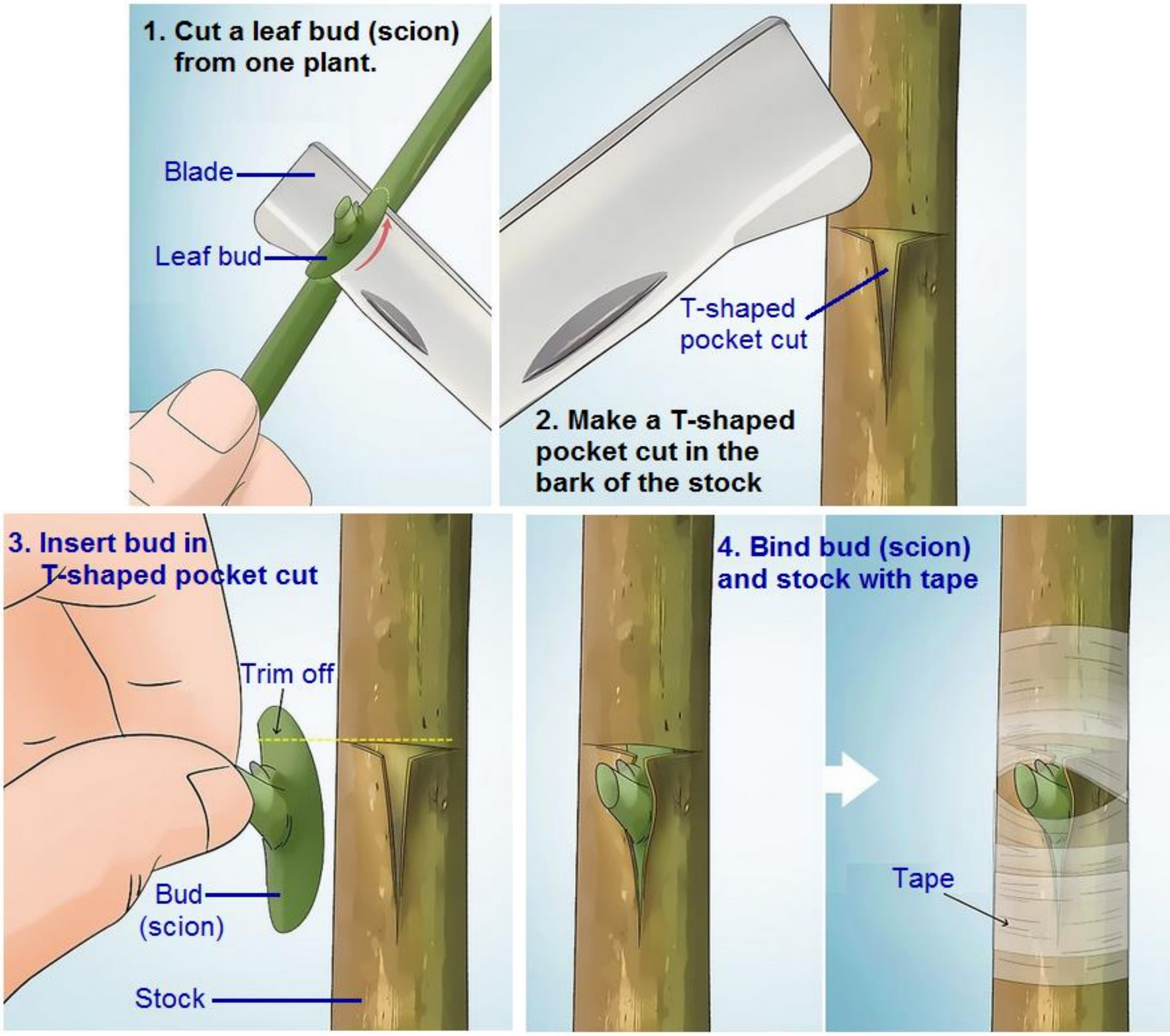


DIAGRAM: Grafting stock and scion of the same plant variety.

(ii) Bud grafting (aka budding)

- Bud grafting is also known as **budding**.
- In bud grafting a leaf bud is transplanted from one tree to another of the same type.
- Unlike stem grafting, which attaches the entire upper part of a plant, bud grafting (budding) only attaches the bud to a different plant.
- The **transplanted bud** is the **scion** and the **plant to which it is transplanted** is the **stock** (rootstock).
- **The following are the stages in bud grafting:**
  1. Cut a leaf bud from one plant. This is your scion
  2. Make a T-shaped pocket cut through the bark of the stock of another plant.
  3. Transplant the bud (scion) into the T-shaped pocket cut on the stock.
  4. Hold the bud (scion) and stock together by binding around with tape.



DIAGRAMS: Stages in budding (bud grafting).
(source: https://www.wikihow.com/Do-Budding-in-Plants#/Image:Do-Budding-in-Plants-Step-4-Version-2.jpg)

The requirements for grafting to be successful:

- Grafting requires:
  a) that cambia of stock and scion must meet and match to fuse and grow together. (Grafting is therefore only suitable for dicots because they have cambium and not suitable for monocots because they do not have cambium).
  b) firm binding at junction to prevent tearing of joined tissue.
  c) that grafting is done in autumn to prevent death from excessive transpiration.
  d) waterproofing the cuts with wax or tape to prevent drying and infection.
  e) related / compatible plant species (e.g. lemons will graft with oranges because they are both citrus, but will not graft with apple since they are not related).

- **The stock and scion are joined and grow into one plant which has the following advantages:**
  1) Neither stock nor scion is altered genetically by grafting. Each provides its desirable characteristics separately to the new plant:
  2) The specifically chosen root **stock** provides the ***vigour***.

3) The **scion** provides the ***quality of the product*** (e.g. flowers or fruits) on the shoot.

ADVANTAGES OF VEGETATIVE REPRODUCTION IN PLANTS

1. Quick. Offspring mature more rapidly.
2. Vegetative structures store large amounts of food leading to rapid growth.
3. Only one parent is needed. This eliminates the need for special mechanisms such as pollination and pollinating agents.
4. All good characteristics are passed on.
5. Useful/desirable traits are preserved as genetically identical offspring are produced.
6. No dispersal so offspring grow in same favourable environment as parent.
7. Seedless plants / plants that do not produce viable seeds can be reproduced by vegetative reproduction e.g. bananas, sugar cane, potatoes and roses.

DISADVANTAGES OF VEGETATIVE REPRODUCTION IN PLANTS

1. There is no mixing/blending of characteristics.
2. Little variation, therefore adaptation to environment is unlikely.
3. Offspring inherit poor/undesirable characteristics (e.g. poor resistance to diseases).
4. If a particular plant clone is susceptible to a certain disease, there is potential to lose the entire crop.
5. Lack of dispersal leads to competition for nutrients, water, light etc.
6. Colonisation of new areas is unlikely since there is no dispersal.

FORM 4: SEXUAL REPRODUCTION IN PLANTS

Form 4: Flowers & Pollination.

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (ATTITUDES, SKILLS AND KNOWLEDGE)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.15.1 Reproduction The flower	• describe the structure and characteristics of wind and insect pollinated flowers • compare the different structural adaptations of insect pollinated and wind pollinated flowers • outline the process of pollination • distinguish between self- and cross pollination • describe the advantages of self and cross pollination	• Flower as a reproductive structure in plants • Wind pollinated and insect pollinated flowers • Pollination • Self and cross pollination • Advantages and disadvantages of self and cross pollination	• Observing insect and wind • pollinated flowers • Comparing insect and a wind pollinated flower • Examining pollen grains using microscopes or hand lenses • Discussing self and cross pollination • Discussing advantages of self and cross pollination	• Flowers • Hand lenses • Microscopes • Pollen grains

SEXUAL REPRODUCTION IN PLANTS

- Flowers are the reproductive organs of plants.
- Most flowers develop seeds that grow into new plants.

Structure and functions of a flower

- You need to be able to describe the **structure** and **functions** of a named **dicotyledonous** flower.

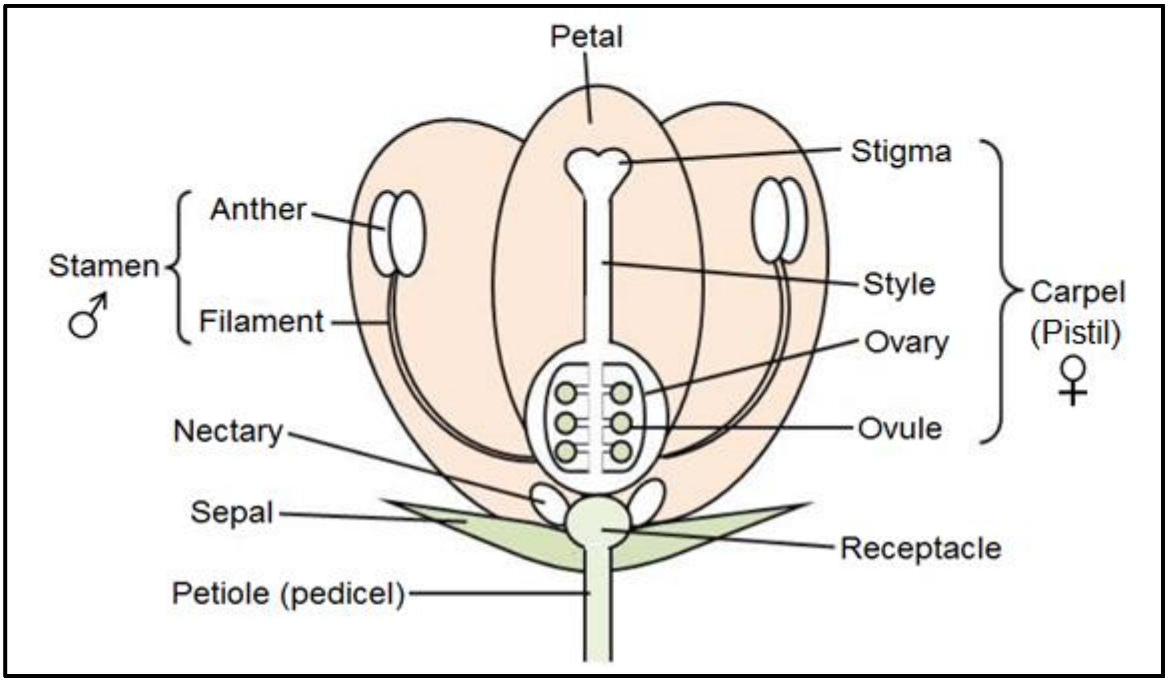


DIAGRAM: Structure of an insect-pollinated dicot flower e.g. bean flower.

- **All of the petals** of a flower are collectively known as the **corolla**.
- **Sepals** are modified leaves collectively called the **calyx**.

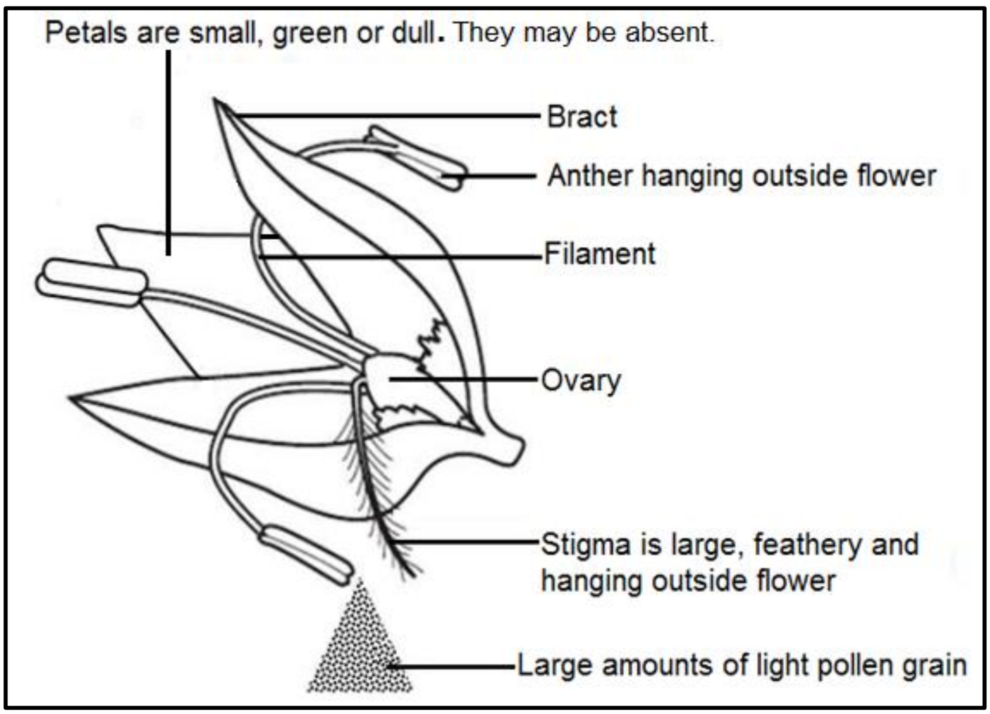


DIAGRAM: Wind-pollinated flower e.g. grass flower.

FUNCTIONS OF PARTS OF A FLOWER		
	Flower part	**Function**
1	Petal	Large, coloured and scented to attract insects/pollinators.
2	Sepal	Protects flower while in bud form.
3	Petiole (stalk/pedicel)	Supports flower to make it easily seen by insects, and to be able to withstand wind.
4	Receptacle	Connects the stalk to the flower and supports the weight of the flower or fruit when it develops.
5	Nectary	Produces nectar, to attract insects.
6	Stamen	Male reproductive part of flower. Made up of anther and filament.
7	Anther	Produce pollen grains in pollen sacs.
8	Pollen grains	Male sex cells.
9	Filament	Supports the anther.
10	Carpel (pistil)	Female reproductive part of flower. Made up of stigma, style and ovary.
11	Stigma	A sticky / feathery surface on which pollen grains are deposited.
12	Style	Links stigma to ovary. Passage through which pollen tube grows.
13	Ovary	Produce ovules.
14	Ovules	Female sex cells. Develop into seeds when fertilised.

Differences between insect and wind pollinated flowers

- **(Characteristics of insect and wind pollinated flowers)**
- **(Structural adaptation of flower parts to mode of pollination)**

Differences between insect- and wind-pollinated flowers		
Feature	**Insect-pollinated flower**	**Wind-pollinated flower**
Petals	▪ Large, brightly coloured and scented to attract insects. ▪ Guidelines for guiding insects into flower.	▪ Small, dull, usually green or absent. ▪ No guidelines.
Stamens	Inside flower so insects must make contact.	Exposed so that wind can easily blow pollen away.
Filaments	Short filaments.	Long filaments.
Anther	Anther inside flower to make contact with insects.	Anther hangs outside flower so wind can blow pollen away.
Stigma	▪ Enclosed to make contact with insects. ▪ Sticky, so that pollen attaches to insects.	▪ Exposed to catch pollen being blown by wind. ▪ Feathery, to catch pollen blown from wind.
Nectary	Nectary present as reward for insects.	Nectary absent.
Pollen Grains	▪ Larger, sticky or spiky to attach to insects. ▪ Smaller amounts.	▪ Smaller, smooth and light so easily carried by wind. ▪ Larger amounts.
Bracts (Modified leaves)	Absent	Sometimes present.

POLLINATION

- This is the transfer of pollen grains from the anther to the stigma.
- Two types of pollination are self-pollination and cross-pollination.
- Self-pollination is the transfer of pollen grains from the anther of one flower to the stigma of the same flower.
- Cross-pollination is the transfer of pollen grains from the anther of one flower to the stigma of a different flower, of the same species.

Agents of pollination

- Agents of pollination include wind, insects, birds and mammals.
- Insect pollinators include bees, butterflies and mosquitoes.

ADVANTAGES AND DISADVANTAGES SELF-POLLINATION

1. **Advantages of self-pollination:**
   a) Purity of plant variety is maintained (desirable characteristics are maintained).
   b) Flowers do not depend on agents of pollination.
   c) Wastage of pollen grains is avoided
   d) Less chances of failure of pollination.
   e) Creates new offspring when pollen grains from other plants are hard to find, especially in geographically isolated areas such as islands.
   f) Does not need to expend (waste) energy on attracting pollinators (Flowers need not produce scent, nectar or be colourful).

2. **Disadvantages of self- pollination:**
   a) Undesirable characteristics cannot be eliminated.
   b) New varieties of plants cannot be produced.
   c) Since all offspring are genetically the same, they may all be wiped out by the same disease or same environmental change.
   d) Continued self-pollination leads to weakening of progeny (Vigour and vitality of the plants decreases with prolonged self-pollination).
   e) Variability and adaptability to changed environment are reduced.

ADVANTAGES AND DISADVANTAGES CROSS-POLLINATION

1. **Advantages of cross pollination:**
   a) Cross pollination leads to the production of new varieties.
   b) More viable seeds are produced.

2. **Disadvantages of cross- pollination:**
   a) Pollination may fail due to distance barrier.
   b) More wastage of pollen grains.
   c) Need to produce more pollen to avoid pollination failure due to distance and pollen wastage.
   d) Increases chances of the occurrence of some unwanted characteristics.
   e) Cannot occur without pollinating agents.

FORM 4: FERTILISATION & SEED DISPERSAL

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (ATTITUDES, SKILLS AND KNOWLEDGE)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.15.1 Reproduction Fertilisation Seeds Fruits Seed dispersal	• outline the process of fertilisation • describe changes that occur after fertilisation in a flower • describe seed dispersal • outline the importance of seed dispersal	• Development of pollen tube **NB. No knowledge of double fertilisation is required** • Formation of seeds and fruits • Various ways of seed dispersal • Importance of seed dispersal - Colonisation of new areas - Reducing overcrowding of plants - Increasing survival chances of plants	• Examining growing pollen tubes • Describing the growth of the pollen tube and its entry into the ovule followed by fertilisation • Observing a variety of fruits and seeds • Relating structure of seed to method of dispersal • Discussing importance of seed dispersal	• Print media • ICT tools • Braille software/Jaws • Fruits • Seeds • Hand lenses

FERTILISATION IN PLANTS

- The pollen grain contains the male nucleus.
- When the pollen grain lands on the stigma of the same species, it absorbs nutrients and germinates forming a pollen tube.
- This pollen tube is nourished by the cells of the style and grows through the style.
- The pollen tube grows down the style, through the ovary wall, and through the micropyle of the ovule.
- The pollen tube carries the male nucleus from the pollen grain to the female nucleus in the ovule.
- Fertilisation (fusion of male and female nuclei) occurs in the ovule to form a diploid zygote.
- If the ovary contains many ovules, each will be fertilised by a different male nucleus brought in by a different pollen tube.

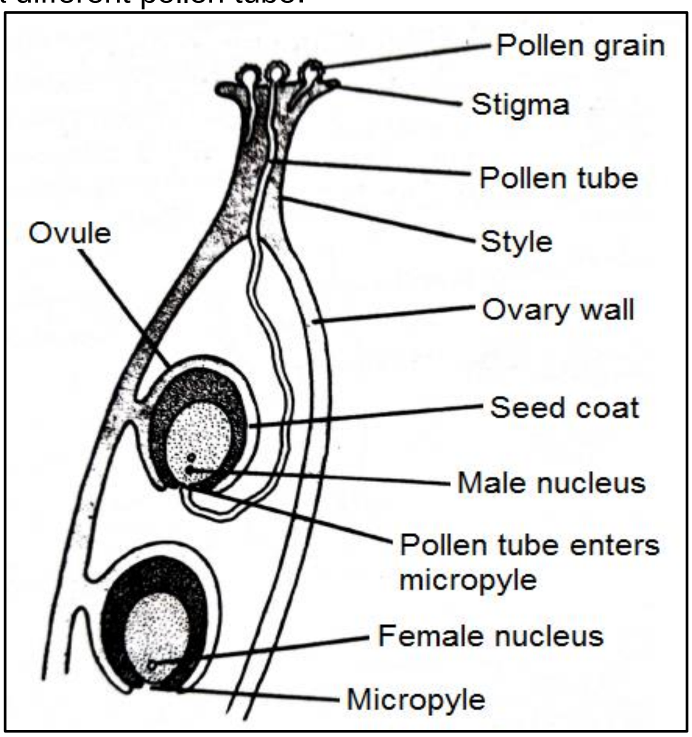


DIAGRAM: Pollen grain germination and fertilisation of ovule.

After fertilisation the following changes take place in a flower:

- The **zygote** develops into an **embryo** (plumule and radicle) and **cotyledons** (food stores).
- The **integuments** (wall of ovule) develop into **testa** (seed coat).
- The **fertilised ovules** become **seeds**.
- The **ovary** develops into a **fruit**.
- The **ovary wall** develops into **pericarp** (fruit wall).
- The **style**, **dries up** and **falls off** leaving a scar.
- The corolla (**petals**), calyx (**sepals**) and stamens **dry up** and **fall off**.
- In some the calyx (**sepals**) persists.

SEED DISPERSAL

- **Seed dispersal** is the **scattering of seeds / fruits** away from the parent plant.
- The flowers produce seeds which can be dispersed by the wind, water, animals or mechanically providing a means of colonising new areas.

5. Wind dispersal

- Wind dispersed fruits and seeds have the following features:

a) Small and light in order to be carried by air currents.
b) Hairy or feathery structures, called a pappus, which increase surface area to catch the wind like a parachute, e.g. dandelion fruit. The fruit counterbalances the pappus.
c) Wing-like structures which catch the wind and increase surface area for buoyancy, e.g. sycamore fruit, jacaranda fruit and crowfruit. When the fruit drops off the tree it spins, slowing down its descent.

- Hairy and wing-like extensions increase surface area of fruits and seeds such that they are easily carried by the wind.
- If caught by the wind the seed will be carried away from the parent plant, reducing competition for nutrients, water and light.

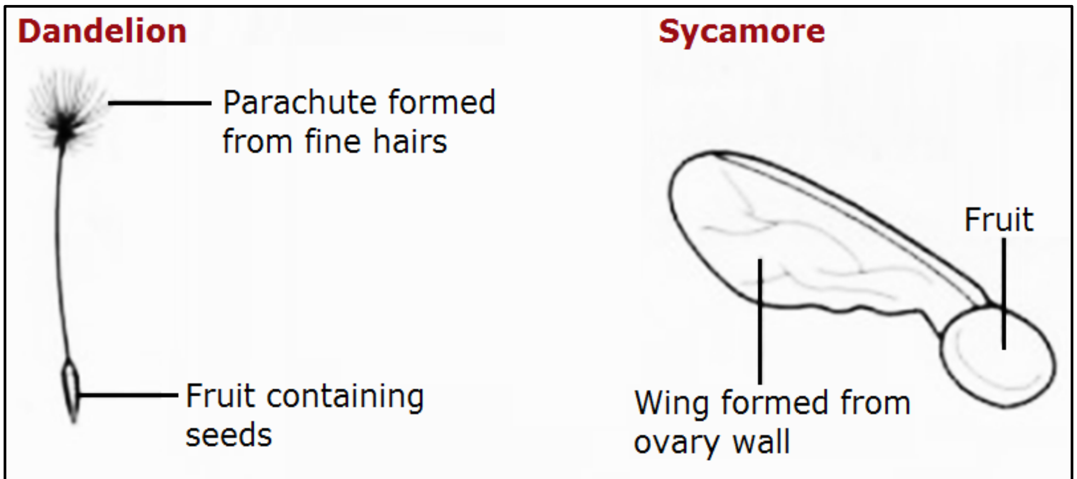


DIAGRAM: Wind-dispersed fruits.

6. Animal dispersal

- There are two main modifications of fruits for animal dispersal: **succulent / fleshy** fruits and **hooked** fruits.

a) **Succulent / fleshy fruits** attract animals because they are brightly coloured, scented and juicy and edible, e.g. tomatoes and guavas.

- When eaten, the seed pass through animal's faeces, which may be a long way from the parent plant. The faeces provide nutrients when the seeds germinate. Such seeds have hard, resistant seed coats which enable them to overcome acids and enzymes in the digestive system.

b) **Hooked fruits** have **hooks** or **spines** which stick on **animal fur** or **on clothes** as it brushes past the parent plant, e.g. black jack and burr grass. Eventually the fruits are brushed off or fall on their own. This disperses the seeds away from the parent plant.

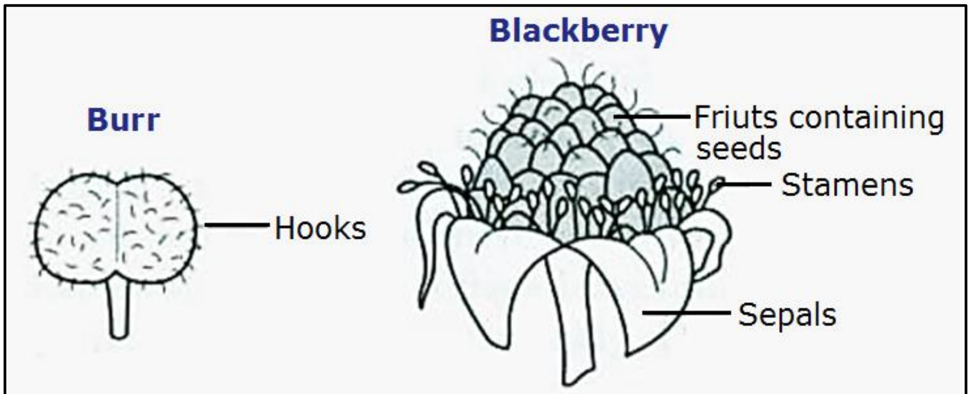


DIAGRAM: Animal-dispered fruits.

7. Self-dispersal mechanism

- Also known as **Explosive mechanism** or **Mechanical (ballistic) dispersal**.
- When dry some seed pods or pericarps (fruit walls) twist and explode to cast away their seeds from the parent plant e.g. the musasa seed pods and pea seed pods.
- Pressure inside the seed pod forces it to open along lines of weakness, twisting, exploding and throwing seeds away from the parent plant.

4. Water dispersal

- Certain fruits/seeds float on water and are carried by waves or currents in rivers or oceans to faraway places.
- Fruits like coconut have fibrous *mesocarp* which is spongy to trap air, the trapped air make the fruit light and buoyant to float on water. (Mesocarp is the middle and fleshy layer of the pericarp or fruit wall).
- Plants like water lily produce seeds whose seed coats trap air bubbles.
- The air bubbles make the seeds float on water and are carried away.
- The pericarp and seed coat are waterproof.

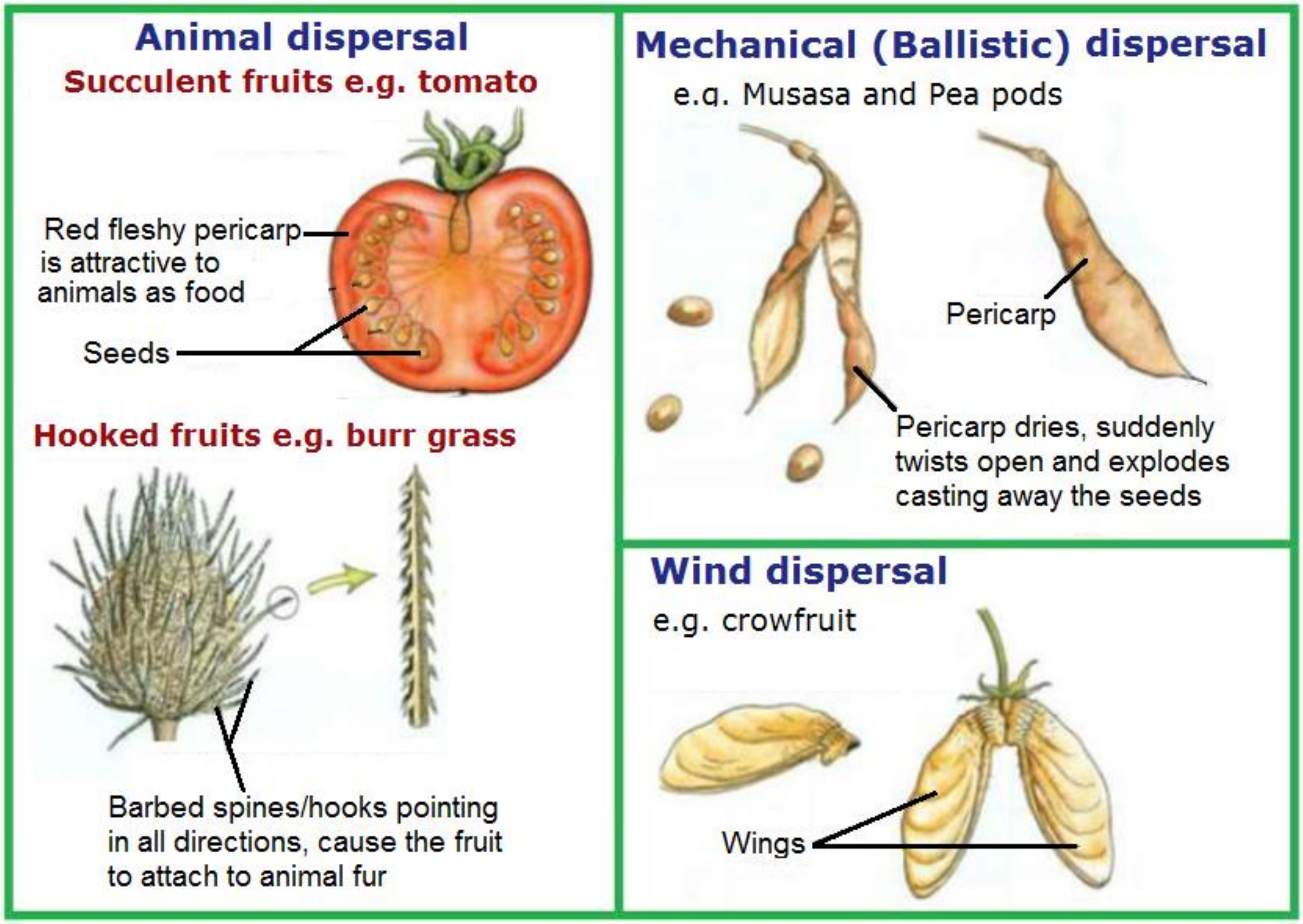


DIAGRAM: Methods of seed / fruit dispersal.

The advantages of fruit or seed dispersal

1. Reduces overcrowding and competition for food and light with parent plant
2. Increases chances for seed germination
3. Enables plants to colonise new and favourable habitats.

The disadvantages of fruit or seed dispersal

1. Dispersal depends on external agents such as wind, water or animals.
2. Can be wasteful: a great number of seed.

Try this question

Describe citing particular examples how seeds are adapted for:

(a) Animal dispersal.

(b) Wind dispersal.

(c) Mechanical /ballistic/ self-dispersal.

(d) Water dispersal.

Solution

(a) Animal dispersal adaptations

- **Fleshy fruits** and **hooked fruits.**
- **Fleshy fruits**: Brightly coloured, scented, juice, edible to attract animals. E.g.

guava and tomato fruits. Eaten by animals. Seeds are hard and have testas that are resistant to digestive enzymes and acids in animal digestive systems. Seeds passed out in faeces away from parent plant.

- **Hooked fruits:** Hooks / spines stick to animal's fur or clothes. the seeds are brushed off or fall on their own

(b) Wind dispersal adaptations

- Small / light for easy carriage by wind.
- Hairy / parachute-like structure, e.g. dandelion fruit, and wing-like structures e.g. jacaranda, to increase surface area for buoyancy / catching wind.

(c) Mechanical /ballistic/ self-dispersal adaptations:

- Seed pods have lines of weakness e.g. musasa pods.
- Seed pods twist when dry. Twisting force opens pod along lines of weakness.
- Explosion occurs. Pressure from inside casts away seeds.

(d) Water dispersal adaptations.

- Fibrous and spongy mesocarp to trap air making fruit light to float on water and to be carried by waves, e.g. coconut fruit.
- Seed coat traps air bubbles making the seeds buoyant / float on water, e.g. water lily seeds.
- The pericarp and seed coat are waterproof to prevent absorption of water during transit e.g. coconut fruit and water lily seeds.

FORM 4: SEED STRUCTURE & GERMINATION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (ATTITUDES, SKILLS AND KNOWLEDGE)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.15.1 Reproduction * Seed structure	• describe the structure of a monocot and dicot seed	• Seed structure– - testa/seed coat - embryo (radicle and plumule) - endosperm - cotyledon - hilum - micropyle - Monocot seeds – maize seeds - Dicot seeds – bean / pea/groundnut	• Identifying different parts external and internal of seeds. • Making labelled drawings of seeds.	• ICT tools • Braille software/Jaws • Variety of fruits and seeds • Scalpel blades • Hand lenses • Bean seeds • Maize seeds

Monocots and dicots

– Flowering plants are classified into two groups based on the number of cotyledons (seed leaves), as monocotyledonous plants (aka monocots) and the dicotyledonous plants (aka dicots).

– A monocot seed has one cotyledon whereas a dicot seed has two cotyledons. (Mono- means one and di- means two).

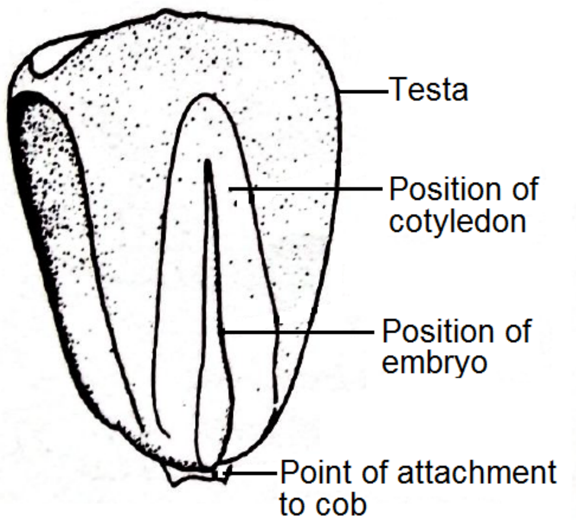
– A seed is a fertilised and mature ovule containing an embryo.

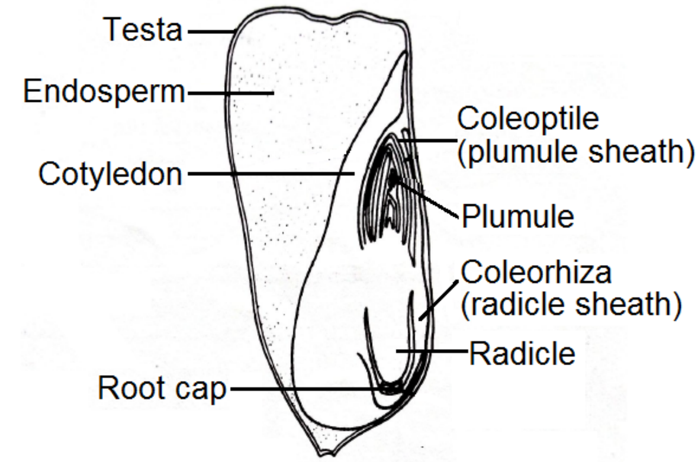
SEED STRUCTURE

Monocot seed e.g. maize seed aka maize grain.

STRUCTURE OF MAIZE GRAIN

External appearance	Longitudinal section

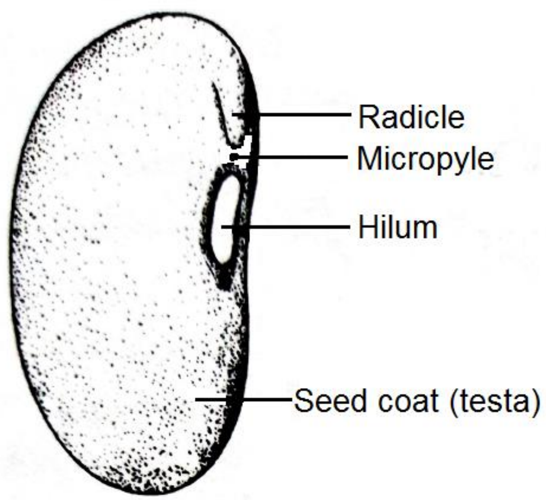


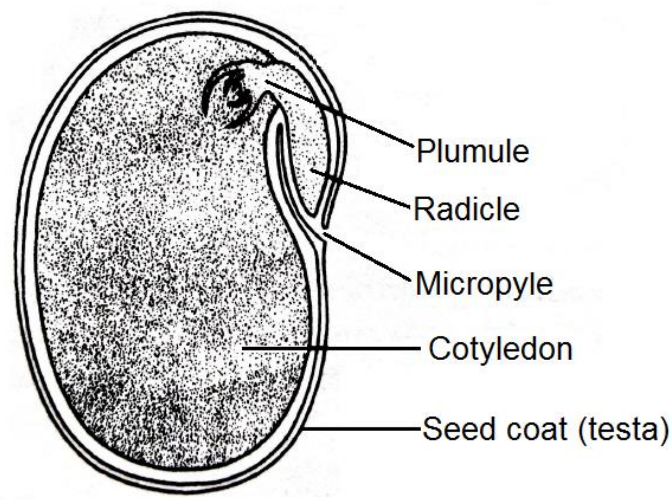


Dicot seed e.g. bean seed, pea seed and groundnut seed.

STRUCTURE OF BEAN SEED

External appearance	Longitudinal section





FUNCTIONS OF PARTS OF A SEED

	Seed part	Description	Function
1	Testa	Tough outer seed coat.	Protect seed from damage due to insects, bacteria and fungi.
2	Hilum	A scar on the testa of a bean seed.	Marks point where the stalk once attached the seed to the pod or ovary wall.
3	Radicle	V-shaped swelling beside the hilum in bean seed.	Grows to form the root after germination.

4	Micropyle	A very small pore between hilum and radicle.	Absorbs water needed for germination.
5	Cotyledons	Fleshy modified seed leaves. Monocot seed e.g. maize seed, has one cotyledon whereas dicot seed e.g. bean seed has two cotyledons.	Act as food stores which supply the nutrition needed for germination.
6	Endosperm	Starch, oil and protein reserve in cereal/monocot grains.	Surrounds/protects and nourishes the embryo in monocots.
7	Embryo	Consists of the plumule and the radicle.	Grows to form the new plant after germination.
8	Plumule	Embryonic shoot.	Grows into the shoot system.
9	Radicle	Embryonic root.	Grows into the root system.
10	Coleoptile	Plumule sheath in a monocot seed.	Pointed protective sheath which protects the emerging shoot tip or plumule.
11	Coleorhiza	Radicle sheath in a monocot seed.	Protective sheath that protects the root tip or the radicle.

Differences between monocot seed and dicot seed

	Monocot seed	**Dicot seed**
1	Example maize seed.	Example bean seed.
2	One cotyledon present in the embryo.	Two lateral cotyledons present in the embryo axis.
3	Cotyledon is small and thin and lacks food materials.	Cotyledons are fleshy and stores more food.
4	Endosperm is present and stores food.	Endosperm is absent and lacks food.
5	Radicle is protected by coleorhiza.	Coleorhiza is absent.
6	Plumule is protected by coleoptile.	Coleoptile is absent.
7	Monocot seed is usually shield-shaped.	Dicot seed is usually kidney-shaped.

PRACTICAL ACTIVITY

Aim: To investigate structural differences between monocotyledonous and dicotyledonous seeds

Materials

- Bean seeds and maize grains which have been soaked overnight.
- Scalpel or razor blades.
- Iodine solution.
- Petri-dish.
- Hand lens.

Procedure

1) Using a scalpel or razor blade make longitudinal sections (LS) of both the bean seed and the maize grain.
2) Observe the LS of the specimens using a hand lens.
3) Note any structural difference between the specimens.
4) Draw the LS of each specimen and label.
5) Put a drop of iodine solution on the cut surfaces of both specimens.
6) Note any differences in colouration with iodine on the surfaces of the two specimens.
7) On your diagrams indicate the distribution of the stain.
8) Account for the difference in distribution of the colouration with iodine in the two specimens.

Observations

Record your observations on:

1. the structural differences between monocot and dicot seeds using a table with appropriate headings, rows and columns.
2. the difference in distribution of colouration with iodine in the two seed specimens.

Conclusion

Make appropriate conclusions on;

1. differences between monocot and dicot seeds.
2. differences in the amount of starch in the two seed specimens.

FORM 4: SEED GERMINATION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (ATTITUDES, SKILLS AND KNOWLEDGE)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.15.1 Reproduction * Germination	• describe the conditions necessary for germination	• Germination - Oxygen - Suitable temperature - Water • Functions of enzymes -Respiration -Hydrolysis of stored food	• Carrying out controlled experiments to investigate conditions necessary for germination	• Pyrogallic acid • Drying agents such as calcium chloride • Refrigerator • Incubator

Germination

- **Seed germination** is the **development** of a **seed** into a **seedling**.
- Germinating seeds use their food stores in cotyledons / endosperm until the seedlings can produce their own food by photosynthesis.

- **Generally seed germination occurs as follows:**

1. At the beginning of germination water is absorbed into the seed through the micropyle and causes the seed to swell.
2. The cells of the cotyledons become turgid and active.
3. They begin to make use of the water and enzymes to dissolve and break down the food substances stored in the cotyledons.
4. The soluble food is transported to the growing plumule and radicle.
5. The plumule grows into a shoot while the radicle grows into a root.
6. The radicle emerges from the seed through micropyle, bursting the seed coat as it does so.

TYPES OF GERMINATION

- **The changes that occur during seed germination are slightly different depending on the type of seed.**
- The nature of germination varies in different seeds.
- During germination the cotyledons may be brought above the soil surface. This type of germination is called **epigeal** germination and occurs in dicot seeds e.g. bean seed.
- If during germination the cotyledons remain underground the type of germination is known as **hypogeal** and occurs in monocot seeds e.g. maize seed.

- **Germination in dicot seed e.g. bean seed is called epigeal germination. It occurs as follows:**

1. Absorption of water through micropyle.
2. Bursting/splitting of testa.
3. Radicle emerges and grows down into soil to form a root system.
4. The hypocotyl (part of embryo below cotyledons) grows upwards pulling cotyledons out of the soil. The plumule emerges between the two cotyledons.
5. Cotyledons open up, turn green to form small leaves and begin to photosynthesize.
6. The plumule forms the first true two green leaves.
7. The food stored in the cotyledons is used for growth and cotyledons shrivel and fall off.
8. The radicle gives rise to the root system while the plumule forms the shoot.
9. At this point, the young plant is called a seedling.

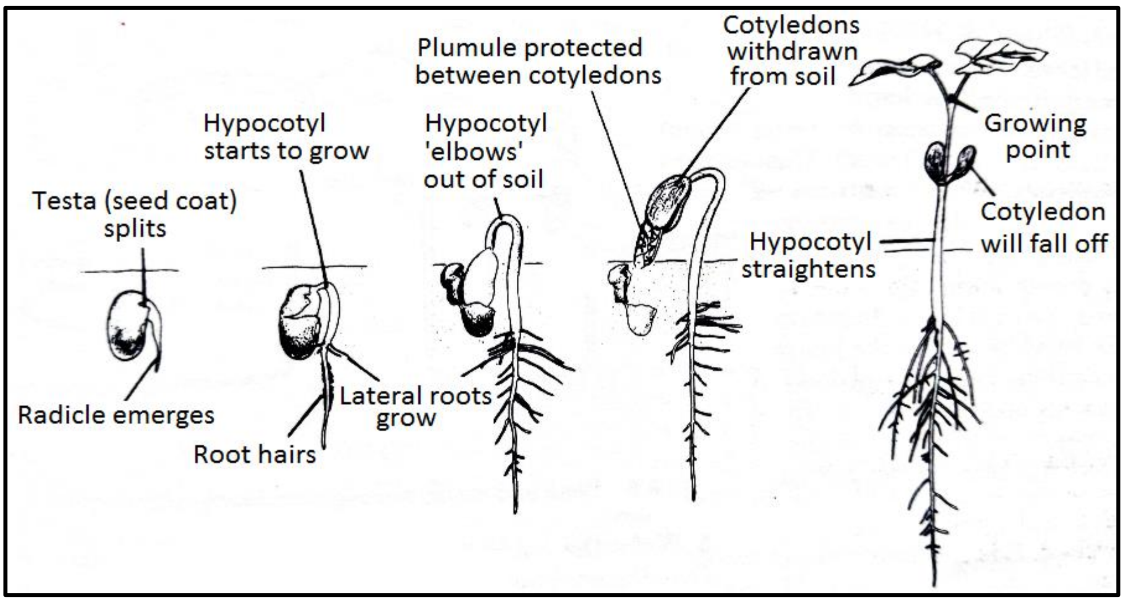


DIAGRAM: Stages in the germination of a bean seed.

- **Germination in dicot seed e.g. maize seed is called hypogeal germination. It occurs as follows:**

1. The seeds absorb water and swells.
2. The radicle grows out from coleorhiza down into soil and forms the primary root.
3. The plumule grows upwards inside the coleoptile.
4. Coleoptile turns green upon being exposed to light and bursts open to expose the first leaves.
5. Adventitious roots grow from the plumule.
6. The food stored in the endosperm is used for growth.

7. The cotyledons remain below the ground.
8. The radicle gives rise to the root system while the plumule forms the shoot.
9. At this point, the young plant is called a seedling.

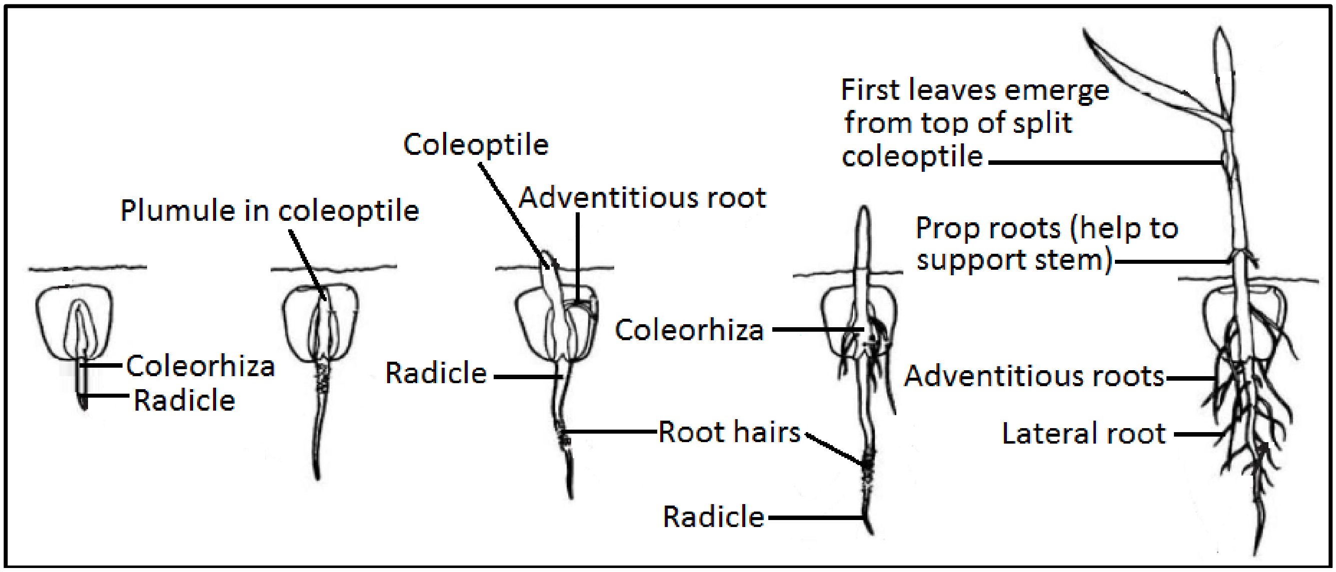


DIAGRAM: Stages in the germination of the maize seed.

Difference between Epigeal and Hypogeal Germination

- **In epigeal germination, the cotyledons emerge out of the soil during germination whereas, in hypogeal germination, the cotyledons remain inside the soil.**
- This means the hypocotyl shows a greater elongation in epigeal germination while the hypocotyl is short in hypogeal germination.
- Epigeal germination occurs in dicot seeds whereas hypogeal germination occurs in monocot seeds.

The conditions necessary for germination

- The conditions necessary for seed germination are **water**, **oxygen** and **optimum temperature**:

	CONDITION	DESCRIPTION
1	**Water**	▪ Water rehydrates the seed, softens the testa and allows all the chemical reactions involved in the growth of the embryo to take place: 1. Water activates the enzymes and provides the medium for enzymes to act and break down the stored food into soluble form. 2. Water hydrolyses and dissolves the food materials. 3. Water is also the medium of transport of dissolved food substances through the various cells to the growing region of the radicle and plumule. 4. Besides, water softens the seed coat which can subsequently burst and facilitate the emergence of the radicle.
2	**Oxygen**	▪ Oxygen is needed for aerobic respiration (oxidation of food substances stored in the seed) so that energy or ATP can be released for cell division and growth of embryo.

3	**Optimum temperature**	▪ Most seeds require a suitable temperature before they can germinate. ▪ Seeds will not germinate at very low or very high temperatures. ▪ The optimum temperature for seeds to germinate is approximately 25°C. It varies from plant to plant. ▪ At higher temperature the protoplasm is killed and the enzymes in the seed are denatured. (Protoplasm is the cytoplasm, organelles and nucleus of the cell). ▪ At very low temperatures the enzymes become inactive. ▪ Therefore, the protoplasm and the enzymes work best within the optimum temperature range. ▪ The rate of germination increases with temperature until it reaches an optimum and decreases thereafter.

Role of enzymes during germination

- Germination is a process controlled by enzymes.
- The functions of enzymes during germination include:
  1. Controlling respiration that releases energy for germination.
  2. Controlling hydrolysis of food stored in the cotyledons and endosperm to nourish the embryo during germination.
     - Enzymes play a vital role during germination in the hydrolysis / breakdown and oxidation of food.
     - Food is stored in seeds in the form of insoluble carbohydrates, lipids and proteins.
     - The insoluble food is converted into a soluble form by the enzymes.
     - Carbohydrates are broken down into glucose by amylase enzyme, lipids into fatty acids and glycerol by lipase, and proteins into amino acids by protease.
     - Enzymes are also necessary for the conversion of hydrolysed products to new plant tissues.

Role of hormones in seed germination

- Several hormones play a vital role in germination since they act as growth stimulators.
- These include gibberellins and cytokinins.
- These hormones also counteract the effect of germination inhibitors.

Controlled experiments to investigate conditions necessary for germination

Design of experiments to investigate factors affecting germination

- Firstly outline the factors needed for seed germination:
- There are 3 essential conditions required for a seed to germinate - a seed will not germinate until these external conditions are met.
- They are:

1. Suitable temperature – for maximum enzyme activity (enzymes work best at an optimum temperature - this requires enzyme and substrate to move around at a faster rate with many successful collisions can occur).
2. Water – (to rehydrate seed and soften the testa and allow the plumule to emerge.
3. Oxygen - (required for aerobic respiration) needed for energy or ATP production.

- The experimental design requires two treatments: The **control** where all factors are present and then the separate **experiment** which omits one of the factors. Therefore you can conclude that if the seed doesn't germinate then that factor omitted must be required for germination.

- The **control** acts as a comparison to the **experiment** and enables you to make a valid conclusion that the missing factor is indeed necessary for germination.
- **How to omit one of the factors:**

1. To remove oxygen you can add alkaline pyrogallol or pyrogallic acid which absorbs oxygen.
2. To remove water don't add water to the experiment.
3. To reduce temperature place in a refrigerator. To increase temperature place in a thermostat controlled incubator.

- The factors that increase the reliability of the experiment include using the same species of plant, same amount of water, same wavelength of light, repeat the investigation several times to ensure results are reliable, etc.
- There must only be one independent variable (thing you change) otherwise this would negatively affect the validity of your results.

1. Experiment to prove oxygen is necessary for germination

Aim: To prove that **oxygen** is necessary for germination.

Materials: Two conical flask marked A and B, pea seeds, water, cotton wool, pyrogallic acid, test tubes.

Procedure

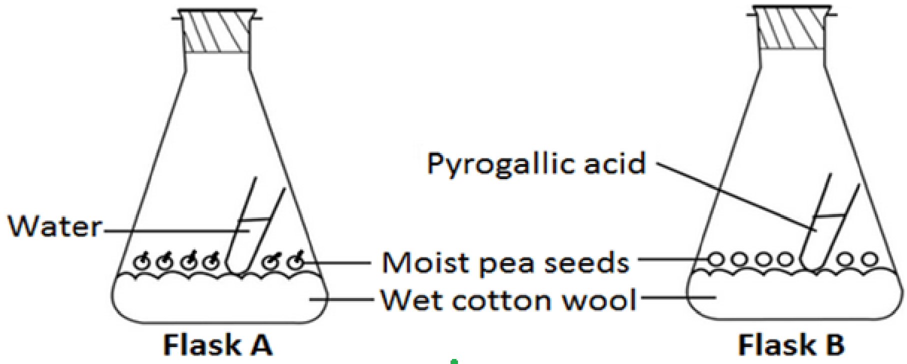
1) Take two conical flask with a cork and mark them A and B respectively.
2) Place a wet cotton wool in each flask with some soaked Pea seeds.
3) Pyrogallic acid absorbs oxygen. Place a test tube of Pyrogallic acid in flask B in such a way that the chemical doesn't drop in the flask.
4) Place a test tube of plain water in flask A.

Observation

The seeds in flask A germinate because of presence of oxygen and seeds in flask B do not germinate because pyrogallic acid absorbs oxygen.

Conclusion

Oxygen is necessary for germination.



- **Control:** All conditions present.
  - * Oxygen present.
  - *Water present.
  - *Suitable temperature.
- **Seeds germinate** since all conditions are present.

- **The experiment:** One condition omitted **No oxygen.**
  - *Water present.
  - *Suitable temperature.
- **No germination** since there is **no oxygen.**
- **Conclusion:** Oxygen is necessary for germination.

DIAGRAM: Experiment to prove that oxygen is necessary for germination.

2. Experiment to prove that suitable temperature is necessary for germination

Aim: To prove that **suitable temperature** is necessary for germination.

Materials: Two flasks marked A and B, pea seeds, water and cotton wool.

Procedure

1) Take two flasks marked A and B respectively.
2) Place some pea seeds on wet cotton wool in both the beakers.
3) Keep flask A at room temperature and B in a refrigerator.
   NB. You may place flask A in a warm closed environment controlled by a thermostat to ensure that temperature remains stable.

Observation

The seeds in flask A germinate and in B do not germinate.

Conclusion

Suitable temperature is necessary for germination.

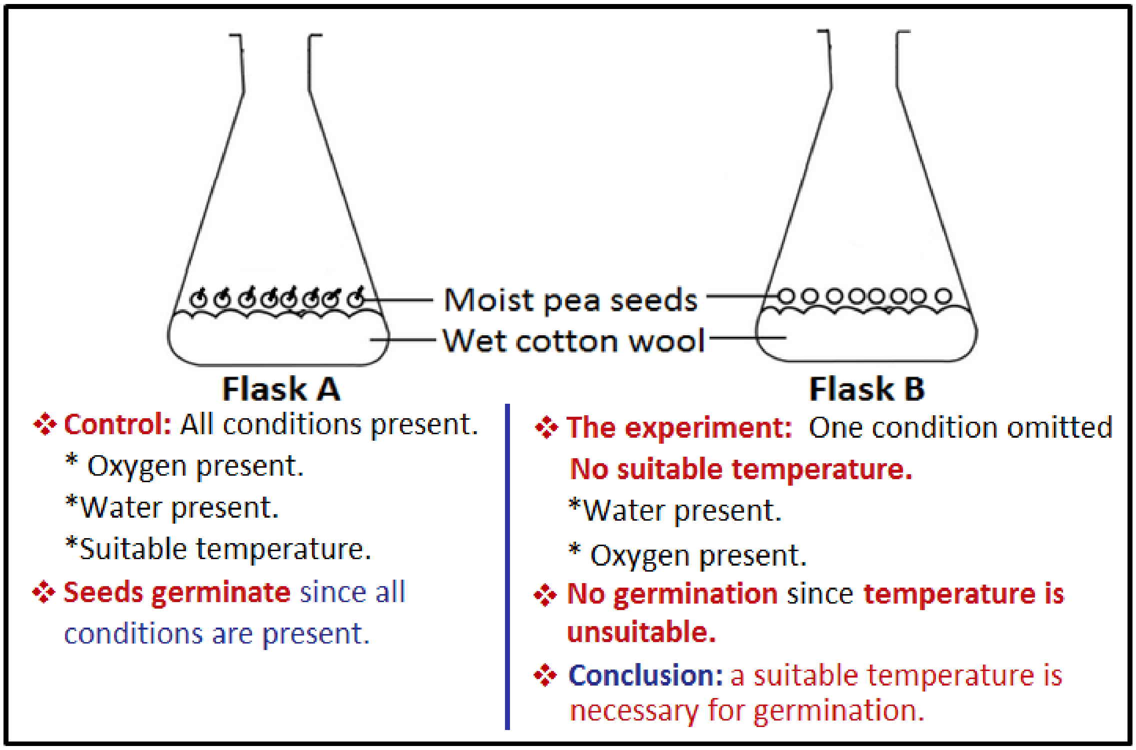


DIAGRAM: Experiment to prove that suitable temperature is necessary for germination.

3. Experiment to prove that water is necessary for germination

Aim: To prove that water is necessary for germination.

Apparatus: Two flasks marked A and B, pea seeds, water and cotton wool.

Procedure

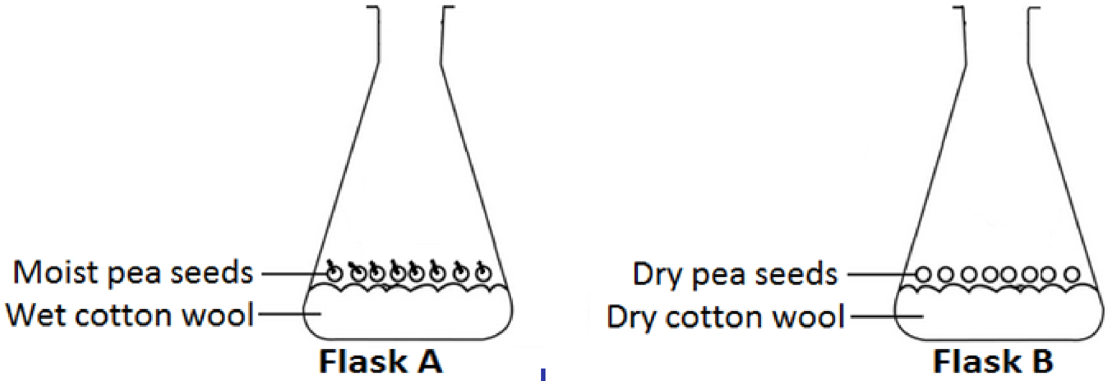
1) Take two flask which are marked A and B respectively.
2) In flask A place some pea seeds on a wet cotton wool and in flask B place some pea seeds on a dry cotton wool.
3) Keep the flasks at room temperature.

Observation

The seeds in flask A germinate and in B do not germinate.

Conclusion

Water is necessary for germination.



- **Control:** All conditions present.
  - *Water present.
  - * Oxygen present.
  - *Suitable temperature.
- **Seeds germinate** since all conditions are present.

- **The experiment:** One condition omitted
  - **No water**
  - * Oxygen present.
  - *Suitable temperature.
- **No germination** since there is **no water.**
- **Conclusion: Water** is necessary for germination.

DIAGRAM: Experiment to prove that water is necessary for germination.

GERMINATION SUCCESS

- Not all seeds germinate when planted. Basically seeds fail to germinate when one or all of the conditions necessary for germination are lacking, namely water, oxygen and favourable temperature. However, there are additional reasons why some seeds fail to germinate.
- **Reasons why some seeds fail to germinate include:**

1. Inadequate water.

   Dry conditions mean the seed doesn't have enough moisture to start the germination process.
2. Lack of oxygen.

   Overwatering results in waterlogged soil that causes the seeds not to have enough oxygen.
3. Temperature not favourable; either very low or very high.

   Enzymes are denatured by very high temperatures.

   Enzymes are inactivated by very low temperatures.

   Protoplasm of seed cells is killed by very high temperatures.
4. Seeds not yet mature or still in dormancy.
5. Planting seeds too deeply causes them to use all of their stored energy before reaching the soil surface. Those planted too shallow may wash away, fail to germinate or be eaten by wildlife.
6. Cloddy and compacted soils that are high in clay will inhibit seed germination and emergence.
7. Soils that form a crust when dry may also prevent effective germination and emergence.

8. Planting seeds in the wrong season.

   The growth of some plants is season-specific. Not all plant seeds can be planted throughout the year because each season has different temperatures. Successful germination of seed of different plants falls within a range of optimum temperatures. E.g. maize seeds germinate poorly in winter whereas wheat seeds germinate very well in winter.

   It is always best-practice to first consult the instructions on the back of the packet of seed. Seed advised for sowing in spring/summer will have a hard time emerging in autumn/winter and vice versa.

9. Loss of viability due to:

   a) depletion of food reserves due to prolonged storage of seeds.

   b) damage by pests and diseases.

   c) seed age.

   d) incorrect storage.

   Only seeds whose embryos are alive and healthy will be able to germinate and grow. Seeds stored for long periods usually lose their viability due to depletion of their food reserves and destruction of their embryo by pests and disease.
   Incorrect storage of seeds, e.g. at high heat, scorching sun and high humidity, can shorten the longevity of a seed. When buying seed from your local store always check the Best-Before date.
   The rate of germination and number of seeds germinating decrease as seed age. Seed past their expiration date may have degraded quality.

- **Germination success is measured in terms of percentage germination and is calculated using this equation:**

$$\text{Percentage germination} = \frac{\textit{Number of seeds which germinated}}{\textit{Number of seeds planted}} \times 100\%$$

Worked example:

A farmer plants in a seed bed 25 rows of 50 seeds. Calculate the percentage germination if 50 seeds fail to germinate.

Solution

Number of seeds planted = 25 x 50

= 1250

Number of seeds which germinated = 1250 – 50

= 1200

$$\text{Percentage germination} = \frac{\textit{Number of seeds which germinated}}{\textit{Number of seeds planted}} \times 100\%$$

$$= \frac{1200}{1250} \times 100\%$$

$$= \underline{96\%}$$

Disadvantages of poor germination

1. Replanting has to be done.
2. Replanting delays the crop leading to late harvesting.

   Late planted crops mature late and fetch poor prices on the market as compared to early planted crops.

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